Belief Records and the Price of Disaster Fear

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Abstract

Index options price rare disasters, yet the disaster distribution they imply need not coincide with the physical distribution measured from macroeconomic data (Backus et al., 2011). We study this disconnect in a complete-market Lucas economy where power-utility investors hold one disaster model at a time and revise it only when its cumulative forecasting record becomes sufficiently unlikely under that model (Ortoleva, 2012). In equilibrium the option-implied disaster intensity factors exactly into the physical disaster law and an endogenous belief multiplier, so inverting option prices through a representative-agent benchmark recovers a belief aggregate, not the physical rate. That belief aggregate affects option prices differently across maturities: short-dated options primarily reflect the contemporaneous cross-section of surviving beliefs—a complacent level any Markov benchmark reproduces—while longer-dated tails increasingly price the revisions to come, whose timing the surviving record governs. The tail term structure therefore varies with that record, revealing an additional state coordinate the near tail omits.

1. Introduction

Index options price rare disasters, and the disaster distribution they imply is not the one the macroeconomic record measures. The physical disaster sizes of Barro (2006), inverted through a frictionless representative-agent kernel, overstate the tail options price on average; yet that same risk-neutral tail spikes far above any physical benchmark after a crisis (Backus et al., 2011; Bollerslev and Todorov, 2011; Andersen et al., 2015b). The surface is pricing investors' disaster *beliefs*, not the physical law—and which feature of those beliefs it prices depends on the maturity. We argue that when those beliefs are revised *lazily*, on a record, this history dependence makes the *term structure* of the implied tail non-Markov: it leaves no mark on the level—a belief multiplier accounts for that—and shows only in the slope. The near-dated tail, the deepest short maturities the disaster-option literature has measured, reflects the current cross-section of beliefs. The term structure of the tail reveals something the cross-section leaves out: the *records* under which those beliefs have survived, the histories of ordinary and surprising data each held view has so far weathered. This paper is about that stratification. Two markets can imply the *same* near tail and carry different forward surfaces, because their beliefs sit on different records; and the slope of the tail, where the records show, is the one option-tail object no model whose state is the current cross-section can produce.

The economy that delivers this is a complete-market Lucas economy with rare consumption disasters and a continuum of CRRA investors, each holding a single model of the disaster rate and revising it lazily—keeping it while the record stays unsurprising under it, abandoning it for the best alternative at a surprise—who from common priors and common data come to disagree endogenously. In equilibrium the risk-neutral disaster intensity factors exactly,

$$U_t \;=\; U^{\mathrm{RA}}\,\kappa_t,$$

into a representative-agent benchmark U^{RA}—the intensity a correct-belief agent prices off the macro-measured physical law, fixed by that law and risk aversion—times a *belief multiplier* κ_t. The kernel does not price the average belief: it aggregates the held models by a wealth-weighted *power mean* of their disaster likelihood ratios, of order $1/\alpha$ equal to the common risk tolerance, so that risk aversion damps belief before it reaches prices and the arithmetic average is only the log-utility knife-edge (Jouini and Napp, 2007; Cvitanić and Malamud, 2011). Inverting the near tail through this kernel returns not the physical intensity $\lambda^\star$ but $\lambda^\star\kappa_t$: what the disaster-option literature recovers is a wealth-weighted aggregate of beliefs—the cross-section S_t—and the multiplier is the wedge between it and the physical law that Baron et al. (2024) and Chen et al. (2012) document.

The split is structural, not an artifact of the disaster setting. The fear factor U_t, the multiplier κ_t, the short rate, and the risk-neutral size shape are functionals of the current cross-section S_t alone. The price–dividend ratio, the level of the disaster premium, and the term structure of the tail are not: they are functionals of a richer state Ψ_t, the cross-section together with the since-adoption records, ages, and revision hazards under which its models have survived. The reason is that a lazily held belief is not a sufficient statistic for its own revision—the since-adoption record governs the revision hazard—so the pricing state *enlarges* from S_t to Ψ_t, and survival history enters the forward surface as a separate coordinate (Teguia, 2026). Two dates with the same near tail therefore carry identical instantaneous prices yet different price–dividend ratios, premium levels, and tail term structures. A *Markov benchmark*—any model whose pricing state is the current cross-section, a representative Bayesian with its sufficient posterior or an affine latent-state filter among them—pins the entire forward surface to that state. What sets lazy revision apart is not a rival belief model but a property its pricing state lacks: Markovianity in the cross-section.

The near tail's level is already informative, and it is shared. A power mean of order below one is pulled toward the smaller likelihood ratios, so a cross-section right *on average* about how often disasters strike still prices the tail strictly below the representative-agent benchmark—dispersion alone is *complacency*, the shortfall growing in the spread of views and in risk aversion. The inverted intensity $\lambda^\star\kappa_t$ then runs below the macroeconomic disaster frequency, the thinner far tail index options imply on average (Backus et al., 2011). This level is not ours to claim: a complete-market economy of heterogeneous *Bayesian* learners clears through the same power-mean aggregator and inherits the same complacency, a property of CRRA preferences and market completeness, not of lazy revision (Jouini and Napp, 2007; Cvitanić and Malamud, 2011). The far-tail gap of Backus et al. (2011), the investor-fears wedge of Bollerslev and Todorov (2011), and the unspanned tail factor of Andersen et al. (2015b) all sit at this front of the curve, the level any Markov benchmark reproduces.

The slope is the paper's falsifiable claim, and it is open—no model in the field produces it, and the data have not yet tested it. The record's imprint grows with maturity—two dates with the same near tail share the front of the curve and part only further out—so the test is a residual one. After controlling for the current option state, survival-record proxies—time since the last disaster, accumulated calm surprise, distance to a rejection boundary—should forecast the slope of the left-tail term structure and longer-horizon tail premia, while any Markov benchmark sets that residual to zero. The raw path-dependence documented in the surface runs the other way—strongest at the short end and *weakening* with maturity (Guyon and Lekeufack, 2023), the current-state-spanned part any Markov model reproduces because S_t already encodes the recent path—so ours is the orthogonal residual that survives the control and *grows* as that raw effect fades.

What the field has measured sits on the shared side of that line (Table III). The persistent unspanned tail factor of Andersen et al. (2015b) and the abrupt, slow-bleeding jump premium of self-exciting models are reproduced by a Markov latent state or a clustering intensity; the record residual is not. The closest neighbor, Dew-Becker et al. (2025), filters a latent value smoothly into a perceived intensity that is its own Markov pricing state, delivering the leverage effect and volatility dynamics we concede; ours needs Ψ_t, so the pricing state, not the shared aggregator, separates the two. The one directional prediction the level itself carries is that the inverted intensity $\lambda^\star \kappa_t$ is countercyclical, below the macroeconomic frequency in calm and rising after disasters.[1]

[1] A countercyclical *priced* intensity is consistent with the procyclical *physical* tail of Gormsen and Jensen (2025), who find the conditional return distribution more left-skewed in good times: their object is the physical

An individual belief does not drift—it stays fixed through calm and jumps only at a rejection—so the cross-section moves in episodic steps, synchronized across investors at a disaster and staggered through calm, and the disagreement driving it is manufactured from a common prior by that staggered timing rather than assumed. Two forces then move the multiplier, both from the concavity of the power mean: a disaster pays the investors who feared it, transferring wealth at a stroke toward the views that anticipated it and lifting the priced tail, while the calm that follows reverses this slowly, as an unsurprising record refutes the complacent only gradually and the mean held intensity falls at rate $-(1/\alpha)\,\mathrm{Var}_{S_t}(\lambda)$. The fear factor is therefore *fast in and slow out*—a sharp post-crisis spike eroding through calm. Left to itself the calm-spell selection would drive the cross-section to agreement—not because revision stops, since a correct-on-average view is rejected and re-adopted forever (Teguia, 2026), but because the survivors converge—so demographic turnover (Blanchard, 1985; Yaari, 1965; Gârleanu and Panageas, 2015) restores the dispersion and delivers a unique stationary law in which the cycle recurs. There the two opposed features of the tail become one: the thinner far tail on average is the multiplier's level through the long calm spells, the post-crisis spike the same level lifted at a stroke.

Two limits bound the interpretation. The calibration of Section 5 is illustrative—the belief block is pinned to a few stylized option-tail targets, for want of disagreement data on disaster intensity—and makes no out-of-sample claim. And preferences are CRRA, not Epstein–Zin: the power-utility choice is what keeps the kernel and the bridge exact, at the cost of the usual smooth-rate/large-premium tension we do not resolve and a safe-rate level that is an artifact of the preference class. Where the intensity-channel sign is pinned a contrary cyclicality refutes the model; where the joint-channel condition is only flagged, that movement could be the condition instead.

Related literature. The paper adds no disaster process to rare-disaster asset pricing (Barro,

distribution, ours the risk-neutral one, and the physical law here is fixed, so the countercyclicality is in what the market prices. Their finding is a challenge to disaster-based models we meet by reconciliation rather than a prediction we reproduce.

2006; Wachter, 2013): the physical law is fixed, and a heterogeneous, lazily revising cross-section moves the priced tail. From heterogeneous-belief pricing and market selection (Jouini and Napp, 2007; Blume and Easley, 2006) it borrows the consensus power-mean kernel and the wealth-reallocation dynamics, applies them to a disaster law learned only from rare events (Chen et al., 2012), and derives the disagreement endogenously. We claim neither the aggregator nor the complacency it implies—both are properties of complete-market CRRA pricing—and locate the contribution in the disaster tail and in the records that move it. The broader theme is the one the companion single-agent paper develops (Teguia, 2026): a large literature lets rare events and slow learning generate *persistent*, belief-driven dynamics (Collin-Dufresne et al., 2016), but in every case the persistence is a property of the *belief*—the posterior is the pricing state, and two markets that share it share their forward surfaces. The enlargement here is different in kind: the coordinate that governs the forward tail is the record under which beliefs have survived, which the cross-section does not encode, so two markets with the *same* cross-section can price the tail's term structure differently.

The mechanism sits among the disaster-learning models without being one of them: they reach persistent post-crisis fear by learning a fixed parameter or filtering a latent state, while here no parameter is learned and no latent state moves, the priced object shifting only as wealth reallocates across an *unconverged*, non-Bayesian cross-section refreshed by turnover. Closest is Dew-Becker et al. (2025), whose filtered intensity is its own Markov pricing state and delivers the leverage effect and volatility dynamics we concede; our object is the disaster intensity, our reversion a wealth effect, and the pricing state—not the shared aggregator—is what separates the two. Against non-Bayesian revision (Ortoleva, 2012; Cho and Kasa, 2015) the contribution is the equilibrium asset-pricing consequences, not the rule; against price-extrapolation (Bastianello and Fontanier, 2026), in which traders mislearn *from prices*, our investors learn the disaster law from the rare-event record, with no price–belief feedback; against robust learning (Hansen and Sargent, 2010), whose countercyclical, history-dependent uncertainty premium is a single agent's pessimistic *distortion* of a sufficient Bayesian posterior and stays Markov in it, ours is a wealth reallocation across *undistorted* heterogeneous beliefs, with the survival record and not the posterior as the state; and where Lanzani (2025) has confidence erode with unexplained evidence, our investors occupy the pole that commits to a single model and tests it.

The paper proceeds as follows. Section 2 sets up the model—primitives, the revision rule, and the equilibrium kernel—and proves the bridge, the instantaneous prices visible in the near tail. Section 3 derives the synchronized belief dynamics, the endogenous emergence of disagreement, the revision hazard, and the law of motion of the pricing state, where the enlargement to Ψ_t is made precise. Section 4 prices the economy, separating the instantaneous prices that load on S_t from the forward-looking prices that load on Ψ_t, and assembles the fast-in/slow-out dynamics around a disaster. Section 5 adds demographic turnover, proves stationarity, and weighs the belief multiplier against the option-implied and macroeconomic evidence. Section 6 concludes; proofs, and a forecast-data signature of lazy revision, are in the appendices.

2. The model and the disaster-fear bridge

We build the economy on one modeling choice: investors agree about everything except the law of disasters. The setting is a continuum-of-agents Lucas tree whose dividend carries Brownian risk and rare downward jumps drawn from a continuous size distribution. Investors share the diffusion—which the continuous record pins down on any interval—and dispute the disaster *law*. In the baseline they dispute its arrival rate λ, the size law held common; disagreement over the size index θ is a transparent extension, developed in closed form in the Internet Appendix. We carry the joint law $\phi = (\lambda, \theta)$ as the general object and read every result in the *intensity channel* ($f_\theta \equiv f$) unless the size extension is flagged. Every quantitative result is intensity-channel; size disagreement is what makes the pricing state infinite-dimensional (Proposition 3) and is otherwise developed in the Internet Appendix. The object to track is the cross-section of held ϕ and how it moves; every downstream object—the belief dynamics, the pricing kernel, the disaster-fear wedge—is a functional of it. Why the disaster law, and not the diffusion, is the admissible thing to disagree about—the diffusion and the realized jumps are pinned down by the consumption path itself, whereas the intensity and the size law are learned only from the rare disasters as they accrue—is established in Appendix A.

A. The aggregate endowment

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions, with $\mathbb{P}$ the physical law. On it live a standard Brownian motion W, a Poisson process N of intensity $\lambda^\star$ independent of W, and an i.i.d. sequence of disaster sizes $\{J_k\}_{k\geq 1}$ independent of (W, N). Write $\{T_k\}_{k\geq 1}$ for the jump times of N and collect the disasters in the integer-valued random measure

$$\mu(dt, dx) = \sum_{k\geq 1} \delta_{(T_k, J_k)}(dt, dx)$$

on $[0, \infty) \times \mathcal{J}$, a marked Poisson process in the sense of Brémaud (1981).

Assumption 1 (Primitives of the endowment): *The drift $\mu_0 \in \mathbb{R}$, the diffusion coefficient $\sigma > 0$, and the disaster intensity $\lambda^\star > 0$ are constants known to the modeler (the fixed physical law, not common knowledge among investors), and the size support $\mathcal{J} = [\underline{J}, \overline{J}] \subset (0, \infty)$ is a known compact interval. The disaster-size law F_J is a Borel probability measure on $\mathcal{J}$ with continuous, strictly positive Lebesgue density f. Aggregate consumption (the Lucas-tree dividend) $C_t > 0$ solves*

$$\frac{dC_t}{C_{t-}} = \mu_0\, dt + \sigma\, dW_t - \int_{\mathcal{J}} \left(1 - e^{-x}\right) \mu(dt, dx), \qquad C_0 > 0, \tag{1}$$

so that, in logs,

$$d \log C_t = \left(\mu_0 - \tfrac{1}{2}\sigma^2\right) dt + \sigma\, dW_t - \int_{\mathcal{J}} x\, \mu(dt, dx) : \tag{2}$$

between disasters C is geometric Brownian, and at T_k it falls by $C_{T_k} = C_{T_k-} e^{-J_k}$, i.e. $\Delta \log C_{T_k} = -J_k$. The filtration $(\mathcal{F}_t)$ is the $\mathbb{P}$-augmentation of the natural filtration of the path C.

Assumption 1 draws a deliberate line between what investors agree on and what they may dispute. Under $\mathbb{P}$ the disaster measure μ has compensator

$$\nu(dt, dx) = \lambda^\star \, dt \, F_J(dx), \tag{3}$$

the arrival rate $\lambda^\star \, dt$ times the size law $F_J(dx)$. Investors share the drift μ_0, the diffusion σ, and the support $\mathcal{J}$, and agree that disasters arrive as a marked Poisson process; what they dispute is the disaster *law* itself—both how often disasters strike and how large they tend to be. The reason the line falls here is informational: the diffusion is common knowledge because the continuous record pins it down on any interval in the high-frequency limit (Appendix A), whereas the disaster law is learned only from the rare disasters as they accrue.

B. The disputed object and the belief rule

An investor holds one dogmatic model of the disaster process: a pair $\phi = (\lambda, \theta)$ consisting of an arrival intensity λ and an index θ of the size law F_θ. Both are parameters of a distribution—the intensity of an arrival law, the index of a size law—and both are disputed. The true process has intensity $\lambda^\star$ and size law F_J; neither is known to the investor, and her held pair is tested against the record as a whole.

The held model is a point $\phi = (\lambda, \theta)$ in the compact parameter set $\Phi := \Lambda \times \Theta$, with $\Lambda = [\underline{\lambda}, \overline{\lambda}] \subset (0, \infty)$ containing $\lambda^\star$ in its interior and $\Theta \subset \mathbb{R}$ a compact interval. Under ϕ the disaster process is a marked Poisson process with intensity λ and i.i.d. marks drawn from F_θ on $\mathcal{J}$, where $\{F_\theta : \theta \in \Theta\}$ has Lebesgue densities f_θ satisfying the identification and regularity conditions of Assumption 4 (Appendix A), verified there for the families used in the paper.

Two facts about the disputed object are used throughout. First, the held and physical laws are mutually absolutely continuous on every finite horizon (Lemma 1), so the likelihood ratios that drive revision are well posed. Second, the truth diverges from a held model ϕ at a per-period Kullback–Leibler rate $\mathcal{K}(\phi)$ (Definition 5) that splits additively into a timing part and a size part and is minimized at a unique pseudo-true model $\phi^\star = (\lambda^\star, \theta^\star)$ (Lemma 2): the held intensity is asymptotically pinned to the truth $\lambda^\star$, while whether the size family is correctly specified is left open and matters only in Section 3.

The disputed primitives are parameters of laws, not realized outcomes: a Dirac mass on a fixed disaster size is falsified by the first continuous draw, whereas disputing the size *law* and the arrival *rate* through their parameters is the faithful object, the one the data reveal only slowly (full support, Assumption 4(i)).

The belief rule is the dynamic hypothesis-testing rule of Ortoleva (2012), applied to the joint model $\phi = (\lambda, \theta)$; the dynamics it generates are derived in Section 3.

Definition 1 (Dynamic Ortoleva rule on the joint model): *All objects are adapted to the public filtration $(\mathcal{F}_t)$, the $\mathbb{P}$-augmentation of the natural filtration of the path C (Assumption 1). Investor i has a permanent surprise threshold $\varepsilon^i \in (0,1)$ and a* prior over models*: a second-order posterior ρ^i_t on the menu Φ, the Bayesian update of a prior ρ_0 over models by the likelihood of the* entire *marked-Poisson record on $[0,t]$,*

$$\rho^i_t(\phi) \;\propto\; \rho_0(\phi)\, e^{-\lambda t}\, \lambda^{N_t} \prod_{k:\, T_k \le t} f_\theta(J_k), \qquad \phi = (\lambda, \theta), \tag{4}$$

which is $\mathcal{F}_t$-measurable and accumulates on the full history. Her held model $\phi_i(t) \in \Phi$ is piecewise constant, equal to the mode of her posterior ρ^i_t over models at her most recent adoption date and held until the next rejection. The rule runs as follows.

Step 0 (initial model—the best given the prior over models). *At $t = 0$ the investor adopts the maximum-a-posteriori model of the prior ρ_0,*

$$\phi_i(0) \;=\; \phi_0 \;\in\; \arg\max_{\phi \in \Phi} \rho_0(\phi), \qquad a^i_0 = 0,$$

the common starting model under the homogeneous start of §C. Write a^i_t for the most recent *adoption date up to* t *and* $\tau_t^i = t - a_t^i$ *for the* regime age.

Step 1 (the test—the since-adoption history against the held model). *The held model is tested* only *on the history accrued since it was adopted: the record on the window* $(a_t^i, t]$*,*

$$\mathcal{R}_t^i \;=\; \big(n_t^i,\ (J_k : a_t^i < T_k \le t)\,\big), \qquad n_t^i \;=\; N_t - N_{a_t^i}, \tag{5}$$

its count and realized sizes, both $\mathcal{F}_t$*-measurable.*[2] *Under the held model* $\phi = (\lambda, \theta)$*—which makes the disaster process a marked Poisson process of intensity* λ *with i.i.d. marks* F_θ *(Assumption 4)—a window of length* Δ *has predictive law* $Q_{\phi,\Delta}$*, the law of the window record* under the held model's own measure $\mathbb{P}^\phi$*, with density (against counting measure on the count and Lebesgue measure on the sizes)*

$$q_{\phi,\Delta}\big(n, (x_1, \ldots, x_n)\big) \;=\; \underbrace{e^{-\lambda\Delta}\,\frac{(\lambda\Delta)^n}{n!}}_{\textit{count: }\mathrm{Poisson}(\lambda\Delta)} \;\cdot\; \underbrace{\textstyle\prod_{j=1}^n f_\theta(x_j)}_{\textit{sizes: i.i.d. } F_\theta}. \tag{6}$$

Departure of the realized record from the held model is measured by the deviance*—the saturated count deviance at the empirical rate* n/Δ *plus the size log-likelihood ratio of the record against its best in-family fit* $\hat\theta$ *(the size MLE on the realized sizes),*

$$D_{\phi,\Delta}(\mathcal{R}) \;=\; \underbrace{\Delta\, d_P\Big(\tfrac{n}{\Delta}\,\Big\|\,\lambda\Big)}_{\textit{count}} \;+\; \underbrace{\sum_k \log\frac{f_{\hat\theta}(J_k)}{f_\theta(J_k)}}_{\textit{size}}, \qquad d_P(a\|\lambda) = a\log\tfrac{a}{\lambda} - a + \lambda, \tag{7}$$

the Poisson count deviance (with $d_P(0\|\lambda) = \lambda$*) plus the parametric size deviance, each non-negative and zero only at a record the held model fits exactly. The* surprise *of a record is its*

[2] The record is recovered from the path by Lemma 3(ii), which rests only on Assumption 1 and is independent of the rule, so the forward reference is non-circular, and the since-birth count N_t enters only through n_t^i. Because the held model is time-homogeneous, the predictive law $Q_{\phi,\Delta}$ below depends on the window only through its length $\Delta = \tau_t^i$, so the test window is never empty once $\tau_t^i > 0$.

predictive deviance tail under $Q_{\phi,\Delta}$,

$$\bar{\varepsilon}(\mathcal{R};\phi,\Delta) \;=\; Q_{\phi,\Delta}\big(\{\mathcal{R}' : D_{\phi,\Delta}(\mathcal{R}') \geq D_{\phi,\Delta}(\mathcal{R})\}\big). \tag{8}$$

The held model passes *the test at* t *iff the realized record is unsurprising,* $\bar{\varepsilon}(\mathcal{R}^i_t;\phi_i,\tau^i_t) > \varepsilon^i$*—equivalently, iff* $\mathcal{R}^i_t$ *lies in the mass-*$(1-\varepsilon^i)$ *typical set, the acceptance region* $A_{\tau^i_t}(\phi_i,\varepsilon^i) = \{\mathcal{R} : D_{\phi_i,\tau^i_t}(\mathcal{R}) < r_{\phi_i,\tau^i_t}(\varepsilon^i)\}$ *with* $r_{\phi,\Delta}(\varepsilon)$ *the* $(1-\varepsilon)$*-quantile of* $D_{\phi,\Delta}$ *under* $Q_{\phi,\Delta}$.

Step 2 (keep, or update and re-select). *As long as the held model passes, it is kept and only the regime age advances* ($\dot{\tau}^i_t = 1$). *It is* rejected *at the first instant the record leaves the typical set,* $\tau^{\mathrm{rev}} = \inf\{t > a^i : \bar{\varepsilon}(\mathcal{R}^i_t;\phi_i,\tau^i_t) \leq \varepsilon^i\}$*, an* $(\mathcal{F}_t)$*-stopping time. At a rejection the investor (i)* updates the prior over models on the full history*—the posterior* ρ^i_t *of* (4) *evaluated at that instant—and (ii)* re-selects *the new model as the tie-broken mode of that posterior. Fix once and for all the lexicographic total order* $\preceq$ *on* Φ*:* $(\lambda,\theta) \preceq (\lambda',\theta')$ *iff* $\lambda < \lambda'$*, or* $\lambda = \lambda'$ *and* $\theta \leq \theta'$*. Let* $\mathfrak{m}(\rho) := \min_{\preceq} \arg\max_{\phi\in\Phi} \rho(\phi)$ *denote the* $\preceq$*-smallest maximiser; it exists and is unique because* $\arg\max_{\phi\in\Phi}\rho$ *is a non-empty compact subset of the compact* Φ *(the posterior is continuous on* Φ*), on which the lexicographic minimum is attained. The investor adopts*

$$\phi'_i \;=\; \mathfrak{m}(\rho^i_t) \;\in\; \arg\max_{\phi\in\Phi} \rho^i_t(\phi), \tag{9}$$

resets the adoption date to that instant (so τ^i *restarts at* 0*), and returns to Step 1.*[3]

The rule's dynamic content is what the rest of the paper turns on. Because the tolerance is held *fixed*, even a correct-on-average model is rejected and re-adopted without end (Lemma 5): revision is recurrent, not terminal. The second-order posterior, updated on the full record and never reset, nonetheless concentrates the re-selected mode at the pseudo-true $\phi^\star$ in the long run (Berk, 1966)—slowly, at the rare-disaster rate of Lemma 3—so recurrent rejection and a

[3]The menu Φ permits re-selection in either direction in each coordinate, so no model is absorbing; since re-selection may return a model near the incumbent, the belief-moving event is the *switch* $\phi'_i \neq \phi_i$, not the rejection itself.

consolidating destination coexist, the trigger being frequentist and the re-selection Bayesian.[4] The since-adoption window gives a held model a fresh sample on which to fail, and the surviving record—not the current belief—is what organizes the dynamics derived in Section 3.

C. The population, and a homogeneous start

A continuum of investors, indexed by $i \in [0, 1]$, watch the same path (1) and share everything in it but one thing. They share the undisputed primitives of Assumption 1—the drift μ_0, the diffusion σ, the support $\mathcal{J}$—and the menu of Assumption 4, but not the physical disaster law $(\lambda^\star, F_J)$, which is known to the modeler and is precisely what they learn and may dispute. The one thing that distinguishes them is a *tolerance for surprise.*

Assumption 2 (One-dimensional heterogeneity; homogeneous start): *Each investor i has a surprise tolerance $\varepsilon^i \in (0, 1)$; the cross-section of tolerances is the primitive (non-random) continuum law G on $(0, 1)$, with continuous and strictly increasing G, so ε^i is investor i's coordinate under G rather than an independent draw. This tolerance is her one permanent type. At $t = 0$ all investors share a common second-order prior $\rho_0 = \rho_0^\Lambda \otimes \rho_0^\Theta$ on $\Phi = \Lambda \times \Theta$, a product of a continuous prior on intensity and one on the size index, with full support and a continuously differentiable log-density; all hold the common model $\phi_0 = (\lambda_0, \theta_0) \in \arg\max_\phi \rho_0(\phi)$, differing only in ε^i.*

[4]The rule is Ortoleva's, applied in real time to the accumulating record as in the model-validation learning of Cho and Kasa (2015), so the rejection time is a first exit; by Sanov's theorem (Dembo and Zeitouni, 1998; Cover and Thomas, 2006) the acceptance region is the empirical law lying within a Kullback–Leibler ball of the held model. Fixed-tolerance testing makes recurrence the *foregone conclusion* of repeated significance testing under continuous monitoring (Anscombe, 1954; Armitage et al., 1969; Robbins, 1970; Robbins and Siegmund, 1970): the standardized record crosses a fixed band infinitely often by the law of the iterated logarithm. Cho and Kasa (2015), who share the test-and-reselect primitive, instead engineer this away with a time-varying threshold and vanishing gain; recurrent regime change also arises as constant-gain escape dynamics (Kasa, 2004; Cho and Kasa, 2008). The consolidating side is classical: under countable additivity a Bayesian cannot be driven to a foregone conclusion (Kadane et al., 1996) and the posterior merges with the truth (Blackwell and Dubins, 1962; Kalai and Lehrer, 1993). That the scalar since-adoption record stays sufficient for the held model at the *endogenous* revision time follows from the exponential-family (Wald) factorization of the disaster-count likelihood, the disaster channel being the last-observation-sufficient member of the families of Küchler and Lauritzen (1989) and Küchler and Sørensen (1989), for which sufficiency survives optional stopping (Küchler and Sørensen, 1998).

The environment and the disputable object are now fixed: a Lucas disaster economy in which investors share the diffusion and the occurrence of disasters but hold rival, identifiable laws $\phi = (\lambda, \theta)$ for the *intensity and size* of disasters, revised only when the rare-event record makes the held law surprising. We now ask what such a cross-section prices, deferring to Section 3 how identical investors come to hold it and how it moves. The object to track is the *risk-neutral disaster-size law*—the distribution of disaster sizes priced into options, the empirical "investor fears" of Bollerslev and Todorov (2011). We build it in three steps: the change-of-measure density each held model induces and the equilibrium pricing kernel, a power-mean aggregate of those densities formed with the *fixed* Negishi multipliers (§D); and the risk-neutral disaster-size law it implies. That law, we show, is the physical law tilted by risk aversion and by a size-belief factor in which the *stochastic* consumption shares weight the held models' likelihood ratios—a tilt that moves with the cross-section while the physical disaster process stays fixed (§E). Preferences are common CRRA and disaster risk is spanned, as in the heterogeneous-belief disaster economy of Chen et al. (2012); the departure is that the beliefs are lazy-Ortoleva held models, not Bayesian, and the dispersion is endogenous (Section 3).

D. The market and the pricing kernel

Each investor prices under her current held model $\phi_i(t-)$ (Definition 1); her belief is a change of measure from the truth with density M_t^i, the point-process likelihood ratio of Brémaud (1981, Ch. VI) (Definition 6).[5]

Assumption 3 (Market and preferences): *A continuum of investors $i \in [0, 1]$ have common CRRA utility $u(c) = c^{1-\alpha}/(1-\alpha)$, $\alpha > 0$ with $\alpha \neq 1$ (the log case $\alpha = 1$ is the continuous*

[5] Between revisions the held model is fixed and the disaster process is marked Poisson with intensity λ_i and size law F_{θ_i}; globally the subjective law is a marked point process with predictable compensator $\lambda_i(t-)\, F_{\theta_i(t-)}(dx)\, dt$ (Brémaud, 1981). Because the intensity is bounded and the per-disaster ratio is continuous and strictly positive on the compact $\mathcal{J} \times \Phi$, M^i is a strictly positive $(\mathcal{F}_t, \mathbb{P})$-martingale with $\mathbb{E}[M_t^i] = 1$ (Appendix C). We adopt the *time-consistent* construction—M_T^i/M_t^i built from the realized held-model process, the agent's own future revisions included—which makes M^i a $\mathbb{P}$-martingale and feeds the Negishi construction of Appendix F; it agrees with the agent's first-person *myopic-dogmatic* view on every instantaneous object, differing only in forward-looking valuation.

extension $u(c) = \log c$*; the disaster results use* $\alpha > 1$*), and common discount rate* $\delta > 0$*. Markets are complete: investors trade a riskless bond, the Lucas tree (a claim to the aggregate endowment* C *of Assumption 1), and a complete set of disaster-contingent (Arrow) claims spanning the jump risk, so there is a unique common state-price density* ξ_t *(Cox and Huang, 1989; Karatzas et al., 1990). Budget (Negishi) multipliers* $\zeta_i > 0$ *satisfy* $\int_0^1 \zeta_i^{-1/\alpha}\, di \in (0, \infty)$*.*
(iv) Symmetric allocation. *We select the symmetric, equal-Pareto-weight equilibrium:* $\zeta_i \equiv \zeta$ *(normalized* $\zeta = 1$*) for every date-0 investor. The allocation is selected in consumption-share space; we do* not *impose equal initial financial wealth, which is inconsistent with a common* ζ *under heterogeneous beliefs (Remark 3). This selection is without loss for what the wedge delivers: its form, the sign and range of its magnitude, and its entire stationary law are invariant or robust to the Pareto weights (shown in the Internet Appendix); the weights fix only which cross-section the kernel reads at a date.*

The pricing weights here are *stochastic* consumption shares, so the share process is part of the state: the consumption-share-weighted cross-section S_t of held models, the wealth-weighted distribution whose emergence from identical investors and law of motion are derived in Section 3.[6]

Proposition 1 (Equilibrium kernel): *Under Assumptions 1–4, and the maintained integrability condition that lifetime utilities and present-value budgets are finite (implied by the transversality condition (T) of Section 4,* $\delta > \sup_S m(S)$*, where* $m(S)$ *is the equilibrium value-growth rate defined there), a competitive equilibrium (Definition 7) exists and is unique; its state-price density is*

$$\xi_t = e^{-\delta t}\, C_t^{-\alpha}\, \Xi_t^{\alpha}, \qquad \Xi_t := \int_0^1 \zeta_i^{-1/\alpha} \big(M_t^i\big)^{1/\alpha}\, di \ \in (0, \infty), \tag{10}$$

and the equilibrium consumption shares are

$$s_t^i := \frac{c_t^i}{C_t} = \frac{\zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha}}{\Xi_t}, \qquad \int_0^1 s_t^i\, di = 1. \tag{11}$$

[6]This is the complete-market, heterogeneous-belief Lucas economy of Basak (2005); Jouini and Napp (2007) and, with disasters, Chen et al. (2012), with one difference: the pricing weights are stochastic consumption shares, which the fixed-weight consensus of Jouini and Napp (2007)—built on exogenous diffusion beliefs—cannot absorb.

The kernel's belief factor is the multiplier-weighted $\frac{1}{\alpha}$-power mean Ξ_t^α of the densities, which is not *the share-weighted arithmetic mean $\langle M\rangle_t := \int_0^1 s_t^i M_t^i\, di$ (the two coincide, up to a price-irrelevant constant, only when the cross-section is degenerate—all M_t^i equal). Under the symmetric allocation (Assumption 3(iv)) $\zeta_i \equiv \zeta$ cancels from (11) and rescales (10) by a constant, so $s_t^i = (M_t^i)^{1/\alpha} / \int_0^1 (M_t^j)^{1/\alpha} dj$.*

The kernel splits into an endowment term $e^{-\delta t} C_t^{-\alpha}$—the representative-agent kernel of the true Lucas economy—times a belief factor Ξ_t^α that the lazy beliefs contribute. The level Ξ_t is a *fixed-weight* (Negishi) power mean of the densities M_t^i; it moves not because the weights move but because the densities do, which is what makes the priced belief object time-varying even though every primitive is fixed. The state-dependent consumption shares s_t^i are the marginal pricing weights of its *local dynamics*—they govern the disaster jump ratio and the calm drift, not the level. Three statistics must be kept distinct: the head-count mean held model $\int \phi_i\, m_t$, the wealth-weighted mean $\int \phi_i s_t^i\, di$, and the priced power mean Ξ_t^α; the kernel prices the last, and the first two are descriptive shadows. The proof (Appendix C) is the complete-market construction: the individual static optimum (martingale method, Cox and Huang, 1989), market clearing, and strict concavity for uniqueness.

E. The disaster-fear factor

Markets are complete, so there is a unique equivalent martingale measure $\mathbb{Q}$, with density process $Z_t = \xi_t \exp(\int_0^t r_s\, ds)/\xi_0$ (r the equilibrium short rate). Since the bank account is continuous, Z and ξ share the same jump ratio, and the risk-neutral disaster law is fixed by that jump ratio alone—the diffusive change of measure does not touch the size law.

Definition 2 (Risk-neutral disaster law, fear factor, and the investor-fears wedge): *Let, at a disaster of size x occurring at t,*

$$g_t(x) := \frac{\Xi_t}{\Xi_{t-}}\bigg|_{J=x} = \int_0^1 s_{t-}^i\, r_i(x;t)^{1/\alpha}\, di = \int_0^1 s_{t-}^i \left(\frac{\lambda_i(t-) f_{\theta_i(t-)}(x)}{\lambda^\star f(x)}\right)^{1/\alpha} di, \tag{12}$$

the share-weighted aggregator whose α-th power $g_t(x)^\alpha$ is the $(1/\alpha)$-power mean of the held models' disaster likelihood ratios (the formulas below use g_t^α). The risk-neutral disaster compensator *is $\nu^{\mathbb{Q}}(dt,dx) = Y_t(x)\,\nu(dt,dx)$ with $Y_t(x) = e^{\alpha x} g_t(x)^\alpha$, i.e.*

$$\nu^{\mathbb{Q}}(dt,dx) = \lambda^\star\, e^{\alpha x}\, g_t(x)^\alpha\, dt\, F_J(dx). \tag{13}$$

As a predictable compensator $\nu^{\mathbb{Q}}$ is evaluated at the left limit: g_t is built from the pre-jump shares s^i_{t-} above, so $g_t, Y_t, U_t, f_t^{\mathbb{Q}}$ enter jump compensators through their predictable $t-$ versions, while S_t (and g_t, U_t) without qualification denote the current post-jump state. Writing $\mathbb{E}_f[h] := \int_{\mathcal{J}} h(y)\, f(y)\, dy$ for expectation under the physical size law, its size density, intensity, and tail are

$$f_t^{\mathbb{Q}}(x) = \frac{e^{\alpha x} g_t(x)^\alpha f(x)}{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]}, \qquad \lambda_t^{\mathbb{Q}} = \lambda^\star\, \mathbb{E}_f\big[e^{\alpha J} g_t(J)^\alpha\big], \qquad \bar{U}_t(x) := \int_x^{\bar{J}} f_t^{\mathbb{Q}}(y)\, dy. \tag{14}$$

The disaster-fear factor *is the risk-neutral disaster intensity $U_t := \lambda_t^{\mathbb{Q}}$; the* frictionless benchmark *is its correct-belief value $U^{\mathrm{RA}} := \lambda^\star\, \mathbb{E}_f[e^{\alpha J}]$ (the value at $g_t \equiv 1$); and the* investor-fears wedge *is the gap from the physical rate,*

$$U_t - \lambda^\star = \lambda^\star\big(\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha] - 1\big).$$

Finally the belief multiplier *is the normalized fear factor,*

$$\kappa_t := \frac{U_t}{U^{\mathrm{RA}}} = \frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]}{\mathbb{E}_f[e^{\alpha J}]} \in (0,\infty),$$

the $e^{\alpha J}$-weighted mean of g_t^α relative to its correct-belief value.

Proposition 2 (The disaster-fear bridge): *Under Assumptions 1–4, in the equilibrium of Proposition 1:*

(i) (tilted law) *the risk-neutral disaster law* (13)–(14) *is the physical size density f tilted by the*

risk-aversion factor $e^{\alpha x}$ *and the belief factor* $g_t(x)^\alpha$*, with fear factor* $U_t = \lambda^\star\,\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$*;*

(ii) (the bridge) *the fear factor factorizes into a preference benchmark* U^{RA} *and a belief multiplier* κ_t*,*

$$U_t \;=\; U^{\mathrm{RA}}\,\kappa_t, \qquad U^{\mathrm{RA}} = \lambda^\star\,\mathbb{E}_f[e^{\alpha J}], \qquad \kappa_t = \frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]}{\mathbb{E}_f[e^{\alpha J}]}; \tag{15}$$

(iii) (belief-driven variation) *holding* $(\alpha, \lambda^\star, F_J)$ *fixed,* U^{RA} *is constant, so*

$$d\log U_t = d\log \kappa_t, \tag{16}$$

and $f_t^{\mathbb{Q}}$*,* $\bar{U}_t$*, and* U_t *are functionals of the belief aggregate* g_t (12) *alone.*

The risk-neutral size law is the BT object (Bollerslev and Todorov, 2011; Breeden and Litzenberger, 1978): the physical F_J bent two ways—by risk aversion $e^{\alpha x}$, which fattens the tail because a representative agent dreads large disasters, and by the belief factor $g_t(x)^\alpha$, the only channel through which beliefs enter the price. That factor is a wealth-weighted *power* mean of the *joint* (intensity-and-size) disaster likelihood ratios—not their head-count or arithmetic mean, and not a mean of size beliefs alone, since intensity beliefs enter even when the size law is common. With $(\lambda^\star, F_J)$ fixed, $d\log U_t = d\log\kappa_t$: any movement in the multiplier maps one-for-one into the fear factor, while a self-re-adoption or offsetting cross-sectional change leaves it unmoved—price moves, physical quantity fixed.

U_t loads on the belief state g_t and on no diffusive quantity—volatility does not enter (14)—the structural counterpart of the unspanned left-tail factor of Andersen et al. (2015b), which predicts equity and variance risk premia but not volatility; the wedge $U_t - \lambda^\star$ is the Bollerslev and Todorov (2011) Investor Fears index (anchors structural, not identical). The smirk prices the pair $(U_t, f_t^{\mathbb{Q}})$, whose normalized shape $\bar{U}_t$ carries the size-dispute information a binary-intensity reading lacks. Only the *variation* of U_t is belief-driven—its level retains the fixed $e^{\alpha x}$ tilt—and these objects sit at the $\tau \to 0$ *front* of a surface.

Index options trade across *maturities*, and a disaster-tail claim of maturity τ prices the expected *path* of the risk-neutral disaster law over $[t, t+\tau]$, not the instant alone—the *forward fear factor*

$$F_t(\tau) := \mathbb{E}\big[U_{t+\tau} \mid \mathcal{F}_t\big] = U^{\mathrm{RA}}\, \mathbb{E}\big[\kappa_{t+\tau} \mid \mathcal{F}_t\big], \qquad \tau \geq 0, \tag{17}$$

the expected risk-neutral disaster intensity at horizon τ, whose origin is the bridge $F_t(0) = U_t = U^{\mathrm{RA}}\kappa_t$. (A maturity-$\tau$ option prices the *risk-neutral* forward intensity $\mathbb{E}^{\mathbb{Q}}[U_{t+\tau} \mid \mathcal{F}_t]$, the option-implied forward tail risk of the affine surface literature (Andersen et al., 2015b); the physical $F_t(\tau)$ equals it at the origin and departs for $\tau > 0$ only by the forward-intensity risk premium, and the survival records enter both, so the term-structure slope's dependence on Ψ_t below is invariant to the measure.) Away from the origin, with U^{RA} fixed, the curve moves with the expected forward multiplier $\mathbb{E}[\kappa_{t+\tau} \mid \mathcal{F}_t]$—the path the cross-section is expected to trace, the belief *dynamics*—so the surface relates to beliefs at every maturity, not only the shortest. Its slope carries the survival *records* Ψ_t of the standing models, which S_t does not encode; the precise statement, and the proof that this record-dependence is absent for every Markov benchmark, is Theorem 1, once the dynamics of Section 3 are in place.

We work throughout in the *intensity channel*: the size law is common ($f_{\theta_i} = f$), so $r_i = \lambda_i/\lambda^\star$ is x-free, the aggregate is the scalar $G_t = \mathbb{E}_{S_t}[(\lambda/\lambda^\star)^{1/\alpha}]$ (Corollary 1), and the multiplier is $U_t/U^{\mathrm{RA}} = G_t^{\alpha}$. This is the object the disaster-option literature inverts, and the setting of every quantitative result. Disagreement over the size index θ is a transparent extension—the aggregator g_t then carries the full joint object and the whole shape $f_t^{\mathbb{Q}}$ moves with the cross-section—which we develop in closed form in the Internet Appendix; the few statements that invoke it are flagged.

The bridge carries the fixed physical law $(\lambda^\star, F_J)$ the macro-disaster literature measures (Barro, 2006) into the moving risk-neutral law $(U_t, f_t^{\mathbb{Q}})$ the option literature measures (Bollerslev and Todorov, 2011; Andersen et al., 2015b), with the belief aggregate g_t the only object on it; κ_t is the *ratio* of the two measurements, equivalently of the inverted intensity $\lambda^\star\kappa_t$ to the physical

$\lambda^{\star}$. Two scope limits hold throughout: the deep disaster tail, not the frequent small jumps of the near-the-money surface; and a multi-year contraction idealized, as in Wachter (2013); Gabaix (2012), by an instantaneous jump.

The factorization's invariance to the Pareto weights, and the robustness of the wedge's sign, range, and stationary magnitude to that allocation, are collected in the Internet Appendix; Remark 5 delimits which belief processes move the factor and which make it flat.

The factorization $U_t = U^{\mathrm{RA}}\kappa_t$ prices *any* cross-section of held beliefs; it does not say where that cross-section comes from or how it moves. How identical investors come to hold dispersed models—synchronized revisions at disasters, staggered relief in calm—and the law of motion of its wealth-weighted distribution S_t, which turns this static tilt into the fast-in/slow-out fear *dynamics*, are the subject of Section 3; and the prices it implies are Section 4.

3. Belief dynamics and the fear-wave law of motion

We turn the revision rule into a law of motion for the wealth-weighted cross-section of held models. The section states three objects—the model each investor holds and tests, the hazard at which she abandons it, and the law of motion they induce on the cross-section—and then reads off their implications. We introduce only what those statements require; the single-agent foundation of the hazard is the companion paper's result (Teguia, 2026), and the deadline structure, the common-destination lemma, and the proofs are in Appendix B.

A. The test object, the revision hazard, and the law of motion

A *model of the world* is a single candidate disaster law $\phi = (\lambda, \theta) \in \Phi = \Lambda \times \Theta$—an arrival rate and a size index—that an investor holds and tests against her *since-adoption record* $\mathcal{R}_t^i$ (5): the count n_t^i of disasters and their realized sizes accumulated since she last revised, read at regime age τ_t^i. She keeps ϕ while that record stays unsurprising under it and abandons it for the menu's best alternative at a surprise; the re-selection target is the mode (9) of the public second-order posterior ρ_t (4), common across investors. Two facts about this object organize what follows, stated here and established in Appendix B. First, the disaster count is informative about the held *intensity*, so the test fires not only at a disaster but also in a long calm, when too few disasters have arrived to be consistent with the held λ_i. Second, the destination factorizes: because both the likelihood and the common prior separate across λ and θ, the joint re-selection mode splits into an intensity mode and a size mode taken separately (Lemma 3(iii)).

The kernel (10) and the bridge (15) aggregate held models by consumption share, so the object the law of motion governs is their wealth-weighted distribution. Three objects now appear together and are worth keeping distinct: the pricing state S_t, the wealth-weighted cross-section any belief model would carry; the public posterior ρ_t, the common destination a reviser re-selects to; and the record-resolved state Ψ_t, the cross-section together with the survival records—the coordinate that governs the forward surface and that S_t does not encode.

Definition 3 (The pricing state): *The* pricing state S_t *is the wealth-weighted law of held models: the probability measure on* $\Phi = \Lambda \times \Theta$ *with* $\int_\Phi h\, dS_t = \int_0^1 h\big(\phi_i(t)\big)\, s_t^i\, di$ *for every bounded* h*. Write* $\bar\lambda_t^S := \int_\Phi \lambda\, dS_t = \int_0^1 s_t^i \lambda_i(t)\, di$ *for the wealth-weighted mean held intensity.*

Definition 4 (The record-resolved state and the post-disaster map): *The* record-resolved state Ψ_t *is the wealth-weighted law of the full per-investor state: the probability measure on* $\Phi \times \mathcal{R} \times [0,\infty) \times (0,1)$ *with*

$$\int h\, d\Psi_t = \int_0^1 h\big(\phi_i(t),\, \mathcal{R}_t^i,\, \tau_t^i,\, \varepsilon^i\big)\, s_t^i\, di$$

for every bounded h*, where* $(\mathcal{R}_t^i, \tau_t^i)$ *are the since-adoption record and regime age of Definition 1, completed by the public posterior* ρ_t*—a deterministic functional of the full public record, common across investors—whose mode is the re-selection destination* $\phi_t^* = \mathfrak{m}(\rho_t)$*. Its held-model marginal is the pricing state,*

$$S_t = \pi_\phi(\Psi_t), \qquad \pi_\phi : (\phi, \mathcal{R}, \tau, \varepsilon) \mapsto \phi.$$

At a disaster of size x, the post-disaster state Ψ_t^x *is obtained from* Ψ_{t-} *by, in order, (a) the share reweight* $s^i_{t-} \mapsto s^i_{t-} r_i(x;t)^{1/\alpha}/g_t(x)$ (D2)*; (b) the record update* $\mathcal{R}^i \mapsto \mathcal{R}^i \oplus x$*; (c) the test/rejection evaluation* (D22)*; and (d) the relocation of every triggered switcher to the common destination* ϕ_t^* (9)*.*

The public posterior is carried alongside Ψ_t because it accumulates on the entire public history and is therefore *not* recovered from the since-adoption records inside Ψ_t. The instantaneous risk-neutral objects $\bar{\lambda}_t^S$, $g_t(\cdot)$, U_t, and $f_t^{\mathbb{Q}}$ read only the marginal S_t; the forward-looking valuation of Section 4 reads the full pair, and we write $v(\Psi_t)$ and $\mathcal{A}_\Psi$ for $v(\Psi_t, \rho_t)$ and $\mathcal{A}_{\Psi,\rho}$, displaying ρ_t only where it does analytic work—the calm boundary map of Lemma 7 and the post-disaster relocation (d) above.

The rate at which an investor abandons her model is the *revision hazard*: the first passage of the deviance $D_{\phi,\tau}(\mathcal{R})$ (7) across its acceptance threshold (Propositions 7 and 12). Between disasters, with the record fixed, the deviance drifts deterministically and meets the threshold at a scheduled *calm deadline* $\Delta^\star(\phi, \varepsilon)$ (B4)—a regime age at which a disaster-free record is rejected for a process expected to deliver disasters; a disaster appends a mark and triggers rejection when the updated record leaves the typical set. The single-agent foundation—that the surviving record, not the current belief, is the sufficient state for this hazard—is the companion paper's result (Teguia, 2026); the record-resolved passage law, which does not close in general, is in Appendix D. In the baseline ($f_\theta \equiv f$, Corollary 1) the deviance reads the record through the count alone, and the hazard closes in the low-dimensional count-and-age state.

Between revisions the cross-section drifts by *selection* (Proposition 10): in calm the share of a below-average-intensity investor grows and that of an above-average one shrinks, while a disaster rewards whoever held a model that fits it—the same selection a Barro (2006)-style truth would impose, here on *held* models. This is the wealth channel that turns the static tilt of Proposition 2 into a moving one. The revisions themselves enter through two flows the hazard defines: a staggered *calm switch flow*, as tolerance-indexed deadlines arrive and each reviser relocates to the common destination $\phi_t^* = \mathfrak{m}(\rho_t)$ (Lemma 4), and a synchronized *disaster jump*, at which every triggered investor relocates to that same destination at once (Proposition 8). The law of motion collects the three.

Proposition 3 (Law of motion of the pricing state): *In the equilibrium of Proposition 1, for every bounded h on Φ the pricing state evolves by:*

(i) (selection drift — continuous, calm)

$$\frac{d}{dt}\langle h, S_t\rangle\Big|_{\mathrm{sel}} = \tfrac{1}{\alpha}\int_\Phi \big(\bar{\lambda}_t^S - \lambda\big)\, h(\phi)\, S_t(d\phi); \tag{18}$$

(ii) (calm switch flow — calm) *the staggered calm-deadline revisers leave their held models and arrive at the common destination $\phi_t^* = \mathfrak{m}(\rho_t)$, the same for every investor revising at t (Lemma 4); aggregated, they contribute*

$$\frac{d}{dt}\langle h, S_t\rangle\Big|_{\mathrm{sw}} = \int_\Phi \big[h(\phi_t^*) - h(\phi)\big]\, \Phi_t^{\mathrm{sw}}(d\phi), \tag{19}$$

with Φ_t^{sw} the calm switch flow (D31) *of Lemma 6—finite-variation, and continuous where the calm-deadline law is atomless; on a flat deadline level set of positive G-mass it carries a predictable finite-variation jump, which by Lemma 7 leaves every traded price continuous; disaster-triggered switches are* not *part of this term and enter only at (iii);*

(iii) (disaster jump — at disaster epochs) *at a disaster of size x the state jumps in two steps—the marginal of the post-disaster map Ψ_t^x of Definition 4. First the shares reweight (Proposition 10(ii)),*

$$\widetilde{S}_t^x(d\phi) = \frac{r(\phi, x)^{1/\alpha}}{g_t(x)}\, S_{t-}(d\phi), \qquad r(\phi, x) = \frac{\lambda f_\theta(x)}{\lambda^\star f(x)}; \tag{20}$$

then every triggered *switcher relocates to the common destination ϕ_t^*, and the final post-disaster marginal S_t is the pushforward of $\widetilde{S}_t^x$ under that relocation. The triggered set is read on the* record-resolved *state Ψ_{t-}, not on Φ—investors sharing a held model ϕ may* *carry different $(\mathcal{R}, \tau, \varepsilon)$ and so need not trigger together—so S_t is not a partition of S_{t-} over Φ, and (20) equals S_t only when the triggered set is empty. Both steps are a single jump at the disaster epoch, not a continuous rate; the continuous calm drift is the sum of (i) and (ii).*

Under size disagreement the disaster tilt reads S_t through the aggregator

$$x \;\longmapsto\; g_t(x) = \int_\Phi \left(\tfrac{\lambda}{\lambda^\star}\right)^{1/\alpha} \left(\tfrac{f_\theta(x)}{f(x)}\right)^{1/\alpha} S_t(d\phi), \tag{21}$$

which by the spanning condition (Assumption 4(v)) no finite collection of linear moments of S_t determines; the instantaneous risk-neutral law $(U_t, f_t^{\mathbb{Q}})$ is therefore infinite-dimensional in S_t.[7]

The derivation—the deadline structure, the finiteness of the switch flow, and the price-continuity of its predictable jumps—is in Appendix B.

B. What the law of motion implies

The three forces give the wedge its *fast-in, slow-out* shape. Only the disaster jump is synchronized—all triggered switchers land on the single common mode ϕ_t^* (Lemma 4), so the repricing is a coordinated wave rather than idiosyncratic noise—and it snaps $\bar{\lambda}_t^S$ up at one instant. Selection and the calm switch flow are its staggered relaxation, both pushing $\bar{\lambda}_t^S$ down in calm (the switch flow under monotone-reselection), only as fast as the surviving disagreement that selection consumes, spread across the population by the deadline dispersion. These signs are the wave priced in Section C; the equations fix only directions, while the wedge's magnitude waits on the stationary cross-section of Section 5.

The law of motion is not autonomous in the cross-section. The instantaneous tail reduces to

[7] This rules out any finite collection of *linear* moments; the reduction is in fact tighter—no *continuous* finite-dimensional statistic of S_t is sufficient for $(U_t, f_t^{\mathbb{Q}})$ under size disagreement (Proposition 11, Appendix D, by Brouwer's invariance of domain). The continuity qualifier is essential: in the weak topology $\mathcal{P}(\Phi)$ is Polish, so a measurable bijection furnishes a discontinuous one-dimensional sufficient statistic.

the scalar G_t in the baseline (Corollary 1), but the switch flow and its destination read the record-resolved Ψ_t, so the *forward* surface depends on Ψ_t even where the instantaneous tail does not—the enlargement the rest of the paper prices, and the coordinate the term-structure test isolates.

A last implication is that the dispersion the cross-section carries is manufactured, not assumed: identical investors watching one path *become* a non-degenerate cross-section.

Proposition 4 (Endogenous emergence of disagreement): *Start from the homogeneous configuration of Assumption 2: at $t = 0$ every investor holds the common model ϕ_0, so $S_0 = \delta_{\phi_0}$ and* $\mathrm{Var}_{S_0}(\lambda) = 0$*. Disagreement is then generated by the rule, not assumed: there is a finite $t > 0$ with* $\mathrm{Var}_{S_t}(\lambda) > 0$ *in the consumption-share measure S_t of Definition 3 (a positive-G-mass split transfers to positive S_t-mass by the absolute continuity established in the proof) whenever either*

(a) (disaster split) *the first disaster T_k precedes the earliest calm deadline—so the population is still homogeneous at T_k^- ($S_{T_k^-} = \delta_{\phi_0}$) and the zero-count depth $\bar{\varepsilon}_k(\phi_0, 0)$ is the operative trigger—the rejecting cohort $\{\varepsilon^i \geq \bar{\varepsilon}_k(\phi_0, 0)\}$ is a strict subset of positive G-mass of the population, and the common destination differs from the incumbent* in intensity*, $\lambda(\phi^{(k)}) \neq \lambda(\phi_0)$ (a size-only move, λ unchanged, leaves* $\mathrm{Var}_S(\lambda) = 0$*); or*

(b) (staggered calm) *the calm-deadline map $\varepsilon \mapsto \Delta^\star(\phi_0, \varepsilon)$ is non-constant on a positive-G-mass set, the calm intensity mode $\tau \mapsto \lambda^{\mathrm{m}}_\tau$ is strictly decreasing (Remark 2), and the realized path stays disaster-free long enough for two positive-G-mass threshold groups to reach their distinct deadlines (so the staggered revisions fire before a disaster resets the count).*

The disaster channel (a) needs only a partial rejection with a moved posterior mode; the calm channel (b) needs the regularity of Remark 2. The point is sharp against the merging benchmark: common priors updated on common data converge—predictive laws merge (Blackwell and Dubins, 1962) and rational learners come to agree (Kalai and Lehrer, 1993)—so persistent disagreement ordinarily needs heterogeneous priors or private signals. Here it needs neither:

identical priors, identical data, and a single non-Bayesian rule whose staggered rejection timing holds the cross-section apart. With common data the population is the exact deterministic transport of the tolerance distribution along the path—the cross-section the image of G, not a continuum average.

4. Prices and the dynamics of the wedge

The kernel of Proposition 1 prices every claim and the fear factor of Proposition 2 identifies its disaster-tail component, so we can read the wedge off observable prices. We proceed in two steps: first the economy at a point in time—short rate, price–dividend ratio, disaster premium—then how those prices move as beliefs revise, the fast-in, slow-out *time series of the wedge*. The static results are state-contingent, valid for any cross-section and free of demographic assumptions; only unconditional averages await the stationary primitive of Remark 6. One distinction organizes the section: the prices split into a part any Markov pricing state already delivers and a part that requires the survival history behind it, which we record once as a decomposition every pricing result inherits.

A. The benchmark decomposition

Write S_t for the current pricing cross-section (Definition 3) and Ψ_t for the record-resolved state that also carries the since-adoption records, ages, and revision hazards (Definition 4), with $S_t = \pi_\phi \Psi_t$ its held-model marginal.

Proposition 5 (The benchmark decomposition: current cross-section versus survival history): *In the intensity channel, every equilibrium price object F_t admits an orthogonal decomposition* *into a part carried by the current cross-section and a part carried by the survival history,*

$$F_t \;=\; \underbrace{B_t(S_t)}_{\text{current cross-section}} \;+\; \underbrace{\mathcal{H}_t(\Psi_t)}_{\text{survival-history residual}}, \qquad B_t := \mathbb{E}[F_t \mid S_t], \quad \mathbb{E}[\mathcal{H}_t \mid S_t] = 0, \tag{22}$$

with B_t a functional of S_t and the residual $\mathcal{H}_t = F_t - \mathbb{E}[F_t \mid S_t]$ a functional of Ψ_t orthogonal to S_t.

(i) (Instantaneous prices carry no residual.) *The fear factor U_t and multiplier κ_t, the risk-neutral disaster law, and the short rate r_t are functionals of S_t alone (Proposition 16); for each, $B_t = F_t$ and $\mathcal{H}_t \equiv 0$.*

(ii) (Forward-looking prices carry a residual that grows with horizon.) *The price–dividend ratio $v(\Psi_t)$, the* level *of the disaster premium, and the forward fear factor $F_t(s) = \mathbb{E}[U_{t+s} \mid \mathcal{F}_t]$ at horizons $s > 0$ have $\mathcal{H}_t \not\equiv 0$. The residual is zero at horizon zero and first order in s, its slope loading on the survival-history coordinates of Ψ_t—the calm deadline-proximity flow Φ_t^{sw} and the disaster-trigger propensity π—and it accumulates with horizon (Propositions 14 and 1).*

(iii) (The residual is the lazy-revision signature.) *For any* Markov benchmark*—a model whose pricing state is Markov in the current cross-section—every forward object is a functional of that state, so $\mathcal{H}_t \equiv 0$. A heterogeneous Bayesian economy is one such benchmark, because each investor's posterior is a sufficient statistic for her own future belief and the wealth-weighted cross-section is therefore itself Markov; an affine latent-state specification and a representative filtering economy are others (the single-agent Bayesian case, where the multiplier itself merges to a constant, is Appendix I). The residual is therefore present for* any *disaster law under lazy revision and* absent *across the entire Markov class, and it is exactly what the term-structure test isolates (Remark 1): conditioning on the current option state removes $B_t(S_t)$ and leaves $\mathcal{H}_t(\Psi_t)$.*

The instantaneous price objects—the short rate, the price–dividend ratio, and the disaster premium—are placed in this decomposition and derived in Appendix E: the short rate and the risk-neutral tail carry no residual, while the price–dividend ratio and the premium level inherit the survival-history residual through $v(\Psi_t)$.

B. The term structure of the tail, and what the measured objects identify

Proposition 16 says the instantaneous tail is read off S_t and the forward-looking objects are not. The next result makes that gap operational. It locates exactly where the record-resolved state enters the *term structure* of the risk-neutral tail, gives the channels their signs, and shows the dependence is a property of the revision rule rather than of the disaster law—absent for any Markov benchmark.

Theorem 1 (Forward tail: level on S_t, term-structure slope on Ψ_t): *Suppose the age structure of Ψ_t admits a density, so the calm flow of held models reaching their deadline $\Delta^\star(\phi,\varepsilon)$ (B4) at t is a finite measure Φ_t^{sw} (Proposition 6). The forward fear factor $F_t(s)$ of (17)—the expected future risk-neutral disaster intensity at horizon s, the term structure that the deep out-of-the-money tail traces across maturities—decomposes as follows. In the intensity channel:*

(i) (Level on S_t.) $F_t(0) = U_t = U^{\mathrm{RA}}\kappa_t$ *is a functional of the pricing state S_t alone (Proposition 2, Proposition 16).*

(ii) (Slope splits, into two history channels.) *The right-hand slope at the origin decomposes into a part measurable in S_t and a part measurable in Ψ_t but not in S_t,*

$$\partial_s^+ F_t(0) = \underbrace{U_t\big[-b_{\mathrm{sel}}(S_t)\big] + \lambda^\star\,\Theta_{\mathrm{rw}}(S_t)}_{\textit{selection drift and disaster reweight: }\sigma(S_t)} + \underbrace{U_t\,c_{\mathrm{sw}}\big(\Phi_t^{\mathrm{sw}}\big)}_{\textit{calm re-leveling}\,\le 0} + \underbrace{\lambda^\star\,\Theta_{\mathrm{sw}}(\Psi_t)}_{\textit{disaster-triggered re-leveling}\,\ge 0},$$

where $b_{\mathrm{sel}}(S_t) \ge 0$ is the calm selection drift of (25) and $\Theta_{\mathrm{rw}}(S_t)$ the disaster reweight compensator (24), both functionals of S_t through g_t. The two Ψ_t-channels are the staggered *revisions: the calm flow*

$$c_{\mathrm{sw}}\big(\Phi_t^{\mathrm{sw}}\big) = \frac{\alpha\,\mathbb{E}_f\big[e^{\alpha J} g_t(J)^{\alpha-1} \int_\Phi \big(r(\phi_t^\star, J)^{1/\alpha} - r(\phi, J)^{1/\alpha}\big)\,\Phi_t^{\mathrm{sw}}(d\phi)\big]}{\mathbb{E}_f\big[e^{\alpha J} g_t(J)^{\alpha}\big]} \;\le\; 0,$$

non-positive under monotone-reselection (scheduled relief, Proposition 7: the calm destination intensity is non-increasing); and the disaster-triggered flow $\Theta_{\mathrm{sw}}(\Psi_t) \ge 0$, the compensator of the relocations fired by a disaster at the record-dependent rate $\lambda^\star\,\pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D22)*, whose destination intensity is non-decreasing in the disaster count (Proposition 6(ii)). Both Φ_t^{sw} and the trigger propensity π are functionals of the record-resolved state Ψ_t—the joint law of held models, ages, and records—and not of its marginal S_t.*

(iii) (Two histories, one cross-section.) *For record-resolved states Ψ_t, Ψ_t' with a common held-model marginal $\pi_\phi \Psi_t = \pi_\phi \Psi_t' = S_t$, the forward curves share their level and their $\sigma(S_t)$-slope, $F_t(0) = F_t'(0) = U_t$; they differ only through the two staggered-revision channels.* Conditional on no disaster over the horizon, *only the calm channel operates and the state with the more imminent deadlines—the larger downward flow Φ_t^{sw}—carries the weakly steeper forward-tail decline, $\partial_s^+ F_t'(0) \le \partial_s^+ F_t(0)$; unconditionally the two channels are oppositely signed, and the net is a cross-sectional condition on the balance of calm-deadline proximity against disaster-trigger propensity. The wedge $F_t(s) - F_t'(s)$ is zero at $s = 0$ and first order in s; its full-horizon law is Markov in Ψ_t, not autonomous in S_t (Proposition 14).*

(iv) (No Markov-state counterpart.) *Replace lazy revision by any Markov benchmark—a model whose pricing state is Markov in the current cross-section. The worked instance is heterogeneous Bayesian learning of the disaster law: each posterior is a sufficient statistic for its own future, so the wealth-weighted cross-section is itself Markov; there is no deadline and no trigger threshold, hence neither Φ_t^{sw} nor Θ_{sw}, and $\partial_s^+ F_t(0)$ is $\sigma(S_t)$-measurable. The forward curve is then pinned by the current cross-section at every horizon (Appendix I); an affine latent-state model and a representative filtering economy give the same conclusion*

through their own Markov states. The slope's dependence on survival history is a property of the revision rule, present for any disaster law and absent across the entire Markov class.

Remark 1 (The observable signature, and what it is not). Theorem 1 yields an over-identifying restriction on the left-tail surface, and fixes the maturity at which it bites. The level of the surface—the short-dated risk-neutral tail—is pinned by the current cross-section S_t, so once one conditions on it the age structure is irrelevant at the short end; the dependence on survival history enters the *slope* and accumulates with maturity. The test is therefore a residual one: regress a forward-tail slope (a long-minus-short tail measure) on variables proxying the two staggered-revision channels—time since the last disaster, accumulated calm surprise, or an estimate of the distance-to-rejection $\Delta^\star - \tau$—*after* controlling for the current short-dated tail, realized volatility, and current skew. The rule predicts a non-zero residual loading, strengthening with maturity, that any Markov benchmark sets to zero.

This residual is a finer object than the raw path-dependence of the implied-volatility surface (Guyon and Lekeufack, 2023; Andrès et al., 2024), and the two must not be conflated. That literature regresses implied volatility on the past return path *without* conditioning on the current pricing state, and finds the explanatory power strongest at short maturities and weakening as maturity grows. In the present model that raw pattern is reproduced through S_t, which itself encodes the recent path—the short-dated tail is the tightest function of S_t, while longer maturities average over future selection and turnover that revert toward the stationary cross-section—and it is *generic*: any Markov benchmark also encodes the past. The rule's signature is the orthogonal residual, history that still moves the forward-tail slope after the current state is removed, which Theorem 1 shows is present under lazy revision and absent across the Markov class.

The three option-tail objects the literature measures are each a functional of the *current* cross-section S_t (Proposition 16): the far-tail level wedge is the level of κ_t (Proposition 2), the investor-fears index is the front wedge $U_t - \lambda^\star$ (Bollerslev and Todorov, 2011), and the unspanned tail factor is U_t itself (Andersen et al., 2015b). Any Markov benchmark reproduces all three, so matching them certifies a sensible time-varying intensity but does not separate lazy revision from the Markov class; the object that does is the forward fear factor's *slope* (Theorem 1), S_t-measurable in level but carrying a Ψ_t-part in slope.

The mapping also reads one of their findings structurally. Andersen et al. (2015b) find the compensation for tail risk reacts far more persistently after a crisis than the underlying risk, and fit it with a persistent latent factor; here that persistence is not a free parameter but the survival history, the fear decaying at the selection rate of Proposition 6 rather than at the speed of the underlying. The discriminating content is once more the slope—a function of the current factor under any Markov persistence, a function of how long the calm has run under lazy revision.

C. The time series of the wedge

A disaster is an impulse; this subsection traces the response of the priced objects assembled above. The engine is market selection on the realized path (Sandroni, 2000; Blume and Easley, 2006; Yan, 2008; Kogan et al., 2006, 2017): a disaster vindicates the fearful at a stroke, the ensuing calm vindicates the complacent only gradually, and the result is a fast jump and a slow, staggered relaxation (Andersen et al., 2015b). Throughout we keep the descriptive mean belief $\bar{\lambda}_t^S = \int_\Phi \lambda \, S_t(d\phi)$ distinct from the priced fear factor U_t: the kernel prices the power mean, not the mean (Section 2).

Proposition 6 (Disaster jump and calm erosion): *In the equilibrium of Proposition 1, with* Var_{S_t} *and* Cov_{S_t} *the variance and covariance under* S_t*:*

(i) (Calm erosion — exact, both calm channels.) *Between disasters the wealth-weighted mean*

belief moves continuously, by selection and the calm switch flow,

$$\frac{d\bar{\lambda}_t^S}{dt} = \underbrace{-\tfrac{1}{\alpha}\,\mathrm{Var}_{S_t}(\lambda)}_{\textit{selection}\,\le\,0} + \underbrace{\int_\Phi \big(\lambda(\phi_t^*) - \lambda\big)\,\Phi_t^{\mathrm{sw}}(d\phi)}_{\textit{calm switch flow (signed below)}}, \tag{23}$$

and, for any bounded h, $\frac{d}{dt}\langle h, S_t\rangle = -\frac{1}{\alpha}\,\mathrm{Cov}_{S_t}(\lambda, h) + \int_\Phi [h(\phi_t^*) - h(\phi)]\,\Phi_t^{\mathrm{sw}}(d\phi)$*: selection erodes the mean belief at a rate equal to its cross-sectional variance, scaled by the risk tolerance* $1/\alpha$*, while the calm switch flow relocates the staggered revisers to the calm destination* ϕ_t^**. On the* disaster-free-since-adoption *part of the switch set the destination intensity is non-increasing (scheduled relief, Proposition 7; the calm MAP of Lemma 4 falls within a calm spell), so* $\lambda(\phi_t^*) \le \lambda$ *and the switch-flow term is non-positive there. For switchers whose since-adoption records already carry non-rejecting disasters (*$n \ge 1$*), a disaster can have raised the public mode, so* $\lambda(\phi_t^*) \le \lambda$ *is not automatic; we state it as the* monotone-reselection *condition under which the switch flow—and with it the calm erosion of* $\bar{\lambda}_t^S$*—is signed. In the intensity channel, and under that condition generally, both calm channels push the mean down.*

(ii) (Disaster jump — intensity channel.) *At a disaster of size* x *the cross-section reweights by (20); the reweight alone raises the mean belief by*

$$\bar{\lambda}_t^S - \bar{\lambda}_{t-}^S = \frac{\mathrm{Cov}_{S_{t-}}\big(\lambda,\, r(\cdot,x)^{1/\alpha}\big)}{\mathbb{E}_{S_{t-}}[r(\cdot,x)^{1/\alpha}]} \;\ge\; 0, \tag{24}$$

≥ 0 *in the intensity channel (*$f_\theta \equiv f$*,* $r(\phi, x) = \lambda/\lambda^\star$*) and strict if* $\mathrm{Var}_{S_{t-}}(\lambda) > 0$*; the triggered* switchers *(Lemma 6) relocate to the common post-disaster mode* ϕ_t^**, whose intensity coordinate is non-decreasing in the disaster count (proved by revealed preference in the appendix). Their contribution to the jump is computed under the* post-reweight *switcher measure (the disaster updates in the order reweight-then-relocate of Definition 4),* $\Delta\bar{\lambda}_t^S|_{\mathrm{switch}} = \widehat{s}_t^{\mathrm{sw}}\big(\lambda(\phi_t^*) - \widehat{\bar{\lambda}}_t^{\mathrm{sw}}\big)$*, where* $\widehat{s}_t^{\mathrm{sw}}$ *and* $\widehat{\bar{\lambda}}_t^{\mathrm{sw}}$ *are the switchers' mass and mean intensity* after *the* $r(\cdot, x)^{1/\alpha}$ *reweight; this* adds *to the jump when* $\lambda(\phi_t^*)$ *exceeds that post-*

reweight mean, a cross-sectional condition we state rather than derive. The unconditional fast-in is the reweight (24) *alone—a covariance signed without reference to the re-selection direction, so the fear jump does not require the disaster to raise the held mode—while the relocation is a conditional amplifier.*

(iii) (The priced fear factor inherits the pattern — intensity channel.) *Through the bridge relation* (16)*, in calm U_t drifts down (and with it the U_t-loading component of the disaster premium, holding the jump ratio ψ_t fixed; see the event-study reading in Section C),*

$$\frac{d\log U_t}{dt} = -\,\frac{\mathbb{E}_f\!\left[e^{\alpha J} g_t(J)^{\alpha-1}\ \mathrm{Cov}_{S_t}\!\left(\lambda, r(\cdot,J)^{1/\alpha}\right)\right]}{\mathbb{E}_f\!\left[e^{\alpha J} g_t(J)^{\alpha}\right]}\ \le\ 0, \tag{25}$$

the selection *part of the calm drift, strict if $\mathrm{Var}_{S_t}(\lambda) > 0$. On a switch-active instant the calm switch flow of (i) adds a further term proportional to $\mathbb{E}_f\left[e^{\alpha J} g_t(J)^{\alpha-1}\int_\Phi (r(\phi_t^*,J)^{1/\alpha} - r(\phi,J)^{1/\alpha})\,\Phi_t^{\mathrm{sw}}(d\phi)\right]$, non-positive under the same monotone-reselection condition as (i) (so that $r(\phi_t^*,\cdot) \le r(\phi,\cdot)$ on the switch set); U_t jumps up at a disaster with g_t. The signs hold in the intensity channel and, generally, whenever λ and $r(\cdot,x)^{1/\alpha}$ are positively associated across the cross-section.*

(iv) (Staggered recovery.) *The calm decay runs until the next disaster or the calm deadlines* $\Delta^\star(\phi,\varepsilon)$ (B4)*, at which revisers re-select a held model that Lemma 4 makes non-increasing on the disaster-free-since-adoption part of the switch set (or under the monotone-reselection condition)—a predictable step, weakly downward there. By Proposition 8 the deadlines are weakly ordered in ε and differ wherever the deadline map is non-constant, so the steps are staggered across that part of the population over a horizon set by $\Delta^\star$, long when ε or the held intensity is small (the dependence on $\lambda^\star$ is indirect—through the held intensity coordinate and the chance a calm spell is interrupted by the next disaster—not a direct property of* (B4)*). The response is fast in (the instantaneous jump of (ii)) and slow out (the drift of (i),(iii) plus staggered deadline steps). In the decomposition of Proposition 5 the fast-in reweight* (24) *and the selection drift* (25) *are the cross-section part $B_t(S_t)$, while*

the calm switch flow Φ_t^{sw}, the disaster-triggered relocation, and the deadline-timed steps $\Delta^\star(\phi, \varepsilon)$ are the survival-history residual $\mathcal{H}_t(\Psi_t)$. The slow-out leg and its staggered shape are therefore the residual: a Markov-in-cross-section benchmark reweights and drifts but has no deadlines and no triggers, so it carries the fast-in and not the staggered slow-out (Appendix I).

Part (i) is market selection in its sharpest local form—the replicator drift in which the mean held belief falls at a rate equal to its own cross-sectional variance scaled by risk tolerance, the transient face of the long-run selection results (Sandroni, 2000; Blume and Easley, 2006; Yan, 2008) specialized to an economy where "accurate" is path-contingent: the complacent are vindicated in calm, the fearful at disasters. The asymmetry is the content—a disaster reweights toward fear and lifts the posterior mode at one instant (fast in), while calm erodes the wave only as fast as the surviving disagreement that selection consumes, staggered across the population by the deadline spread (slow out)—the model counterpart of the persistent post-crisis tail factor of Andersen et al. (2015b). The signs are intensity-channel statements—in the joint channel they need λ and $r(\cdot, x)^{1/\alpha}$ positively associated, which we flag rather than assume—and the *magnitudes* (the jump size, the decay rate $\frac{1}{\alpha}\operatorname{Var}_S(\lambda)$, the horizon $\Delta^\star$) wait on the stationary cross-section of Section 5.

The three price functionals move together at a disaster, each driven by the same jump in the fear factor: the disaster premium's U_t-component rises and drifts down through calm at the rate (25) (Proposition 15); the short rate moves by $dr_t = d\bar{\lambda}_t^S - dU_t$, falling *iff* the priced-tilt response dominates the mean-intensity response (Proposition 13); and the excess-variance term, generated by movement of the cross-section (Internet Appendix), bursts when the repricing ratio ψ_t amplifies the jump. Under $dU_t > d\bar{\lambda}_t^S$ and in the intensity channel one belief object thus delivers a premium jump up, a safe-rate jump down, and a variance burst, each relaxing slowly through the staggered calm recovery. The cycle's *shape* is pinned by the primitives; its *scale*, and the demographic refresh that makes it stationary, is the work of Section 5. A forecast-data signature—lazy revision read as state-dependent information rigidity, and its place relative to the under- and over-reaction evidence—is developed in the Internet Appendix.

D. What history prices, and what it does not

The state that prices disasters is not what the market believes but a pair: what it believes, and how long it has believed it without rechecking. The first coordinate is the cross-section S_t. The second is the cross-section of *fragility*: how each holding's since-adoption record stands against the model it holds (Teguia, 2026), hence how near it is to revision. Two markets can share their beliefs and their near-dated tail and still differ here—one has just re-anchored, records fresh, deadlines far; the other has held the same fear through a long calm, records aged, deadlines near. They price the forward tail apart. Belief fragility is priced, and current belief does not recover it (Proposition 3).

This is economics, not bookkeeping, because of what fragility cannot stand for. The disaster law $(\lambda^\star, F_J)$ is fixed and known, so all the record does here is govern revision. It prices the tail—it moves the forward fear factor and the valuation $v(\Psi_t)$—yet forecasts nothing about how often or how hard disasters strike. That is the content of the non-Markov bridge: a market-memory variable orthogonal to every fundamental forecast, yet priced. No filtering model can do this. There, history moves prices only by moving a belief about fundamentals, so the two move together; here they part.

Three things follow. Through a continuing calm the same near-dated tail relaxes faster going forward where the fear has aged, nearer the deadline that revises it down (Theorem 1). A disaster need not reprice two markets of identical beliefs alike: the reprice runs through the holdings it tips into revision at once, read from the records and not from S_t (Proposition 6), so whether more revision deepens or cushions the fall is a cross-sectional condition, not a sign the model fixes. And the effect reaches the level, not the slope alone: $v(\Psi_t)$ and the premium read the whole record.

A renewal process is Markov once one tracks age, so one might ask whether "(beliefs, age)" is a Markov state in disguise. It is not. The record resets at every rejection, and both the revision hazard and its destination read the full record and the public posterior, not a scalar clock. We claim the dimensionality only where the mathematics forces it—the proof admits no finite sufficient statistic—and the economics where it bites: a few measures of fragility, time since the last disaster, accumulated calm surprise, distance to a deadline. The dissociation, not the dimension, is the result.

One boundary. Fragility is priced; it does not feed back on the rule that makes it. Whether a long quiet breeds a more violent revision—fragility itself building in good times, not merely its price—is a separate question, left to other work. This is the pricing face of one fact about the rule: because rejection never stops—the calm deadline is finite even at the truth, so a correct-on-average holding is revised forever (Lemma 5)—fragility keeps accumulating and never collapses into the belief.

5. The wedge and the option-tail evidence

The belief cross-section alone swings the priced disaster intensity several-fold—*above and below* the representative-agent benchmark—with no physical parameter moving. This section establishes the stationary economy in which that swing lives and measures it against the option-tail evidence. The event study of Section C and any unconditional moment first presuppose a *stationary* cross-section, which the present primitives do not supply: posterior consistency drives an infinitely-lived investor's held model to the Kullback–Leibler point $\phi^{\star}$ and the fear dispersion dies out (Remark 6). We add the one primitive that sustains it—demographic turnover—prove the cross-section is then stationary, and read the quantitative content as comparative statics: physical and preference parameters from the macro-disaster literature, the belief block varied and then, in the illustrative calibration of Section B, pinned to a few stylized option-tail targets.

We restore a nondegenerate stationary cross-section with the device the rare-disaster literature uses for the purpose: a standard *demographic turnover*, perpetual-youth overlapping generations with fair annuities (Yaari, 1965; Blanchard, 1985; Gârleanu and Panageas, 2015). Each investor faces a constant death hazard $\Omega > 0$; newborns arrive at the same rate, drawn from a fixed type law Π over (ε, ϕ_0) with ϕ_0-marginal Π_ϕ, and enter holding their drawn model with a reset test clock while inheriting the common public posterior. Because young cohorts hold ϕ_0 and have not yet revised, turnover continually replenishes the disagreement that selection consumes. We state the primitive as Assumption 5 and specialize to a finite menu $\Phi = \{\phi_1, \ldots, \phi_K\}$ for the stationary and quantitative exercise; the assumption, the equilibrium it defines, and all proofs are in Appendix F.

Two results carry the economy of Sections 2–4 into this stationary world, and the quantitative reading needs only their conclusions. First, *the kernel keeps its form* (Proposition 17): with fair annuities the death hazard cancels from every survivor's Euler equation and enters only through the composition of the living, so $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^{\alpha}$ holds over the living cross-section, the short rate (E1) is unchanged—level and all—and turnover acts only by adding a mean-reversion of the cross-section toward the newborn law Π_ϕ at rate Ω. Every *instantaneous* pricing relation of Section 4 therefore transfers verbatim, evaluated at the turnover-stationary S_t, while the forward-looking valuation keeps its form with the cross-section dynamics augmented by this mean-reversion. Second, *that cross-section is stationary* (Proposition 18): in the limiting regime where the common posterior has concentrated ($\mathrm{MAP}(\rho_t) = \phi^\star$, so the re-selection destination is calendar-invariant), a finite menu and a constant inflow of fresh types make S_t a time-homogeneous functional of a stationary, ergodic disaster clock, with a unique nondegenerate stationary law whose unconditional moments equal long-run time averages.

The quantitative reading turns on two summary properties of the stationary cross-section. The stationary *dispersion*—and through Proposition 6 the amplitude of the fear wave—is governed by the newborn law Π_ϕ and the turnover rate Ω: more dispersed births leave more disagreement, and Ω sets how fast fresh cohorts replace the selected cross-section. Whether the stationary cross-section is *on average complacent*—the level reading we take to the option evidence below—is likewise a parameter condition on Π_ϕ, not a consequence of the primitives.

A. Parameters and quantitative reading

We discipline the physical and preference parameters with the macro-disaster evidence and hold them fixed throughout; the belief block (Π, Φ) is not separately observed, since disaggregated disagreement data on disaster intensity do not exist. We read it two complementary ways. The comparative statics below *vary* the belief dispersion to trace the mechanism, using the option numbers only as a reference, not as inputs. The calibration of Section B instead *pins* a three-parameter belief block to three stylized option-tail targets; this is an illustrative consistency check—whether one mechanism with near-orthogonal parameters can span the option-tail evidence—not a fit to the option surface and not an out-of-sample prediction. No survey data enter.

Before beliefs enter, risk aversion alone inflates the disaster-state pricing: a single contraction of size b carries the kernel weight $e^{\alpha J}$ with $J = -\log(1-b)$. At $\alpha = 3.5$ this multiplier is 1.4 at $b = 0.10$, 2.3 at the Barro–Ursúa mean $b = 0.21$, 3.3 at $b = 0.29$, and 6.0 at $b = 0.40$; the power-law tail (Barro and Jin, 2011) pushes the f-average $\mathbb{E}_f[e^{\alpha J}]$ above the value at the mean size.[8] The representative-agent fear factor is $U^{\mathrm{RA}} = \lambda^\star\, \mathbb{E}_f[e^{\alpha J}]$: physical disasters of frequency 3% are priced as if several times more frequent. This effect is the standard rare-disaster channel; our contribution is what the belief cross-section does to it.

The belief cross-section moves the risk-neutral tail several-fold. Under the binary menu, the fear factor is $U_t = G_t^\alpha\, U^{\mathrm{RA}}$ with $G_t = S_t^{hi}(\lambda_{hi}/\lambda^\star)^{1/\alpha} + (1 - S_t^{hi})(\lambda_{lo}/\lambda^\star)^{1/\alpha}$ (D17). With the

[8] The power-law parameters describe the Barro and Jin (2011) law of the gross contraction $z = 1/(1-b) = e^J$ ($\gamma \approx 7.2$, $z_{\min} = 1/(1-0.10) = 1.11$), not a Pareto on b. Under it $\mathbb{E}_f[b] = 1 - \gamma/[(\gamma+1)z_{\min}] = 0.21$ (the Barro–Ursúa mean) and $\mathbb{E}_f[e^{\alpha J}] = \mathbb{E}[z^\alpha] = \gamma z_{\min}^\alpha/(\gamma - \alpha) \approx 2.82$ are jointly delivered; a Pareto on b would imply a smaller mean (about 0.12) and is not what is used.

Block	Parameter	Value	Source
Physical	disaster intensity $\lambda^\star$	0.03/yr	Barro and Ursúa (2008), consumption disasters (10%)
	mean disaster size $\mathbb{E}_f[b]$	0.21	Barro and Ursúa (2008)
	size law f	power law ($\gamma \approx 7.2$, $b_{\min} = 0.10$)	Barro and Jin (2011)
	trend, diffusion (μ_0, σ)	0.025/yr, $0.02/\sqrt{\text{yr}}$	Barro (2006); Barro and Ursúa (2008)
Preferences	risk aversion α	3.5	Barro (2006); Barro and Ursúa (2008) (equity premium)
	discount δ	0.03/yr	standard
Demography	death hazard Ω	0.02/yr (50-yr horizon)	Blanchard (1985); Gârleanu and Panageas (2015)
Beliefs *(free)*	menu $\{\lambda_{lo}, \lambda_{hi}\}$	varied (e.g. $\{0.01, 0.06\}$)	illustrative; not matched
	newborn law Π	varied	illustrative; not matched

Table I: Parameters. The physical and preference blocks are disciplined by the macro-disaster literature; the belief block is illustrative and varied. Disaster sizes are consumption contractions for the unlevered tree; for comparison Barro (2006) report $\lambda^\star \approx 0.017$/yr at a 15% GDP threshold with mean size 0.29. At the illustrative menu the kernel inputs evaluate to $\mathbb{E}_f[e^{\alpha J}] = 2.82$ and the representative-agent fear factor to $U^{\text{RA}} = 0.085$/yr.

illustrative $\{\lambda_{lo}, \lambda_{hi}\} = \{0.01, 0.06\}$ around $\lambda^\star = 0.03$,[9] the ratio U_t/U^{RA} runs from 0.33 when the cross-section is fully complacent ($S^{hi} = 0$) to 2.0 when fully fearful ($S^{hi} = 1$), passing through 0.91 at $S^{hi} = \frac{1}{2}$: the belief cross-section alone swings the priced disaster intensity roughly six-fold, *above and below* the representative-agent benchmark, with no change in any physical parameter. The amplitude of this swing is governed by the menu spread and the stationary law of S^{hi}, i.e. by the free belief block through Proposition 18.

The same multiplier, written for a general cross-section in the intensity channel (Corollary 1), is $U_t/U^{\text{RA}} = G_t^\alpha$ with $G_t = \mathbb{E}_{S_t}[(\lambda/\lambda^\star)^{1/\alpha}]$ (D7). Replacing the two-point menu by a continuum

[9] The two menu points straddle but do not include $\lambda^\star$; the long-run pinning of the held intensity to $\lambda^\star$ (Lemma 2) requires $\lambda^\star$ to lie in the menu, as in the continuum $[\lambda_{lo}, \lambda_{hi}]$ used in the next paragraph, so the binary menu is a pricing illustration rather than a learning environment. Under demographic turnover the cross-section is in either case a refreshed mixture, not a point converged on $\lambda^\star$.

of held intensities filling $[\lambda_{lo}, \lambda_{hi}] = [0.01, 0.06]$ and varying the *shape* of the cross-section gives, at $\alpha = 3.5$, $\lambda^\star = 0.03$, the illustrative values:

cross-section of held λ	mean belief $\bar{\lambda}^S$	multiplier U_t/U^{RA}
Beta$(2, 5)$, complacent-skewed	0.024	0.78
Uniform$[0.01, 0.06]$	0.035	1.09
Beta$(2, 2)$, symmetric	0.035	1.12
Beta$(5, 2)$, fearful-skewed	0.046	1.51
Beta$(8, 2)$, very fearful	0.050	1.66

The continuum fills the interval $[\lambda_{lo}/\lambda^\star, \lambda_{hi}/\lambda^\star] = [0.33, 2.0]$ between the binary extremes; in the intensity channel it does not raise that ceiling (the ceiling-lift of Remark 4 is a size-channel statement), but it makes the multiplier a smooth, dispersion-sensitive functional of the cross-section. These Beta cross-sections are illustrative instantaneous evaluations of the general intensity-channel multiplier (Corollary 1); they are not stationary laws of the turnover economy, whose stationarity theorem (Proposition 18) is stated under the finite menu. That sensitivity is exact: since $\alpha > 1$ and $y \mapsto y^{1/\alpha}$ is strictly concave, Jensen's inequality applied to $\lambda/\lambda^\star$ under S_t gives

$$\frac{U_t}{U^{\mathrm{RA}}} = G_t^\alpha = \left(\mathbb{E}_{S_t}[(\lambda/\lambda^\star)^{1/\alpha}]\right)^\alpha \;\leq\; \frac{\bar{\lambda}_t^S}{\lambda^\star}, \tag{26}$$

with equality only when the cross-section is degenerate. Two consequences. First, a cross-section whose wealth-weighted mean belief equals the truth, $\bar{\lambda}_t^S = \lambda^\star$, still prices the tail *below* the representative-agent benchmark—dispersion alone is complacency: at an equal-mass two-point spread $\{\lambda^\star - s, \lambda^\star + s\}$ the multiplier is 0.96 at $s = 1\%/\text{yr}$ and 0.82 at $s = 2\%/\text{yr}$. Second, fear ($U_t > U^{\mathrm{RA}}$) requires $\bar{\lambda}_t^S$ to exceed $\lambda^\star$ by enough to overcome this Jensen discount, which a sufficiently severe disaster delivers by reweighting wealth toward and concentrating the cross-section on the fearful models (Proposition 3). Equation (26) is the analytic core of the chronic-complacency/disaster-spike asymmetry: the kernel prices the power mean of beliefs, which sits below the wealth-weighted arithmetic mean.

Selection is slow relative to disasters. The selection part of the calm erosion of Proposition 6(i), $d\bar{\lambda}^S/dt|_{\text{sel}} = -\frac{1}{\alpha}\operatorname{Var}_S(\lambda)$, reallocates wealth at the rate $\frac{1}{\alpha}(\lambda_{hi} - \lambda_{lo}) \approx 0.014$/yr—the decay of the high-to-low wealth *odds*, an e-folding of about 70 years (a heuristic scale; the full relaxation also carries the calm switch flow and demographic turnover). Disasters recur every $1/\lambda^\star \approx 33$ years and cohorts turn over every $1/\Omega \approx 50$ years. Because selection is *slower* than disaster recurrence, the fear share does not fully relax between typical disasters: the stationary cross-section carries a persistently positive, fluctuating fear share, and with it a non-degenerate, time-varying disaster premium—the model counterpart of the chronic equity premium and the slow post-crisis reversion of the tail factor (Andersen et al., 2015b).

These magnitudes reconcile two option facts that look opposed. The same multiplier $U_t/U^{\text{RA}} = G_t^\alpha$—below one in calm, above one after disasters—carries the economy between the modest option-implied disasters Backus et al. (2011) document and the large, spiking tail premium of Bollerslev and Todorov (2011) and Andersen et al. (2015b): the literatures measure it at different points of the fear cycle, against different anchors, and are not in conflict. The countercyclical, below-macro prediction is about the inverted intensity $\lambda^\star\kappa_t$, not the raw U_t, which carries the risk-aversion premium $\mathbb{E}_f[e^{\alpha J}]$ on top.

B. Confronting the option-tail evidence

The closed-form objects above line up, one for one, with three model-free option-tail literatures, so the model can be taken to that evidence without estimating the belief process. We hold the physical block at its macro-disaster calibration (Table I), which delivers the representative-agent benchmark $U^{\text{RA}} = \lambda^\star\mathbb{E}_f[e^{\alpha J}] = 0.085$/yr in closed form, and pin a three-parameter belief block—newborn dispersion, disaster-trigger mass, and calm relaxation rate—to three stylized targets drawn from that evidence. The targets are nearly orthogonal to the parameters: dispersion sets the calm level of κ_t, the trigger sets the disaster jump, and the relaxation rate sets the post-crisis half-life. The intensity rows of Table II are closed-form implications of the fear factor $U_t = U^{\mathrm{RA}}\kappa_t$ (15) and the finite-menu reduction of Corollary 1; the premium rows are *structural* consequences—the dominance of the disaster over the diffusive premium and the loading of the premium on U_t rather than σ—that additionally involve the price–dividend jump ratio ψ_t, a Feynman–Kac functional of the full record-resolved state rather than a closed form (Proposition 14), and are read off its qualitative behavior rather than a solved level, as the row notes below record.

The three anchors map directly (Table II): Backus et al. (2011)'s modest option-implied disaster law is $\kappa_t < 1$ through calm, so the option-implied intensity sits below U^{RA} and near $\lambda^\star$; Bollerslev and Todorov (2011)'s time-varying fear premium is $U_t > \lambda^\star$ on average, with the disaster premium nearly the whole equity premium since the diffusive $\alpha\sigma^2$ is negligible; and Andersen et al. (2015b,a)'s unspanned, persistent, premium-forecasting tail factor is κ_t, which jumps at disasters, erodes slowly in calm (25), and enters the premium through U_t while σ is held fixed.

Table II: Confronting the option-tail evidence. The physical block is fixed at the macro-disaster calibration (Table I), giving $U^{\mathrm{RA}} = 0.085$/yr; a three-parameter belief block is pinned to the calm level, the crisis jump, and the persistence. The model entries are closed-form in the fear factor $U_t = U^{\mathrm{RA}}\kappa_t$, the premium rows also carrying the jump ratio ψ_t.

Evidence	Documented finding	Model (closed form)
Backus et al. (2011)	options imply a more modest disaster law than the macro record	calm $\bar{\kappa} = 0.35 < 1$; option-implied $U_t \approx \lambda^\star$ in calm, below the benchmark U^{RA}
Bollerslev and Todorov (2011)	the risk-neutral jump tail exceeds the statistical one (a fear premium), time-varying	mean $U_t/\lambda^\star = 1.19 > 1$; spikes to 3.2–4.8$\times$ at and after crises
Bollerslev and Todorov (2011)	rare-event compensation is a large fraction of the equity premium	$\alpha\sigma^2 = 0.14\%$/yr diffusive $\Rightarrow \mathrm{DRP}_t$ (E37) is essentially the whole premium
Andersen et al. (2015b)	the tail factor predicts the equity premium; volatility does not	exact here: DRP_t loads on U_t, σ constant $\Rightarrow$ zero volatility loading
Andersen et al. (2015a)	the tail factor is far more persistent than volatility	calm half-life of U_t (25) ≈ 22 months $\gg$ that of volatility

Two of the five rows rest on a belief-free *identity*. The diffusive premium is mechanically small and fixed: $\alpha\sigma^2 = 0.14\%/\text{yr}$ at the Table I diffusion, so by (E36) the disaster premium DRP_t accounts for essentially all of the equity premium wherever it is sizable relative to that part, as at the calibration; the *share* itself is not belief-free, since the level of DRP_t moves with ψ_t and the full pricing state. The predict-the-premium-not-volatility pattern is an identity in this model: DRP_t loads on U_t (E37) while σ is constant, so the fear factor forecasts the premium with a nonzero loading and volatility forecasts it with exactly zero—the Andersen et al. (2015b) separation, an exact zero loading in the model (σ constant) rather than a fitted coefficient. The other three rows—the calm level $\bar{\kappa}$, the crisis jump, and the half-life—are the calibrated targets, and they are stylized: published estimates vary with sample and method, so we match orders of magnitude, not point values.

This is a disciplined confrontation, not a structural estimation: we show that the closed-form fear factor is consistent with the published model-free moments, not that the belief process is identified by them. The one feature the belief block cannot repair is the safe rate. Under power utility the safe rate sits at the high CRRA *level* (Section A)—its realized dynamics are modest, but the level is the standard risk-free-rate artifact at $\alpha = 3.5$; that is a preference limitation, removed only by an Epstein–Zin extension, and not something the belief calibration can absorb.

One model-free finding runs the other way. Gormsen and Jensen (2025) estimate the *physical* return distribution from option prices and find it more left-skewed and fat-tailed in good times—a *procyclical* conditional tail risk they read as hard to square with disaster models in which the physical law is elevated in bad times. This economy is not such a model: the physical law $(\lambda^\star, F_J)$ is fixed, and what is countercyclical is the *priced* multiplier κ_t, not the world's disaster risk. The link is tighter than coexistence—the complacency that holds $\kappa_t < 1$ through calm *is* an under-pricing of the tail in precisely the spells where, on their evidence, the conditional distribution is fattest—so the gap between the thin priced tail and the fat physical tail is the belief wedge, bracketed from opposite ends: its priced side the modest option-implied disasters of Backus et al. (2011) the calibration matches, its physical side the procyclical return tail Gormsen and Jensen measure. The mechanism turns this into a discriminating prediction—the physical-minus-priced gap should widen through calm and collapse at disasters, developed in the Internet Appendix—so Gormsen and Jensen's fact is a second empirical handle on the price of disaster fear, not an embarrassment to it.

6. Conclusion

This paper has built a complete-market disaster economy in which a continuum of CRRA investors disagree about the disaster law and revise it lazily, and has shown that the risk-neutral disaster tail factors exactly into a frictionless representative-agent benchmark and a belief multiplier, $U_t = U^{\mathrm{RA}} \kappa_t$ (Proposition 2). The mechanism is one line: the kernel prices the *power* mean of beliefs, which sits below the wealth-weighted *arithmetic* mean, so a cross-section correct on average still prices the option tail below the benchmark in calm—dispersion alone is complacency—and reprices it upward in coordinated jumps when a disaster proves the fearful right. Lazy revision makes this a fast-in/slow-out fear factor (Proposition 6)—synchronized disaster jumps, slow staggered calm relaxation as selection reallocates wealth—with a unique stationary cycle once demographic turnover refreshes the disagreement (Propositions 17, 18).

The reading is that the option surface measures beliefs, not the physical disaster law: a moving priced tail over a fixed quantity of disaster risk. It reconciles the modest option-implied disasters of Backus et al. (2011) with the spiking risk-neutral tails of Bollerslev and Todorov (2011) and Andersen et al. (2015b) as one belief cycle, predicting that the representative-agent-inverted intensity $\lambda^{\star} \kappa_t$ is countercyclical—below the macroeconomic frequency in calm, rising after disasters. And it leaves an over-identifying mark: because a held belief is not a sufficient statistic for its own revision, the forward-looking surface is read off the records under which current beliefs have survived, not the beliefs alone, and forecasts that surface even after the contemporaneous option state is controlled for—a residual that any Markov benchmark sets to zero (Proposition 5), the cleanest line separating lazy revision from the whole class.

The limitations are explicit: CRRA preferences fix the safe-rate *level* at the high-α artifact of that class (removable, we conjecture but do not prove, by an Epstein–Zin extension); the disaster-dynamics signs are intensity-channel results under a positive-association condition we flag rather than assume; and the belief block is calibrated against the option-tail moments rather than estimated, since disagreement data on disaster *intensity* do not exist. A fully structural calibration awaits such a measurement, and the nearest handle is the gap between the priced tail and the *physical* return tail, whose procyclical higher-moment risk (Gormsen and Jensen, 2025) is the far side of the same wedge—the surface under-pricing the tail in the very calm spells where the conditional distribution is fattest. Generating that procyclical tail from the primitives, rather than only reading it against them, is the natural next step this fixed-law economy poses but does not settle. What the wedge between the option surface and the macroeconomic record measures, throughout, is not a mismeasurement of either, but the price of disaster fear.

Notation

Primitives (the fixed physical law)

C_t	aggregate consumption, the Lucas-tree dividend
$\mu_0,\ \sigma$	drift and diffusion of log consumption
$\lambda^\star$	physical disaster intensity
$F_J,\ f$	physical disaster-size law and density on $\mathcal{J} = [\underline{J}, \overline{J}]$
$T_k,\ J_k$	time and size of the kth disaster

Beliefs and the revision rule

$\phi = (\lambda, \theta)$	a held model: disaster intensity λ, size index θ
$\Phi = \Lambda \times \Theta$	model space; F_θ the belief size law
$\theta^\star$	pseudo-true size index (minimizes $\mathrm{KL}(F_J \| F_\theta)$)
$\varepsilon^i,\ G$	investor i's surprise tolerance; its cross-section law
$\rho_t,\ \phi_t^\star$	common public posterior; re-selection destination $\phi_t^\star = \mathfrak{m}(\rho_t)$
$\mathcal{R}_t^i,\ \tau_t^i$	investor i's since-adoption record and regime age

Preferences and the pricing kernel

α	coefficient of relative risk aversion
s_t^i	investor i's wealth (consumption) share
Ξ_t	equilibrium pricing kernel (state-price density)
$g_t(x)$	share-weighted aggregator; g_t^α is the $(1/\alpha)$-power mean of the disaster likelihood ratios

The fear factor and the belief multiplier

$\lambda_t^{\mathbb{Q}},\ f_t^{\mathbb{Q}}$	risk-neutral disaster intensity and size density
U_t	disaster-fear factor, $U_t = \lambda_t^{\mathbb{Q}}$
U^{RA}	frictionless benchmark $\lambda^\star\, \mathbb{E}_f[e^{\alpha J}]$ (correct-belief value, at $g_t \equiv 1$)
κ_t	belief multiplier U_t/U^{RA}; the bridge is $U_t = U^{\mathrm{RA}}\kappa_t$

Appendix A. Proofs for Section 2

This appendix states and proves Lemma 1, Lemma 2, and Lemma 3. The first is a direct computation from the change-of-measure density; the second is the Kullback–Leibler rate calculation; and the third invokes the high-frequency threshold theory of Mancini (2009), whose hypotheses we verify by name before assembling the lemma.

We record the identification and regularity conditions on the belief family deferred from Section B, verified below for the families used in the paper.

Assumption 4 (The belief family): *The size family $\{F_\theta : \theta \in \Theta\}$ of Section B, with Lebesgue densities f_θ on $\mathcal{J}$ ($\Theta \subset \mathbb{R}$ a compact interval, and $\Lambda = [\underline{\lambda}, \overline{\lambda}] \subset (0, \infty)$ containing $\lambda^\star$ in its interior), satisfies:*

(i) Full support: *$f_\theta(x) > 0$ and continuous for all $x \in \mathcal{J}$, $\theta \in \Theta$.*

(ii) Regularity: *$\theta \mapsto \log f_\theta(x)$ is continuously differentiable with score $\dot{\ell}_\theta(x) := \partial_\theta \log f_\theta(x)$, and the Fisher information $I(\theta) := \int_{\mathcal{J}} \dot{\ell}_\theta(x)^2 \, F_\theta(dx)$ is finite and strictly positive on Θ.*

(iii) Identifiability: *$\theta \neq \theta' \Rightarrow F_\theta \neq F_{\theta'}$.*

(iv) Strict KL identifiability: *$\theta \mapsto \mathrm{KL}(F_J \| F_\theta)$ has a unique minimizer $\theta^\star$ on Θ.*

(v) Spanning: *for every $\alpha > 0$, the family $\{(f_\theta/f)^{1/\alpha} : \theta \in \Theta\}$ is linearly independent in $L^1(\mathcal{J}, f)$—no nontrivial finite linear combination vanishes f-a.e.*

(vi) Smooth pseudo-true point: *$\theta^\star \in \operatorname{int}\Theta$; for each $x \in \mathcal{J}$, $\theta \mapsto \log f_\theta(x)$ is twice continuously differentiable on a neighbourhood of $\theta^\star$, with $|\dot{\ell}_\theta(x)|$ and $|\partial_\theta^2 \log f_\theta(x)|$ dominated there by an envelope $\Psi(x)$ satisfying $\mathbb{E}_{F_J}[\Psi] < \infty$ and $\mathbb{E}_{F_J}[\Psi^2] < \infty$; and, writing $M(\theta) := \mathbb{E}_{F_J}[\log f_\theta(J)]$, the pseudo-Hessian $H := -M''(\theta^\star)$ is strictly positive and $B := \mathbb{E}_{F_J}[\dot{\ell}_{\theta^\star}(J)^2] < \infty$.*

The conditions invoked in Assumption 4 play the following roles.

Condition (iv) is a separate identification requirement from (iii), neither implying the other: where (iii) asks injectivity of $\theta \mapsto F_\theta$, (iv) asks uniqueness of the Kullback–Leibler projection of the truth onto the family. It holds, for instance, whenever $\{F_\theta\}$ is a regular exponential family in θ (where $\theta \mapsto \mathrm{KL}(F_J \| F_\theta)$ is strictly convex), and it is what makes the pseudo-true model a point rather than a set. It is invoked wherever a single pseudo-true point is needed—the pseudo-true model $\phi^\star$ of (A3), the smooth pseudo-true/asymptotic-normality setup of Assumption 4(vi), and the concentration limits of Remark 6 and Proposition 18; without it each such statement reads with $\arg\min_\theta \mathrm{KL}(F_J \| F_\theta)$ in place of a point. Condition (v) is the priced-aggregator analogue of identifiability: it ensures distinct *finite* size-belief mixtures leave distinct imprints on the risk-neutral tail (finite-mixture separation; it does not by itself give injectivity of the full mixture map over arbitrary continuum-supported states, which would need a completeness or transform-uniqueness condition). It holds for the families used here by distinct-exponent independence, not by analyticity alone. For the power-law size family $f_\theta(x) \propto x^{-(\theta+1)}$ on $\mathcal{J}$, $(f_\theta/f)^{1/\alpha} = c_\theta\, x^{-(\theta+1)/\alpha} f(x)^{-1/\alpha}$, which reduces to the monomial $c_\theta\, x^{(\theta_0-\theta)/\alpha}$ when $f = f_{\theta_0}$ lies in the family; in the misspecified case the common factor $f^{-1/\alpha}$ is shared across θ and so leaves independence intact—distinct indices give distinct real exponents, linearly independent on any non-degenerate interval (their Wronskian is a nonzero Vandermonde determinant in the exponents); the same holds for any family reducible to distinct-exponent or regular-exponential form. It is used only in Proposition 3 (the infinite-dimensionality of the pricing state). Condition (vi) is the standard regularity under which a possibly-misspecified maximum-likelihood estimator is $\sqrt{n}$-asymptotically normal (White, 1982; van der Vaart, 1998); it strengthens (ii) and, under correct specification, has its sandwich collapse ($H = B = I(\theta^\star)$), recovering the Fisher-information form of (ii). It is used only in Lemma 3(iii).

Conventions carried from Section B. The rejection criterion is the surprise $\bar{\varepsilon}$ (8): the held model is rejected iff $\bar{\varepsilon} \leq \varepsilon^i$, the boundary counted as a rejection. The closed-threshold form $\{D_{\phi,\Delta} \geq r_{\phi,\Delta}(\varepsilon)\}$ used in later expressions is shorthand for this p-value criterion; the two coincide exactly when D is atomless at $r_{\phi,\Delta}(\varepsilon^i)$, as for the continuous calm deviance—so the deterministic calm deadline $\Delta^\star$ where the calm deviance crosses r is the attained rejection time—and differ only at a count atom, where $\bar{\varepsilon} \le \varepsilon^i$ governs. The test is two-sided: the deviance is large whether disasters run too frequent or too rare, and whether the realized sizes pull $\hat{\theta}$ above or below θ. Two divergence rates also stay distinct: the full KL (A2) fixes second-order posterior concentration, the pseudo-true model (A3), and survival (Lemma 2), while the since-adoption rejection deviance that triggers revision drifts at the Poisson KL in timing and at the *excess* in-family size divergence relative to the pseudo-true fit, which can vanish at $\theta^\star$ even when the full size KL is positive—so the two coincide only when the size family is correctly specified.

We first record the change-of-measure and divergence facts.

Lemma 1 (Equivalence of held and physical laws on finite horizons): *Under Assumptions 1 and 4, for every $\phi = (\lambda, \theta) \in \Phi$ and every $t > 0$, $\mathbb{P}|_{\mathcal{F}_t} \sim \mathbb{P}^\phi|_{\mathcal{F}_t}$, with*

$$\frac{d\mathbb{P}^\phi}{d\mathbb{P}}\Big|_{\mathcal{F}_t} = \exp\!\big(-(\lambda - \lambda^\star)t\big) \prod_{k:\, T_k \le t} \frac{\lambda\, f_\theta(J_k)}{\lambda^\star\, f(J_k)} \;\in\; (0, \infty) \quad \mathbb{P}\text{-}a.s. \tag{A1}$$

Proof of Lemma 1. Imported result (marked-Poisson change of measure; Brémaud, 1981, Ch. VI; Jacod and Shiryaev, 2003, III.3). If $\mathbb{P}, \mathbb{P}^\phi$ are the laws on $(\Omega, \mathcal{F}_t)$ of marked Poisson processes with (intensity, mark law) $(\lambda^\star, F_J)$ and (λ, F_θ), $\lambda^\star, \lambda \in (0, \infty)$, and if $F_J \sim F_\theta$, then $\mathbb{P}|_{\mathcal{F}_t} \sim \mathbb{P}^\phi|_{\mathcal{F}_t}$ with Radon–Nikodym density (A1). *Hypotheses in our model.* (1) $\lambda^\star > 0$ (Assumption 1) and $\lambda \in [\underline{\lambda}, \overline{\lambda}] \subset (0, \infty)$ (Assumption 4); (2) f and f_θ are continuous and strictly positive on the common compact support $\mathcal{J}$ (Assumption 1; Assumption 4(i)), so F_J and F_θ have identical null sets, i.e. $F_J \sim F_\theta$. Both hypotheses hold, so the import applies and yields (A1). Finally N has finitely many jumps on $[0, t]$ $\mathbb{P}$-a.s. (Assumption 1), so the product is a finite product of strictly positive factors and the exponential is strictly positive; hence the density lies in $(0, \infty)$ $\mathbb{P}$-a.s., and $\mathbb{P}, \mathbb{P}^\phi$ share their null sets.

On the full filtration. Lemma 1 is stated on the consumption-path filtration $\mathcal{F}_t$, which carries both the Brownian component σW and the marked-Poisson disaster component. The diffusion (μ_0, σ) is undisputed (common knowledge, Assumption 1): every held law $\mathbb{P}^\phi$ shares the law of the continuous part with $\mathbb{P}$ and differs only in the disaster law (λ, F_θ). Because the Brownian and marked-Poisson components are independent under each law, the Radon–Nikodym density on $\mathcal{F}_t$ factors as the (identity) density of the unchanged Brownian law times the marked-Poisson density (A1); the change of measure acts only on the jump factor, and the equivalence $\mathbb{P}|_{\mathcal{F}_t} \sim \mathbb{P}^\phi|_{\mathcal{F}_t}$ holds on the full filtration. $\square$

Definition 5 (Per-period Kullback–Leibler divergence rate): *For a held model $\phi = (\lambda, \theta) \in \Phi$, let $\mathbb{P}^\phi$ be its law—the marked Poisson process of intensity λ with i.i.d. marks F_θ (Assumption 4)—and $\mathbb{P}$ the physical law (Assumption 1). The* per-period Kullback–Leibler rate *is the divergence of the truth from the held model, measured on the unit interval $\mathcal{F}_1$,*

$$\mathcal{K}(\phi) \;:=\; \mathrm{KL}\big(\mathbb{P} \,\big\|\, \mathbb{P}^\phi\big)\big|_{\mathcal{F}_1} \;=\; \mathbb{E}_{\mathbb{P}}\Big[\log \tfrac{d\mathbb{P}}{d\mathbb{P}^\phi}\Big|_{\mathcal{F}_1}\Big],$$

the $\mathbb{P}$-expectation of the log-likelihood ratio of the two laws restricted to $\mathcal{F}_1$.

Lemma 2 (Divergence rate: decomposition and pseudo-true model): *Under Assumptions 1 and 4, $\mathcal{K}(\phi) \in [0, \infty)$ for every $\phi \in \Phi$ and decomposes additively into a timing part and a size part,*

$$\mathcal{K}(\phi) \;=\; \underbrace{\Big[\lambda^\star \log \tfrac{\lambda^\star}{\lambda} - \lambda^\star + \lambda\Big]}_{\textit{timing}} \;+\; \underbrace{\lambda^\star \,\mathrm{KL}\big(F_J \,\big\|\, F_\theta\big)}_{\textit{size}}, \qquad \mathrm{KL}\big(F_J \,\big\|\, F_\theta\big) = \int_{\mathcal{J}} \log \frac{f(x)}{f_\theta(x)}\, F_J(dx). \tag{A2}$$

Each term is nonnegative; the timing term is zero only at $\lambda = \lambda^\star$, and the size term only when $f_\theta = f$, F_J-a.e. $\mathcal{K}$ is continuous on the compact Φ and attains its minimum, and because the two coordinates separate the minimizer factorizes,

$$\phi^\star = (\lambda^\star, \theta^\star), \qquad \theta^\star = \arg\min_{\theta \in \Theta} \mathrm{KL}\big(F_J \,\big\|\, F_\theta\big), \tag{A3}$$

with $\theta^\star$ unique by Assumption 4(iv). Finally $\mathcal{K}$ is a genuine per-unit-time rate: on any horizon $\mathrm{KL}(\mathbb{P}\|\mathbb{P}^\phi)\big|_{\mathcal{F}_t} = t\,\mathcal{K}(\phi)$*.*

Proof of Lemma 2. Step 0 (the log-density is well defined, finite, and integrable). By Lemma 1, $\mathbb{P}|_{\mathcal{F}_1} \sim \mathbb{P}^\phi|_{\mathcal{F}_1}$, and evaluating (A1) at $t = 1$ and inverting,

$$\frac{d\mathbb{P}}{d\mathbb{P}^\phi}\Big|_{\mathcal{F}_1} = \exp\big((\lambda - \lambda^\star)\big) \prod_{k:\,T_k \le 1} \frac{\lambda^\star f(J_k)}{\lambda\, f_\theta(J_k)}.$$

Each factor is strictly positive: $\lambda, \lambda^\star \in (0,\infty)$ gives $e^{(\lambda-\lambda^\star)} > 0$; $f(J_k) > 0$ and $f_\theta(J_k) > 0$ (Assumption 4(i)) give $\lambda^\star f(J_k)/(\lambda f_\theta(J_k)) > 0$; and $N_1 < \infty$ $\mathbb{P}$-a.s. (Assumption 1) makes the product a finite product of strictly positive factors. Hence $d\mathbb{P}/d\mathbb{P}^\phi|_{\mathcal{F}_1} \in (0,\infty)$ $\mathbb{P}$-a.s. For integrability, the integrand of $\mathcal{K}(\phi)$ is, by (A4) below, $\log \frac{d\mathbb{P}}{d\mathbb{P}^\phi}|_{\mathcal{F}_1} = (\lambda - \lambda^\star) + \sum_{k:T_k\le 1} \log \frac{\lambda^\star f(J_k)}{\lambda f_\theta(J_k)}$, so, with $B := \sup_{(x,\theta)\in\mathcal{J}\times\Theta} \big|\log \frac{f(x)}{f_\theta(x)}\big| < \infty$ (finite because $(x,\theta) \mapsto \log \frac{f(x)}{f_\theta(x)}$ is continuous on the compact $\mathcal{J}\times\Theta$, Assumption 4(i), by Weierstrass) and $C := \big|\log \frac{\lambda^\star}{\lambda}\big| + B < \infty$, the triangle inequality gives

$$\Big|\log \tfrac{d\mathbb{P}}{d\mathbb{P}^\phi}\Big|_{\mathcal{F}_1}\Big| \;\le\; |\lambda - \lambda^\star| + \sum_{k:\,T_k\le 1} \Big(\big|\log \tfrac{\lambda^\star}{\lambda}\big| + \big|\log \tfrac{f(J_k)}{f_\theta(J_k)}\big|\Big) \;\le\; |\lambda - \lambda^\star| + C\,N_1.$$

Taking $\mathbb{P}$-expectations and using $\mathbb{E}[N_1] = \lambda^\star < \infty$,

$$\mathbb{E}_{\mathbb{P}}\Big|\log \tfrac{d\mathbb{P}}{d\mathbb{P}^\phi}\Big|_{\mathcal{F}_1}\Big| \;\le\; |\lambda - \lambda^\star| + C\,\mathbb{E}[N_1] = |\lambda - \lambda^\star| + C\lambda^\star < \infty,$$

so the integrand is $\mathbb{P}$-integrable and $\mathcal{K}(\phi) = \mathbb{E}_{\mathbb{P}}\big[\log \frac{d\mathbb{P}}{d\mathbb{P}^\phi}\big|_{\mathcal{F}_1}\big] = \mathrm{KL}\big(\mathbb{P}\,\|\,\mathbb{P}^\phi|_{\mathcal{F}_1}\big)$ is finite and well defined—the Kullback–Leibler divergence of the physical law from the held-model law, written throughout with the single ratio $d\mathbb{P}/d\mathbb{P}^\phi$.

Step 1 (the integrand). By Lemma 1, inverting (A1) at $t = 1$ and taking logs, the integrand of $\mathcal{K}$ is, with each jump factor split into a timing and a size piece,

$$\log \frac{d\mathbb{P}}{d\mathbb{P}^{\phi}}\Big|_{\mathcal{F}_1} = (\lambda - \lambda^\star) + \sum_{k:\, T_k \le 1} \Big(\log \frac{\lambda^\star}{\lambda} + \log \frac{f(J_k)}{f_\theta(J_k)} \Big). \tag{A4}$$

Step 2 (expectation of a jump sum). For any g with $\mathbb{E}_{F_J}|g| < \infty$, condition on $N_1 = n$; under $\mathbb{P}$ the marks are then i.i.d. F_J and independent of the count (Assumption 1), so

$$\mathbb{E}_{\mathbb{P}}\Big[\sum_{k:\, T_k \le 1} g(J_k) \,\Big|\, N_1 = n \Big] = n\, \mathbb{E}_{F_J}[g(J)],$$
$$\mathbb{E}_{\mathbb{P}}\Big[\sum_{k:\, T_k \le 1} g(J_k) \Big] = \sum_{n \ge 0} \mathbb{P}(N_1 = n)\, n\, \mathbb{E}_{F_J}[g] = \mathbb{E}[N_1]\, \mathbb{E}_{F_J}[g] = \lambda^\star\, \mathbb{E}_{F_J}[g],$$

using $\mathbb{E}[N_1] = \lambda^\star$ on the unit interval.

Step 3 (apply Step 2 to the split term). With $g(x) = \log \frac{\lambda^\star}{\lambda} + \log \frac{f(x)}{f_\theta(x)}$ (integrable by Step 0),

$$\mathbb{E}_{\mathbb{P}}\Big[\sum_{k:\, T_k \le 1} \log \tfrac{\lambda^\star f(J_k)}{\lambda f_\theta(J_k)} \Big] = \lambda^\star \log \frac{\lambda^\star}{\lambda} + \lambda^\star \int_{\mathcal{J}} \log \frac{f(x)}{f_\theta(x)}\, F_J(dx) = \lambda^\star \log \frac{\lambda^\star}{\lambda} + \lambda^\star\, \mathrm{KL}(F_J \| F_\theta). \tag{A5}$$

Step 4 (assemble). Taking $\mathbb{E}_{\mathbb{P}}$ of (A4) ($\lambda - \lambda^\star$ is its own expectation) and substituting (A5), then regrouping the timing terms,

$$\mathcal{K}(\phi) = (\lambda - \lambda^\star) + \lambda^\star \log \tfrac{\lambda^\star}{\lambda} + \lambda^\star\, \mathrm{KL}(F_J \| F_\theta) = \Big[\lambda^\star \log \tfrac{\lambda^\star}{\lambda} - \lambda^\star + \lambda \Big] + \lambda^\star\, \mathrm{KL}(F_J \| F_\theta),$$

which is (A2).

Signs. Write $T(\lambda) = \lambda^\star \log \frac{\lambda^\star}{\lambda} - \lambda^\star + \lambda = \lambda^\star \log \lambda^\star - \lambda^\star \log \lambda - \lambda^\star + \lambda$. Differentiating term by term in λ ($\lambda^\star \log \lambda^\star$ and $-\lambda^\star$ constant),

$$T'(\lambda) = -\lambda^\star \cdot \tfrac{1}{\lambda} + 1 = 1 - \frac{\lambda^\star}{\lambda}, \qquad T''(\lambda) = \frac{d}{d\lambda}\Big(1 - \frac{\lambda^\star}{\lambda} \Big) = \frac{\lambda^\star}{\lambda^2}. \tag{A6}$$

By (A6) T' vanishes only at $\lambda = \lambda^\star$, and $T''(\lambda) = \lambda^\star/\lambda^2 > 0$ on $\Lambda \subset (0, \infty)$, so T is strictly convex with unique minimiser $\lambda = \lambda^\star$ and $T(\lambda^\star) = 0$; hence the timing term is ≥ 0, zero only at $\lambda = \lambda^\star$. By Gibbs' inequality $\mathrm{KL}(F_J \| F_\theta) \geq 0$, zero iff $f_\theta = f$, F_J-a.e.; hence the size term is ≥ 0 with the stated equality. Both terms being ≥ 0, $\mathcal{K} \geq 0$.

Continuity, minimiser, factorisation. T is continuous on Λ (bounded away from 0). For the size term, the KL integrand $x \mapsto \log\big(f(x)/f_\theta(x)\big)$ is continuous in θ (Assumption 4(i)) and dominated by the integrable envelope B of Step 0 (which bounds the *log-ratio*, not $\log f_\theta$ alone), so by dominated convergence $\theta \mapsto \mathrm{KL}(F_J \| F_\theta)$ is continuous on the compact Θ; thus $\mathcal{K}$ is continuous on the compact Φ and attains its minimum (extreme-value theorem). Since $\mathcal{K}(\lambda, \theta) = T(\lambda) + \lambda^\star \mathrm{KL}(F_J \| F_\theta)$ separates additively, the joint minimiser factorises into $\arg\min_\lambda T = \{\lambda^\star\}$ and $\arg\min_\theta \mathrm{KL}(F_J \| F_\theta) = \{\theta^\star\}$, the latter a singleton by Assumption 4(iv); this is (A3).

Rate. The marked Poisson record has stationary independent increments (Assumption 1), so the densities multiply across disjoint unit blocks: on $\mathcal{F}_t$ with integer t, $\log \frac{d\mathbb{P}}{d\mathbb{P}^\phi} = \sum_{j=1}^t \log \frac{d\mathbb{P}}{d\mathbb{P}^\phi}|_{(j-1,j]}$, each block term i.i.d. with mean $\mathcal{K}(\phi)$ by Steps 1–8; hence $\mathrm{KL}(\mathbb{P} \| \mathbb{P}^\phi)|_{\mathcal{F}_t} = t\,\mathcal{K}(\phi)$, and the same expectation computation with $\mathbb{E}[N_t] = \lambda^\star t$ extends it to all $t > 0$. Thus $\mathcal{K}$ is a per-unit-time rate. $\square$

The imported results, and our hypotheses. Mancini (2009, Thm. 3.1 and Cor. 3.2) consider $dX_t = a_t\,dt + \sigma_t\,dW_t + dJ_t$ with J a finite-activity jump process $J_t = \sum_{j=1}^{N_t} \gamma_j$, observed on a grid of mesh h over a fixed horizon. Under (1) a pathwise bound on the drift integral, which holds whenever a is càdlàg; (2) $\int_0^T \sigma_s^2\,ds < \infty$ with σ pathwise bounded, which holds whenever σ is càdlàg; (3) a deterministic threshold $r(h)$ with $r(h) \to 0$ and $h \log(1/h)/r(h) \to 0$; and (4) $\mathbb{P}\{\Delta N_t \neq 0,\ \gamma_{N_t} = 0\} = 0$, the threshold a.s. isolates the jump-free intervals, $\mathbf{1}\{\Delta_i N = 0\} = \mathbf{1}\{(\Delta_i X)^2 \leq r(h)\}$ for small h, and the truncated realized variation is consistent for the integrated variance,

$$\operatorname*{plim}_{h \to 0} \sum_i (\Delta_i X)^2\, \mathbf{1}\{(\Delta_i X)^2 \leq r(h)\} = \int_0^T \sigma_s^2\,ds. \tag{A7}$$

Mancini (2009, Thm. 3.7) adds that if J is a *compound Poisson* process, if σ is adapted, continuous, and *independent of W and N* (no leverage), with $\mathbb{E}[\int_0^T \sigma_s^2\, ds] < \infty$, then the truncation estimator of each realized size, $\hat{\gamma}^{(i)} = \Delta_i X\, \mathbf{1}\{(\Delta_i X)^2 > r(h)\}$, satisfies

$$\sqrt{n} \sum_i \big(\hat{\gamma}^{(i)} - \gamma^{(i)}\big) \mathbf{1}\{\Delta_i N \geq 1\} \xrightarrow{d} MN\Big(0,\ T \int_0^T \sigma_s^2\, dN_s\Big). \tag{A8}$$

Our endowment (2) is the special case $a_t \equiv \mu_0 - \frac{1}{2}\sigma^2$, $\sigma_t \equiv \sigma$, $J_t = -\sum_{k: T_k \leq t} J_k$ with N Poisson($\lambda^\star$) and marks $\{J_k\}$ i.i.d. F_J. We verify each hypothesis. The drift is constant, hence càdlàg, so (1) holds; σ is constant, hence càdlàg with $\int_0^T \sigma^2\, ds = \sigma^2 T < \infty$, so (2) holds; take $r(h) = h^{0.99}$, for which $r(h) \to 0$ and $h \log(1/h)/r(h) = h^{0.01} \log(1/h) \to 0$, so (3) holds; the marks satisfy $J_k \in [\underline{J}, \overline{J}]$ with $\underline{J} > 0$, so $\gamma_{N_t} = -J_{N_t} \neq 0$ a.s. and (4) holds. For (A8): J is compound Poisson by construction; σ is constant, hence adapted, continuous, and (being deterministic) independent of W and N, so the no-leverage hypothesis holds; and $\mathbb{E}[\int_0^T \sigma^2 ds] = \sigma^2 T < \infty$. All hypotheses of both results hold in our model.

What the path identifies, and what it does not. A held model can differ from the truth in the diffusion coefficient, the arrival intensity, or the size law, and these have different fates. The diffusion coefficient and the realized disaster sizes are *pathwise functionals* of the record: recovered, in the high-frequency limit, from the path on any interval, so a wrong diffusion is refuted on every interval. The intensity and the size *law* are not functionals of any finite record; they are properties of the generating law, learned only from the disasters that accrue at the rare-disaster rate. That asymmetry is what makes the joint (λ, θ) the admissible object of persistent disagreement.

In the high-frequency limit the path delivers the diffusion coefficient and every realized disaster size exactly (Mancini, 2009; Aït-Sahalia and Jacod, 2009), leaving no room to disagree past the first interval; in the long-horizon limit the intensity and the size law are still being learned, at the common rate $(\lambda^\star t)^{-1/2}$, because both are inferred from a sample that grows only as fast as disasters arrive (Lemma 3). A disagreement about σ cannot survive a single interval;

a disagreement about (λ, θ) survives for a length of time set by $1/\lambda^\star$. That is the property a disputed primitive must have. The intensity coordinate is the object of the companion foundation paper (Teguia, 2026); here it is carried jointly with the size law, the two learned in parallel from the same rare disasters.

Lemma 3 (Identification asymmetry): *Under Assumptions 1 and 4, observe the path of C on $[0, t]$.*

(i) (Diffusion: pathwise.) *The continuous quadratic variation of $\log C$ satisfies, for every $t \geq 0$ and $h > 0$,*

$$[\log C]^c_{t+h} - [\log C]^c_t = \sigma^2 h \qquad \textit{exactly,} \tag{A9}$$

and is recovered from a discrete record on any fixed $[s, s+h] \subseteq [0, t]$ as the probability limit, as the mesh shrinks, of the truncated realized variation. A held diffusion coefficient $\neq \sigma$ is contradicted on every interval.

(ii) (Realized disasters: pathwise.) *The disaster times $\{T_k\}$ and sizes $J_k = -\Delta \log C_{T_k}$ are the discontinuities of $\log C$ and their magnitudes; from a discrete record they are recovered consistently, at rate $\sqrt{n}$ in the mesh (n the number of high-frequency observation intervals, not the disaster count N_t). The recovered $\{J_k\}_{k:T_k \leq t}$ is an i.i.d. sample from F_J of size $N_t \sim \mathrm{Poisson}(\lambda^\star t)$.*

(iii) (The disaster law: statistical, and slow in both coordinates.) *Neither the intensity nor the size law is a functional of the record on $[0, t]$; both are identified only through the realized disaster record—for the intensity, the Poisson count over exposure time, so that the informative zero-count spells enter as well, not only the arrivals—and the information is block-diagonal between them.* Intensity: *the count N_t is a sufficient statistic for λ in the Poisson observation on $[0, t]$, with Fisher information t/λ; the maximum-likelihood estimator $\hat{\lambda}_t = N_t/t$ has $\mathrm{Var}(\hat{\lambda}_t) = \lambda^\star/t$, so its* relative *error is $(\hat{\lambda}_t - \lambda^\star)/\lambda^\star = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big)$ (the standard-deviation-normalized error is $O_{\mathbb{P}}(1)$).* Size law: *let $\hat{\theta}_t$ be the size (pseudo-*

)maximum-likelihood estimator on the realized sizes; under the regularity of Assumption 4(vi) it is asymptotically normal at rate $n^{-1/2}$ conditional on $N_t = n$ (Appendix A), and with $N_t \sim \text{Poisson}(\lambda^\star t)$ this pins the rate at

$$\hat{\theta}_t - \theta^\star = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big). \tag{A10}$$

Both errors vanish only as $t \to \infty$, and slowly when $\lambda^\star$ is small, since both are governed by the disaster count $N_t \sim \text{Poisson}(\lambda^\star t)$. By full support $f_\theta(J_k) > 0$ at every realized size, so a held model is never contradicted by an individual disaster—only by the accumulated record.

Proof of Lemma 3. (i). By (2), $\log C_t = \log C_0 + \int_0^t (\mu_0 - \frac{1}{2}\sigma^2)\, ds + \sigma W_t - \sum_{k:T_k \le t} J_k$. The drift $\int_0^\cdot (\mu_0 - \frac{1}{2}\sigma^2)\, ds$ is of finite variation, hence contributes nothing to the quadratic variation; the disaster sum $\sum_{k:T_k \le \cdot} J_k$ is a finite-activity pure-jump process, hence contributes nothing to the *continuous* quadratic variation. The only continuous component of $\log C$ is σW, so its continuous quadratic variation is

$$[\log C]^c_t = [\sigma W]_t = \sigma^2 [W]_t = \sigma^2 t,$$

the last equality using $[W]_t = t$. Therefore

$$[\log C]^c_{t+h} - [\log C]^c_t = \sigma^2(t+h) - \sigma^2 t = \sigma^2 h,$$

which is (A9). Mancini's truncated realized variation is a *fixed-horizon, shrinking-mesh* object, so we fix any interval $[t, t+\delta]$ of positive length δ and partition it with mesh $\Delta \to 0$ (*not* $\delta \to 0$). With the hypotheses verified above, (A7) applied on $[t, t+\delta]$ with mesh-dependent threshold $r(\Delta)$ gives

$$\operatorname*{plim}_{\Delta\to 0} \sum_{i:\,[t_{i-1},t_i]\subseteq[t,t+\delta]} (\Delta_i \log C)^2\, \mathbf{1}\{(\Delta_i \log C)^2 \le r(\Delta)\} = [\log C]^c_{t+\delta} - [\log C]^c_t = \sigma^2\delta,$$

the interval length held fixed while the mesh vanishes. Hence $\sigma^2 = (\sigma^2\delta)/\delta$ is recovered on every fixed interval, so σ is a pathwise functional of the continuous record and two investors who disagree about the diffusion coefficient are contradicted on every interval.

(ii). By Assumption 1, N is Poisson with finite intensity, so on $[0,t]$ it has finitely many jumps a.s.; by (2) the only discontinuities of $\log C$ are at the T_k, where $\Delta \log C_{T_k} = -J_k$, so the times and sizes are read off the path as its discontinuities and their magnitudes. From a discrete record, the jump-time estimator $\widehat{\Delta_i N} = \mathbf{1}\{(\Delta_i \log C)^2 > r(h)\}$ recovers the jump intervals (Mancini Thm. 3.1) and the size estimator $\hat{\gamma}^{(i)}$ recovers each realized size: in this finite-activity setting each jump is eventually isolated in its own mesh interval, so the truncated increment over that interval recovers the size with a Brownian error that vanishes as the mesh shrinks; the rate-$\sqrt{n}$ mixed-normal limit (A8) is the joint fluctuation of these componentwise recoveries, not a substitute for them. The marks are i.i.d. F_J and independent of N by Assumption 1, and $N_t \sim \text{Poisson}(\lambda^\star t)$, so $\{J_k\}_{k:T_k\le t}$ is an i.i.d. F_J-sample of size $N_t \sim \text{Poisson}(\lambda^\star t)$.

(iii). By (i)–(ii), the record on $[0,t]$ determines σ and the realized sizes exactly in the high-frequency limit, but the size *law* enters only through the i.i.d. sample $\{J_k\}_{k\le N_t}$; the diffusion and drift carry no information about F_J. Condition on $N_t = n$ and let $\hat{\theta}_t \in \arg\max_{\theta\in\Theta} \sum_{k\le n} \log f_\theta(J_k)$ be the size maximum-likelihood estimator.

Achieved rate. Imported result (asymptotic normality of the possibly-misspecified maximum-likelihood estimator; White, 1982, Thm. 3.2; van der Vaart, 1998, Thm. 5.39). For i.i.d. $J_1,\dots,J_n \sim F_J$ with pseudo-true value $\theta^\star = \arg\max_{\theta\in\Theta} M(\theta)$, $M(\theta) := \mathbb{E}_{F_J}[\log f_\theta(J)]$, interior and unique, and under twice-continuous differentiability of $\theta \mapsto \log f_\theta$ with first and second derivatives dominated by an F_J-square-integrable envelope and nonsingular pseudo-Hessian

$H := -M''(\theta^\star)$,

$$\sqrt{n}\,\big(\hat\theta_t - \theta^\star\big) \;\xrightarrow{d}\; N\big(0,\, H^{-1}BH^{-1}\big), \qquad B := \mathbb{E}_{F_J}\big[\dot\ell_{\theta^\star}(J)^2\big]. \tag{A11}$$

Hypotheses in our model. The pseudo-true $\theta^\star$ is the unique maximiser of M. By the definition of the Kullback–Leibler divergence and linearity of $\mathbb{E}_{F_J}$,

$$\mathrm{KL}(F_J\|F_\theta) = \mathbb{E}_{F_J}\Big[\log \tfrac{f(J)}{f_\theta(J)}\Big] = \mathbb{E}_{F_J}[\log f(J)] - \mathbb{E}_{F_J}[\log f_\theta(J)] = \mathbb{E}_{F_J}[\log f(J)] - M(\theta),$$

and $\mathbb{E}_{F_J}[\log f(J)]$ does not depend on θ, so minimising $\mathrm{KL}(F_J\|F_\theta)$ in θ is the same as maximising $M(\theta)$: $\arg\max_\theta M = \arg\min_\theta \mathrm{KL}(F_J\|F_\theta) = \{\theta^\star\}$ by Assumption 4(iv); interiority of $\theta^\star$, the C^2 and domination conditions, $H > 0$, and $B < \infty$ are Assumption 4(vi). All hypotheses hold, so (A11) applies and, given $N_t = n$, $\hat\theta_t - \theta^\star = O_{\mathbb{P}}(n^{-1/2})$. When the size family contains F_J (correct specification) $H = B = I(\theta^\star)$ and the asymptotic variance is $I(\theta^\star)^{-1}$; the order $n^{-1/2}$ is the same either way. The lemma asserts only this rate—an upper bound on the estimation error—so no efficiency or lower-bound claim is made or required.

From count to calendar time. Since $N_t \sim \mathrm{Poisson}(\lambda^\star t)$ has $\mathbb{E}[N_t] = \lambda^\star t$ and $N_t/(\lambda^\star t) \to 1$ a.s. as $t \to \infty$ (strong law for the Poisson process), substituting $n = N_t$ gives

$$\sqrt{N_t}\,\big(\hat\theta_t - \theta^\star\big) = O_{\mathbb{P}}(1), \qquad \text{hence} \qquad \hat\theta_t - \theta^\star = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big),$$

which is (A10); the rate is governed by the realized disaster count $\lambda^\star t$, so it vanishes only as $t \to \infty$ and slowly when $\lambda^\star$ is small. For the intensity, the count N_t is a sufficient statistic for λ in the Poisson observation on $[0, t]$, with Fisher information t/λ. The maximum-likelihood estimator is $\hat\lambda_t = N_t/t$, and since $N_t \sim \mathrm{Poisson}(\lambda^\star t)$ has $\mathrm{Var}(N_t) = \lambda^\star t$,

$$\mathrm{Var}(\hat\lambda_t) = \mathrm{Var}\Big(\frac{N_t}{t}\Big) = \frac{\mathrm{Var}(N_t)}{t^2} = \frac{\lambda^\star t}{t^2} = \frac{\lambda^\star}{t}, \qquad \mathrm{Var}\Big(\frac{\hat\lambda_t - \lambda^\star}{\lambda^\star}\Big) = \frac{\mathrm{Var}(\hat\lambda_t)}{(\lambda^\star)^2} = \frac{1}{\lambda^\star t},$$

so the *relative* error is $(\hat{\lambda}_t - \lambda^\star)/\lambda^\star = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big)$, the same disaster-count rate. The count and the marks are independent (Assumption 1), so the information is block-diagonal between λ and θ and the two are learned in parallel. Finally, by Assumption 4(i), $f_{\theta_i}(x) > 0$ for all $x \in \mathcal{J}$, so every realized $J_k \in \mathcal{J}$ has strictly positive density under any held model and is never a zero-probability event: no single mark *falsifies* a held model by impossibility. Test rejection is a separate matter—the deviance can cross its threshold at a disaster epoch, including the first, when the updated record leaves the typical set—so full support rules out falsification by impossibility, not rejection by the test. $\square$

Appendix B. Proofs for Section 3

Sufficiency of the since-adoption record. For an exponential-family size law with natural sufficient statistic T, the pair $\big(n, \sum_j T(x_j)\big)$ is jointly sufficient for (λ, θ) by the Fisher–Neyman factorisation criterion, so the deviance (7) depends on the record only through it (the predictive density (6) additionally carries the marks' base-measure factors, which cancel in the deviance ratio); for a general family the test reads a goodness-of-fit statistic of (6) and no sufficiency is claimed. It is recorded for completeness—it is why exponential families are computationally special—but no result in the main text relies on it.

Proof of Proposition 4. By Proposition 9, $m_t = \mathsf{T}_{t\#}G$ with $\mathsf{T}_t(\cdot)$ deterministic, so the held-model marginal is the image of G under the threshold-to-trajectory map. Under (a): at T_k the rejecting cohort relocates to the common $\phi^{(k)}$ (Proposition 8) while the non-rejecting cohort of positive G-mass retains ϕ_0; since the destination differs in intensity by hypothesis, $\lambda(\phi^{(k)}) \neq \lambda(\phi_0)$, the held-model marginal places positive mass on two distinct intensities, whence $\mathrm{Var}_{S_{T_k}}(\lambda) > 0$. (We invoke only distinctness: the revealed-preference monotonicity of Lemma 4 is a fixed-count comparison and does not by itself sign $\lambda(\phi^{(k)}) - \lambda(\phi_0)$ once the elapsed calm time at T_k enters.) Under (b): along the zero-count history the reviser of threshold ε adopts the calm mode $\lambda^{\mathrm{m}}_{\Delta^\star(\phi_0,\varepsilon)}$ at her deadline; with the deadline map non-constant on a positive-mass set and $\tau \mapsto \lambda^{\mathrm{m}}_\tau$ strictly decreasing, two positive-mass groups of thresholds adopt distinct intensities before the first disaster, whence $\mathrm{Var}_{S_t}(\lambda) > 0$ for t past the earliest such deadline. *From head count to consumption shares.* Each case produces a split of positive G-mass, hence positive head-count mass m_t (Proposition 9); the variance claim is in the consumption-share measure S_t. The two are mutually absolutely continuous: with the symmetric allocation (Assumption 3(iv)) the share is $s^i_t = (M^i_t)^{1/\alpha}/\int (M^j_t)^{1/\alpha}\, dj$, and M^i_t is bounded and bounded away from zero on the compact menu: by (C1), with $N_t < \infty$ (Assumption 1) the density is a finite product of reweights $r_i = \lambda_i f_{\theta_i}/(\lambda^\star f)$—each continuous and strictly positive on the compact $\Phi \times \mathcal{J}$ (Assumption 4), hence in a fixed positive range—times the intensity factor $e^{-(\lambda_i-\lambda^\star)t}$ accrued over the current held-model segment, bounded for $\lambda_i \in [\underline{\lambda}, \overline{\lambda}]$ (a revision only relabels ϕ_i and restarts the segment, preserving the two-sided bound); so there are $0 < \underline{c} \le \bar{c} < \infty$ with $\underline{c}\, m_t(A) \le S_t(A) \le \bar{c}\, m_t(A)$ for every Borel A. A positive-m_t-mass split into two intensity classes is therefore a positive-S_t-mass split. In either case the dispersion is strictly positive at a finite t, from the homogeneous start. □

Lemma 4 (The second-order posterior is common): *For every pair of investors i, j and every t, $\rho^i_t = \rho^j_t =: \rho_t$. The common posterior is continuous in t between disasters and jumps at each disaster epoch; during a disaster-free spell its maximum-a-posteriori intensity $\lambda^{\mathrm{m}}_t :=$* $\min \arg\max_\lambda \max_\theta \rho_t(\lambda, \theta)$ *(the intensity coordinate of the tie-broken mode $\mathfrak{m}(\rho_t)$ of Definition 1) is non-increasing in t.*

Proof of Lemma 4. By (4), $\rho^i_t(\phi) \propto \rho_0(\phi)\, L_t(\phi)$ with $L_t(\phi) = e^{-\lambda t}\lambda^{N_t} \prod_{k:T_k \le t} f_\theta(J_k)$. The prior ρ_0 is common across investors (Assumption 2) and L_t is a function of the common path $(t, N_t, \{J_k\})$ alone, with no dependence on i or ε^i; hence $\rho^i_t = \rho^j_t =: \rho_t$. Continuity in t follows from L_t and its normalizer: $L_t(\phi)$ is continuous in t off the disaster epochs and jumps at them, and the normalizing constant $Z_t = \int_\Phi \rho_0(\phi) L_t(\phi)\, d\phi$ is continuous and strictly positive *on disaster-free intervals* (a continuous, strictly positive integrand on the compact Φ) and jumps at each disaster epoch. At a disaster of size J_k, $L_t(\lambda,\theta)$ is multiplied pointwise by the model-specific factor $\lambda f_\theta(J_k)$, while the normalizer Z_t is multiplied by its scalar posterior-predictive average $\int_\Phi \lambda f_\theta(J_k)\,\rho_{t-}(d\phi)$, *not* by any model-dependent factor; the posterior $\rho_t = \rho_0 L_t/Z_t$ therefore jumps precisely because the numerator updates model by model and the denominator by this common scalar average, and is continuous between disasters. This gives the continuity-between-disasters and disaster-jump claims of the statement.

For the monotonicity, fix a disaster-free interval, on which N_t and the sizes are constant. By the product prior $\rho_0 = \rho_0^\Lambda \otimes \rho_0^\Theta$ (Assumption 2), the log-posterior separates,

$$\log \rho_t(\lambda,\theta) = \underbrace{\Big[\log \rho_0^\Lambda(\lambda) - \lambda t + N_t \log \lambda\Big]}_{=:\ell_t^\Lambda(\lambda)} + \underbrace{\Big[\log \rho_0^\Theta(\theta) + \textstyle\sum_k \log f_\theta(J_k)\Big]}_{\theta\text{-only}} + \text{const.}$$

The θ-only term does not depend on λ, so $\max_\theta \log \rho_t(\lambda,\theta) = \ell_t^\Lambda(\lambda) + \text{const}$ and the maximum-a-posteriori intensity is $\lambda_t^{\mathrm{m}} = \min \arg\max_{\lambda\in\Lambda} \ell_t^\Lambda(\lambda)$. The maximizer set is non-empty because ℓ_t^Λ is continuous (hence upper semicontinuous) on the compact interval Λ, so $\lambda_t^{\mathrm{m}} = \min \arg\max_{\lambda\in\Lambda} \ell_t^\Lambda(\lambda)$ is well defined. We prove monotonicity directly, by revealed preference. Fix $t_2 > t_1$ in the disaster-free interval and write $\lambda_1 := \lambda_{t_1}^{\mathrm{m}}$, $\lambda_2 := \lambda_{t_2}^{\mathrm{m}}$. Each is optimal at its own date:

$$\ell_{t_1}^\Lambda(\lambda_1) \geq \ell_{t_1}^\Lambda(\lambda_2), \qquad \ell_{t_2}^\Lambda(\lambda_2) \geq \ell_{t_2}^\Lambda(\lambda_1).$$

Adding the two inequalities and regrouping,

$$\big[\ell_{t_2}^\Lambda(\lambda_2) - \ell_{t_1}^\Lambda(\lambda_2)\big] - \big[\ell_{t_2}^\Lambda(\lambda_1) - \ell_{t_1}^\Lambda(\lambda_1)\big] \;\geq\; 0. \tag{B1}$$

Since N_t and the sizes are constant on the interval (write N for the common count), the prior and count terms of ℓ^Λ cancel in the difference at any fixed λ:

$$\ell_{t_2}^\Lambda(\lambda) - \ell_{t_1}^\Lambda(\lambda) = \big[\log \rho_0^\Lambda(\lambda) - \lambda t_2 + N\log\lambda\big] - \big[\log \rho_0^\Lambda(\lambda) - \lambda t_1 + N\log\lambda\big] = -\lambda(t_2 - t_1). \tag{B2}$$

Evaluating (B2) at $\lambda = \lambda_2$ and $\lambda = \lambda_1$, the left-hand side of (B1) equals

$$\big[-\lambda_2(t_2 - t_1)\big] - \big[-\lambda_1(t_2 - t_1)\big] = -(t_2 - t_1)(\lambda_2 - \lambda_1).$$

Thus

$$-(t_2 - t_1)(\lambda_2 - \lambda_1) \;\geq\; 0,$$

and since $t_2 > t_1$ this forces $\lambda_2 \leq \lambda_1$, i.e. $\lambda^{\mathrm{m}}_{t_2} \leq \lambda^{\mathrm{m}}_{t_1}$. Hence λ^{m}_t is non-increasing in t through the disaster-free spell. $\square$

Proposition 7 (Law of motion; scheduled intensity relief): *Fix investor i. Between adoptions the held model ϕ_i is constant and the regime age advances at unit rate, $\dot{\tau}^i_t = 1$; the revision time is the first exit of the record from the typical set,*

$$\tau_i^{\mathrm{rev}} = \inf\big\{t > a^i : \; \bar{\varepsilon}(\mathcal{R}^i_t; \phi_i, \tau^i_t) \leq \varepsilon^i\big\}. \tag{B3}$$

Two exit channels generate it. (Calm.) *If no disaster occurs after adoption, the record is the zero-count history, whose surprise $\bar{\varepsilon}$ tends to 0 as the age grows—the zero-count record becomes atypical of a process expected to deliver disasters; hence there is a finite, deterministic calm deadline*

$$\Delta^\star(\phi_i, \varepsilon^i) = \inf\big\{\Delta > 0 : \; \bar{\varepsilon}\big((0, \varnothing); \phi_i, \Delta\big) \leq \varepsilon^i\big\} < \infty, \tag{B4}$$

at which the held model is rejected absent any disaster; the two-argument $\Delta^\star(\phi, \varepsilon)$ denotes this zero-count case. (Disaster.) *At a disaster the record jumps—count $+1$ and a new size—and $\bar{\varepsilon}$ is re-evaluated, so a disaster is an exit if it lands the record outside the typical set. The revision time is the first exit of the* updated *record—the earlier of the current calm deadline and the next disaster that lands outside the typical set; after a non-rejecting disaster the count is $n \geq 1$, so the relevant calm deadline is the first passage of the deviance for that updated record, not the original zero-count deadline. At a calm rejection reached along a spell that is* disaster-free since adoption*, the re-selected intensity satisfies $\lambda^{\mathrm{m}}_{\tau_i^{\mathrm{rev}}} \leq \lambda_i$; an intervening non-*

rejecting disaster updates the record to count $n \geq 1$ and can raise the mode, so this scheduled relief is the disaster-free statement, relative to the start of the current disaster-free spell.

Proof of Proposition 7. By Definition 1 the held model equals the second-order mode fixed at the last adoption and is unchanged until the next rejection, while the regime age advances at unit rate; this gives constancy and $\dot{\tau}_t^i = 1$. By Definition 1 a rejection is the first time the deviance exceeds its threshold, equivalently the first time $\bar{\varepsilon} < \varepsilon^i$, which is (B3).

Calm channel. Suppose no disaster occurs after adoption, so the record at age $\Delta = \tau_t^i$ is the zero-count history $(0, \varnothing)$. Its deviance is the pure count term, by (7) and $d_P(0\|\lambda) = \lambda$,

$$D_{\phi_i,\Delta}\big((0,\varnothing)\big) = \Delta\, d_P(0\|\lambda_i) = \lambda_i\Delta,$$

and by (8) its surprise is the predictive mass of records at least as deviant, which Markov's inequality (valid as $D \geq 0$) bounds by the mean deviance under the held model,

$$\bar{\varepsilon}\big((0,\varnothing);\phi_i,\Delta\big) = Q_{\phi_i,\Delta}\big(\{\mathcal{R}' : D_{\phi_i,\Delta}(\mathcal{R}') \geq \lambda_i\Delta\}\big) \leq \frac{\mathbb{E}_{Q_{\phi_i,\Delta}}\big[D_{\phi_i,\Delta}\big]}{\lambda_i\Delta}. \tag{B5}$$

We show the bound vanishes as $\Delta \to \infty$. Under $Q_{\phi_i,\Delta}$ the count is $n \sim \text{Poisson}(m)$ with $m := \lambda_i\Delta$, and given n the sizes are i.i.d. F_{θ_i}; by (7) the deviance splits as $D = c(n) + \Lambda_n$ into the count deviance $c(n) = n\log(n/m) - n + m \geq 0$ (with $c(0) = m$) and the size deviance $\Lambda_n = \sum_{k=1}^n \log\frac{f_{\hat{\theta}}(J_k)}{f_{\theta_i}(J_k)} = \sup_{\theta\in\Theta}\sum_{k=1}^n \log\frac{f_{\theta}(J_k)}{f_{\theta_i}(J_k)} \geq 0$, and we bound each mean.

Count mean: bounded. Write the count deviance as $c(n) = n\log(n/m) - n + m = n\log n - n\log m - n + m$. Taking the Q-expectation with $n \sim \text{Poisson}(m)$, so $\mathbb{E}[n] = m$ and hence $\mathbb{E}[-n+m] = 0$,

$$\mathbb{E}[c(n)] = \mathbb{E}[n\log n] - (\log m)\,\mathbb{E}[n] + \mathbb{E}[-n+m] = \mathbb{E}[n\log n] - m\log m. \tag{B6}$$

For the first term, the $n = 0$ summand is zero and $n/n! = 1/(n-1)!$, so shifting the index by

$j = n - 1$ (with $M \sim \text{Poisson}(m)$),

$$\begin{aligned}\mathbb{E}[n \log n] &= \sum_{n \geq 1} e^{-m} \frac{m^n}{n!} \, n \log n = \sum_{n \geq 1} e^{-m} \frac{m^n}{(n-1)!} \log n \\ &= m \sum_{j \geq 0} e^{-m} \frac{m^j}{j!} \log(j+1) = m \, \mathbb{E}[\log(M+1)]. \end{aligned} \tag{B7}$$

Substituting (B7) into (B6) and combining the logarithms,

$$\mathbb{E}[c(n)] = m \, \mathbb{E}[\log(M+1)] - m \log m = m \, \mathbb{E}\Big[\log \tfrac{M+1}{m}\Big] \leq m \log \tfrac{m+1}{m} \leq m \cdot \tfrac{1}{m} = 1,$$

the first inequality by Jensen (log concave, $\mathbb{E}[M+1] = m+1$) and the second by $\log(1+x) \leq x$. Thus $\mathbb{E}[c(n)] \leq 1$ for every $\Delta > 0$.

Size mean: $o(m)$. The map $(\theta, x) \mapsto \log f_\theta(x)$ is continuous on the compact $\Theta \times \mathcal{J}$ (Assumption 4(i)), hence bounded, so $B := \sup_{\theta, \theta' \in \Theta, \, x \in \mathcal{J}} \big|\log \frac{f_\theta(x)}{f_{\theta'}(x)}\big| < \infty$ and $0 \leq \Lambda_n \leq Bn$. We invoke the uniform strong law of large numbers in the form underlying maximum-likelihood consistency (Wald, 1949): if Θ is compact, $\theta \mapsto \log f_\theta(x)$ is continuous for each x, and the envelope is F_{θ_i}-integrable, then

$$\sup_{\theta \in \Theta} \Big| \tfrac{1}{n} \textstyle\sum_{k \leq n} \log \frac{f_\theta(J_k)}{f_{\theta_i}(J_k)} - \big(-\mathrm{KL}(F_{\theta_i} \| F_\theta)\big) \Big| \xrightarrow{\text{a.s.}} 0 \qquad (n \to \infty). \tag{B8}$$

Each hypothesis holds in our model: Θ is compact (Assumption 4), $\log f_\theta(x)$ is continuous in θ (Assumption 4(i)–(ii)), and the constant envelope B is trivially F_{θ_i}-integrable. Since $\sup_\theta \big(-\mathrm{KL}(F_{\theta_i} \| F_\theta)\big) = 0$, attained at $\theta = \theta_i$, (B8) gives $\Lambda_n / n = \sup_\theta \frac{1}{n} \sum_{k \leq n} \log \frac{f_\theta}{f_{\theta_i}}(J_k) \to 0$ a.s., with $0 \leq \Lambda_n / n \leq B$. As $\Delta \to \infty$, $m \to \infty$ and $n \to \infty$ a.s., and $n/m \to 1$ a.s. with $\{n/m\}$ uniformly integrable ($\mathbb{E}[(n/m)^2] = 1 + 1/m$ is bounded). Hence $\Lambda_n / m = (\Lambda_n / n)(n/m) \to 0$ a.s. and is dominated by the uniformly integrable $B \, (n/m)$, so by Vitali's convergence theorem $\mathbb{E}[\Lambda_n]/m \to 0$.

Conclusion. Combining the two means, $\mathbb{E}_{Q_{\phi_i, \Delta}}[D] / (\lambda_i \Delta) \leq 1/m + \mathbb{E}[\Lambda_n]/m \to 0$, so by

(B5) $\bar{\varepsilon}\big((0,\varnothing);\phi_i,\Delta\big) \to 0$ as $\Delta \to \infty$. As $\varepsilon^i > 0$, the set $\{\Delta > 0 : \bar{\varepsilon}((0,\varnothing);\phi_i,\Delta) < \varepsilon^i\}$ is non-empty, so its infimum—the calm deadline (B4)—is finite, $\Delta^\star(\phi_i,\varepsilon^i) < \infty$. By definition of the infimum, $\bar{\varepsilon} \geq \varepsilon^i$ for every $\Delta < \Delta^\star$, so the held model survives up to $\Delta^\star$ and is rejected there: $\Delta^\star$ is the first crossing, and no monotonicity of $\bar{\varepsilon}$ in the age is needed. The count contributes atoms to the deviance, so $\Delta \mapsto \bar{\varepsilon}((0,\varnothing);\phi_i,\Delta)$ may step downward across ε^i and need not attain it; the closed-threshold convention of Definition 1—reject at the first age with $\bar{\varepsilon} \leq \varepsilon^i$—fixes the boundary, and rejection occurs exactly at the infimum $\Delta^\star$. It is deterministic, since $\bar{\varepsilon}((0,\varnothing);\phi_i,\cdot)$ is a function of (ϕ_i,Δ) alone. Finally the deadline is non-increasing in ε^i: a larger threshold enlarges the rejection set $\{\Delta : \bar{\varepsilon} < \varepsilon^i\}$, lowering its infimum—which, as all investors share ϕ_0 at the start, is what staggers calm revision across the population.

Disaster channel. At a disaster the record gains a count and a size, so the deviance jumps and is re-evaluated; the record exits the typical set there iff the post-disaster deviance exceeds the threshold. The revision time is the earlier of the calm deadline and the first such disaster.

Relief (disaster-free spell). Consider a calm rejection reached along a spell disaster-free since adoption a^i, with re-selected intensity the mode $\lambda^{\mathrm{m}}_{\tau_i^{\mathrm{rev}}}$ and incumbent the mode at adoption $\lambda_i = \lambda^{\mathrm{m}}_{a^i}$. By Lemma 4 the mode is non-increasing *through a disaster-free spell*, so $\lambda^{\mathrm{m}}_{\tau_i^{\mathrm{rev}}} \leq \lambda_i$. If instead a non-rejecting disaster intervenes it appends a mark and can raise the mode—Lemma 4 bounds the mode only along disaster-free motion—so the bound then holds relative to the mode at the start of the *current* disaster-free spell, not necessarily to λ_i. This is the scheduled relief that erodes the held intensity *between* disasters; disasters themselves enter through the disaster reweight and relocation, not the calm relief. □

Proposition 8 (Synchronization at disasters; staggered calm): *At a disaster epoch T_k every investor's record receives the same increment at the same instant. Within the cohort of investors sharing a held model ϕ and adoption date a at T_k^- (hence the same record and predictive law), the post-disaster surprise depth is a common value $\bar{\varepsilon}_k(\phi,a)$, so the investors of that cohort* *who revise are exactly those with $\varepsilon^i \geq \bar{\varepsilon}_k(\phi, a)$; and* every *investor who revises at T_k, across all cohorts, adopts the common mode $\phi^{(k)} = \mathfrak{m}(\rho_{T_k})$ (Definition 1). Revision at a disaster is thus synchronized in timing and destination, with a cohort-specific trigger. Away from disasters, investor i revises at her calm deadline $a^i + \Delta^\star(\phi_i, \varepsilon^i)$, evaluated on the record accrued since her current adoption (a post-adoption disaster that does not itself trigger rejection resets the deadline on the updated, positive-count record; the zero-count expression is the disaster-free special case)—a time* non-increasing *in ε^i (Proposition 7), so calm revisions are staggered across thresholds wherever that deadline map is non-constant. Starting from the homogeneous configuration of Assumption 2, both channels generate dispersion rather than assume it—the synchronized disaster split and the staggered calm relief each disperse the held intensity, under conditions made precise in Proposition 4 (see also Remark 2).*

Proof of Proposition 8. At a disaster T_k the path increments by a single count and the common size J_k, which enters every investor's record simultaneously. Two investors of the same cohort—same held ϕ and same adoption date a at T_k^-—have, by construction, the identical record on $(a, T_k]$ and hence the identical post-disaster record; since the deviance (7) and the surprise (8) are deterministic functions of $(\mathcal{R}, \phi, \Delta)$, both share a common post-disaster surprise $\bar{\varepsilon}_k(\phi, a)$. By Definition 1 cohort member i rejects iff $\varepsilon^i \geq \bar{\varepsilon}_k(\phi, a)$. Every investor who rejects at T_k, of any cohort, re-selects $\arg\max_\phi \rho^i_{T_k}(\phi)$, which by Lemma 4 is the common posterior ρ_{T_k}; the deterministic tie-break $\mathfrak{m}$ of Definition 1 then selects the single common point $\phi^{(k)} = \mathfrak{m}(\rho_{T_k}) \in \arg\max_\phi \rho_{T_k}(\phi)$. The destination depends on the common full-history posterior alone—not on the reviser's held model or adoption date—so it is common to all revisers, while the trigger $\bar{\varepsilon}_k(\phi, a)$ is constant only within a cohort.

In calm, by Proposition 7 investor i revises at $a^i + \Delta^\star(\phi_i, \varepsilon^i)$, a deadline *non-increasing* in ε^i; investors of a common cohort therefore revise at times ordered (weakly) by ε^i—staggered across thresholds with *distinct* deadlines, and simultaneous on any level set where $\Delta^\star(\phi, \cdot)$ is flat. From the homogeneous start (Assumption 2) all hold ϕ_0 at $t = 0$ with common second-order prior; wherever the deadline map $\Delta^\star(\phi_0,\cdot)$ is non-constant across thresholds (Remark 2) the population begins to disperse in calm before any disaster, and each disaster induces a synchronized split—dispersion is generated, not assumed. $\square$

Remark 2 (When the held-model marginal has continuum support). The size-only design produced a finitely supported held-model marginal because all revision was synchronized at disasters. The calm channel can enlarge the support: if, on a sub-interval of $(0,1)$, the calm deadline $\varepsilon \mapsto \Delta^\star(\phi_0,\varepsilon)$ is continuous and strictly monotone and the calm intensity mode $\tau \mapsto \lambda^{\mathrm{m}}_\tau$ is strictly decreasing, then the first-rejection cohort adopts a continuum of distinct intensities and the held-model marginal is not finitely supported. In pure calm $\ell^\Lambda_\tau(\lambda) = \log \rho^\Lambda_0(\lambda) - \lambda\tau$ has strictly decreasing differences in (λ,τ)—for $\lambda' > \lambda$, $\ell^\Lambda_\tau(\lambda') - \ell^\Lambda_\tau(\lambda) = \log \frac{\rho^\Lambda_0(\lambda')}{\rho^\Lambda_0(\lambda)} - (\lambda'-\lambda)\tau$ is strictly decreasing in τ from the linear $-\lambda\tau$ term alone—so by Topkis's theorem the maximiser $\lambda^{\mathrm{m}}_\tau$ is non-increasing in τ, with no condition on ρ^Λ_0. The *strict* interior decline is the separate, stronger property: it holds when ρ^Λ_0 is strictly log-concave (the first-order condition then has a strictly decreasing interior root), while at the boundary $\lambda^{\mathrm{m}}_\tau$ can plateau (no interior maximizer is assumed). The strict monotonicity of $\Delta^\star(\phi_0,\cdot)$ requires, in *addition*, that the zero-count surprise $\bar\varepsilon$ be strictly decreasing in the age—a deadline-regularity condition not implied by log-concavity alone, which we impose where this remark is invoked. Absent either regularity the support may remain finite; the claim is scoped to these conditions and is not used downstream.

Proposition 9 (The cross-section is a pushforward of G): *Fix the common path. Let $\mathsf{T}_t(\varepsilon)$ be the held model and regime age that Definition 1 produces at t for an investor of threshold ε, started from ϕ_0 at age 0. Then T_t is a deterministic measurable map and the cross-sectional distribution of (held model, regime age) is the pushforward*

$$m_t \;=\; \mathsf{T}_{t\#}\, G, \tag{B9}$$

with no law of large numbers invoked: G is the primitive continuum type law of Assumption 2, not a random sample, so the transport is exact. The held-model marginal jumps at disasters, *where all revisers move to the common $\phi^{(k)}$, and changes at the staggered calm deadlines $a^i + \Delta^\star(\phi_i, \varepsilon^i)$, where the reviser of threshold ε moves to the common mode prevailing at her deadline.*

Proof of Proposition 9. Fix the common path. By Definition 1 and Proposition 7, given the path an investor's revision times are determined by her threshold ε alone—the calm deadline (B4) and the disaster rejections $\{T_k : \varepsilon > \bar{\varepsilon}_k\}$ are deterministic functions of ε—and the re-selected models are the common modes prevailing at those instants (Lemma 4), also path-determined. Hence $\varepsilon \mapsto \mathsf{T}_t(\varepsilon) = (\phi(t;\varepsilon), \tau(t;\varepsilon))$ is a deterministic map of the threshold. It is Borel measurable. On $[0,t]$ there are a.s. finitely many disasters; *between* consecutive disasters an investor may undergo several calm rejections, each resetting her regime, but for a fixed $\varepsilon > 0$ the calm deadline is bounded below, $\Delta^\star(\phi, \varepsilon) \geq \delta(\varepsilon) > 0$ (the fresh-record surprise starts at one and falls continuously to $\varepsilon < 1$), so only finitely many calm resets occur in $[0,t]$. Thus $\varepsilon \mapsto \mathsf{T}_t(\varepsilon)$ is a *finite* composition of measurable maps, alternating calm and disaster steps: the calm deadline $\varepsilon \mapsto \Delta^\star(\phi, \varepsilon)$ is monotone in ε (Proposition 7), hence Borel, and the calm re-selected mode is a measurable function of the path-determined record at the deadline; the disaster-rejection threshold at T_k is not a population constant but the state-dependent surprise depth $\bar{\varepsilon}_k\big(\phi(T_k^-;\varepsilon), a(\varepsilon)\big)$, a measurable function of the (already measurable) pre-disaster held model and adoption date, so $\{\varepsilon : \varepsilon > \bar{\varepsilon}_k(\cdot)\}$ is Borel as the preimage of a measurable map even though it need not be a fixed sub-interval of $(0,1)$; and the post-disaster mode $\phi^{(k)} = \mathfrak{m}(\rho_{T_k})$ is a measurable function of the common record. A finite alternating composition of these measurable maps is Borel—without asserting that $\phi(t;\cdot)$ is constant on threshold intervals, which the staggered calm deadlines and the state-dependent $\bar{\varepsilon}_k$ may violate. Since investors differ only in the permanent threshold, whose population distribution is G (Assumption 2), and all share the one path, each investor's state is the deterministic image $\mathsf{T}_t(\varepsilon^i)$ of her type; the cross-sectional distribution of states is therefore the image measure $m_t = \mathsf{T}_{t\#}G$ by construction—there is no idiosyncratic randomness to average, so no law of large numbers is invoked. This is (B9). The held-model marginal jumps at disasters, where all revisers move to the common $\phi^{(k)}$ (Proposition 8), and changes at the calm deadlines $a^i + \Delta^\star(\phi_i, \varepsilon^i)$, where the reviser of threshold ε moves to the common mode prevailing at her deadline (Lemma 4). $\square$

A long-run property of the rule. The pricing-state dynamics of Section 3 and the fragility reading of Section D rest on one fact about the rule: revision never stops, even for a correct model. We record it here, self-contained.

Lemma 5 (Recurrence of revision): *Fix a tolerance $\varepsilon \in (0,1)$ and run the rule of Definition 1 under the physical law $\mathbb{P}$ (Assumption 1). For any held model $\phi = (\lambda, \theta) \in \Phi$:* (i) *the holding time τ^{rev} is finite $\mathbb{P}$-almost surely; and* (ii) *re-adopting the held model after each rejection, it is rejected infinitely often, $\mathbb{P}$-almost surely. Consequently, when the menu contains the data-generating model—$\lambda^\star \in \operatorname{int}\Lambda$ and $F_J = F_{\theta^\star}$ for some $\theta^\star \in \Theta$—a correct-on-average model $\phi = (\lambda^\star, \theta^\star)$ is rejected and re-adopted forever: past a finite (a.s.) date every regime begins at the correct model and is nonetheless abandoned again.*

Proof. Hold $\phi = (\lambda, \theta)$ adopted at age 0; write $n_\Delta = N_{a+\Delta} - N_a$ for the since-adoption count over a window of length Δ, $M_\Delta := n_\Delta - \lambda^\star \Delta$ for the count centered at the physical rate, and $D_\Delta := D_{\phi,\Delta}(\mathcal{R})$ for the deviance (7). By Definition 1 the model is rejected at the first age at which its surprise $\bar{\varepsilon}(\mathcal{R}; \phi, \Delta)$ (8)—the predictive deviance tail under the held model's own law $Q_{\phi,\Delta}$—falls to ε or below; it therefore suffices to exhibit a finite age at which the realized deviance outgrows the bulk of that predictive law, so that $\bar{\varepsilon} \to 0$. The count channel does this: the size deviance in (7) is nonnegative, so $D_\Delta \geq D^{\mathrm{c}}_\Delta := \Delta\, d_P(n_\Delta/\Delta \| \lambda)$. The acceptance threshold it must clear is bounded in Δ: under $Q_{\phi,\Delta}$ the deviance D_Δ converges in distribution to a fixed chi-square-type law—its count part to a scaled χ^2_1 by the central limit theorem, its size part to a scaled χ^2_1 by Wilks's theorem—so the $(1-\varepsilon)$-quantile $r_{\phi,\Delta}(\varepsilon)$ stays bounded.

Step 1 (a finite exit, $\mathbb{P}$-a.s.). Under $\mathbb{P}$ the disaster count is a Poisson process of rate $\lambda^\star$, so the empirical rate $n_\Delta/\Delta \to \lambda^\star$ almost surely. If the held intensity is wrong, $\lambda \neq \lambda^\star$, then $D^{\mathrm{c}}_\Delta = \Delta\, d_P(n_\Delta/\Delta \| \lambda) \sim \Delta\, d_P(\lambda^\star \| \lambda)$ grows linearly, since $d_P(\lambda^\star \| \lambda) > 0$, and clears the bounded threshold in finite time. If the held intensity is correct, $\lambda = \lambda^\star$, the centered count

$M_\Delta = n_\Delta - \lambda^\star\Delta$ has i.i.d. unit-age increments of mean 0 and variance $\lambda^\star$ and so obeys the law of the iterated logarithm (Hartman and Wintner, 1941), $\limsup_{\Delta\to\infty} M_\Delta/\sqrt{2\lambda^\star\Delta\log\log\Delta} = 1$ $\mathbb{P}$-a.s. Expanding d_P about $\lambda^\star$ ($d_P(\lambda^\star\|\lambda^\star) = 0$, $\partial_a d_P|_{\lambda^\star} = 0$, $\partial_a^2 d_P = 1/a$) gives $D^{\mathrm{c}}_\Delta \sim M_\Delta^2/(2\lambda^\star\Delta)$, hence $\limsup_{\Delta\to\infty} D^{\mathrm{c}}_\Delta/\log\log\Delta = 1$, and the count deviance again clears the bounded threshold at a finite age—a fixed band crossed in finite time even at the truth, the boundary-crossing behind the foregone conclusion of fixed-level sequential testing (Robbins and Siegmund, 1970); the disaster-free record is the transparent instance, rejected at the deterministic calm deadline $\Delta^\star(\phi,\varepsilon)$ (B4), where $\Delta\, d_P(0\|\lambda) = \lambda\Delta$ crosses $r_{\phi,\Delta}(\varepsilon)$. In either case the realized record leaves the typical set—equivalently $\bar{\varepsilon} \to 0$—at a finite age, so $\tau^{\mathrm{rev}} < \infty$ $\mathbb{P}$-a.s. This is (i).

Step 2 (infinitely often). Each rejection time is an $(\mathcal{F}_t)$-stopping time, so by the strong Markov property the model re-adopted at it monitors a fresh since-adoption record, and the k-th holding time depends only on the model adopted at the k-th rejection; by Step 1 each is finite $\mathbb{P}$-a.s., correct model or not. The menu permits re-selection in either coordinate direction (Definition 1), so no model is absorbing and each rejection is followed by a further regime. Rejections arrive through two channels, and any bounded interval contains finitely many of each. A *calm* rejection cannot occur before the calm deadline, which is bounded below uniformly—$\Delta^\star(\phi,\varepsilon) \geq \delta(\varepsilon) > 0$, continuous in ϕ on the compact menu with the zero-count surprise equal to one at age zero—so an interval of length L holds at most $\lfloor L/\delta(\varepsilon)\rfloor$ of them. A *disaster* rejection consumes a disaster, and the disaster process is non-explosive, so an interval holds no more disaster rejections than its almost-surely finite disaster count. The holding time itself is *not* bounded below—an early disaster can leave the typical set at once—but the crash channel is rate-limited by non-explosiveness, not by a floor; the two channels together place finitely many rejections in any bounded interval. The rejection times therefore increase to infinity: the held model is rejected infinitely often, $\mathbb{P}$-a.s. This is (ii).

Step 3 (correct-on-average). When $\lambda^\star \in \operatorname{int}\Lambda$ and $F_J = F_{\theta^\star}$ with $\theta^\star \in \Theta$, the model $(\lambda^\star, \theta^\star)$ lies in Φ, and the second-order posterior ρ_t (4)—the Bayesian update on the full marked-Poisson record over the identified menu (Assumption 4)—concentrates on it under $\mathbb{P}$—the re-selection posterior merges with the truth (Blackwell and Dubins, 1962)—so its mode $\mathfrak{m}(\rho_t)$ equals $(\lambda^\star, \theta^\star)$ for all t past a finite, a.s. date. From that date every re-selection (9) adopts the correct model, which by the correct-intensity case of Step 1 is rejected and re-adopted again. A correct-on-average model is thus revised forever. $\square$

Appendix C. Proofs for the equilibrium kernel and the disaster-fear bridge

The belief density is a positive martingale. We verify the claim after Definition 6. Under $\mathbb{P}$ the disaster measure μ has $(\mathcal{F}_t, \mathbb{P})$-compensator $\nu(dt, dx) = \lambda^\star\, dt\, F_J(dx)$ (3). The held model $\phi_i(t-) = (\lambda_i(t-), \theta_i(t-))$ is predictable (left-continuous and piecewise constant, Definition 1) and proposes the compensator $\nu^i(dt, dx) = \lambda_i(t-)\, dt\, F_{\theta_i(t-)}(dx)$. The process (C1) is the Radon–Nikodym density of the marked-point-process law with compensator ν^i against that with ν (Brémaud, 1981, VI, Theorems 2–4), which is a positive $(\mathcal{F}_t, \mathbb{P})$-local martingale solving $dM_t^i = M_{t-}^i \int_{\mathcal{J}} \big(r_i(x;t) - 1\big)\, (\mu - \nu)(dt, dx)$. We check the two hypotheses of Brémaud (1981, VI, T4) under which it is a true martingale with $\mathbb{E}[M_t^i] = 1$. *(a) Bounded intensity.* $\Lambda = [\underline{\lambda}, \overline{\lambda}] \subset (0, \infty)$ (Assumption 4), so $\int_0^t \lambda_i(s-)\, ds \le \overline{\lambda}\, t < \infty$. *(b) Bounded likelihood ratio.* The map $(x, \phi) \mapsto r(x; \phi) = \lambda f_\theta(x)/(\lambda^\star f(x))$ is continuous and strictly positive on the compact $\mathcal{J} \times \Phi$ (full support, Assumption 4(i)), hence $0 < \underline{r} := \min_{\mathcal{J}\times\Phi} r \le r_i(x;t) \le \overline{r} := \max_{\mathcal{J}\times\Phi} r < \infty$. Conditions (a)–(b) give $\mathbb{E}[\int_0^t \int_{\mathcal{J}} |r_i(x;s) - 1|\, \nu(ds, dx)] \le (\overline{r} + 1)\, \overline{\lambda}\, t < \infty$, so the stochastic integral is a true martingale and $\mathbb{E}[M_t^i] = 1$ (Brémaud, 1981, VI, T4). Positivity is immediate from $r_i > 0$. $\square$

Independent increments and the time-consistent construction. The product form $\prod_k r_i(J_k; T_k)$ is the independent-increment factorization of the Poisson record—each disaster contributing its own likelihood ratio, with no cross-disaster interaction *given the held-model* *path*; revisions, which change ϕ_i, are the only channel coupling disasters. The same structure makes the since-adoption segment likelihood the *conditional* Radon–Nikodym increment given the pre-adoption past, so testing the held model on its since-adoption record (Definition 1) is exact as conditional testing—a property of the Poisson record's independent increments, not of general locally equivalent laws. The ratio M_T^i/M_t^i is built from the predictable held-model *process* $\phi_i(s-)$ over $(t, T]$, so it carries the model in force at each instant, those the agent adopts at her own future revisions included. This is the time-consistent reading: investor i's intertemporal valuation $\mathbb{E}_t^i$ is taken under the law of her realized model process—equivalently, she prices in a rational-expectations equilibrium along the common path while holding, at each instant, her current model. It is what makes M^i a $\mathbb{P}$-martingale and what the static complete-market (Negishi) construction of Appendix F uses; a *myopic-dogmatic* density—pricing as if the current model $\phi_i(t)$ were permanent until an unforeseen rejection—would not be a $\mathbb{P}$-martingale across the agent's revision epochs, and would not support that construction. That stance is nonetheless the agent's own first-person perspective, and its departure from dynamic consistency is bounded by the rarity of those rejections (Weinstein, 2017). The two readings agree on every *instantaneous* object: the short rate and the fear factor depend on the current model $\phi_i(t-)$ alone; the kernel *level* $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^{\alpha}$ carries the cumulative densities through Ξ_t, but both stances build Ξ_t from the same realized model process. They differ only in *forward-looking* valuation—the price–dividend ratio—where the time-consistent construction delivers the equilibrium price; we adopt it throughout.

Definition 6 (Belief density): *Investor i's* belief density *is the Radon–Nikodym derivative of her held-model law $\mathbb{P}^{\phi_i}$ against the physical law $\mathbb{P}$ on $\mathcal{F}_t$,*

$$M_t^i = \frac{d\mathbb{P}^{\phi_i}}{d\mathbb{P}}\bigg|_{\mathcal{F}_t} = \exp\!\Big(-\int_0^t \big(\lambda_i(s-) - \lambda^\star\big)\, ds\Big) \prod_{k:\, T_k \le t} r_i(J_k; T_k), \qquad r_i(x;t) := \frac{\lambda_i(t-)\, f_{\theta_i(t-)}(x)}{\lambda^\star\, f(x)}, \tag{C1}$$

with $M_0^i = 1$: the intensity-wedge drift times the per-disaster likelihood ratios r_i. Investor i's conditional expectation is $\mathbb{E}_t^i[X] = \mathbb{E}_t[(M_T^i/M_t^i)X]$.

Remark 3 (The allocation is selected, not derived). Under complete markets with heterogeneous beliefs, equal initial wealth and a common multiplier cannot both hold. The CRRA optimum is $c_t^i = \zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha}(e^{-\delta t}/\xi_t)^{1/\alpha}$, so ζ_i is pinned by the budget $\zeta_i^{-1/\alpha}\,\mathbb{E}\big[\int_0^\infty \xi_t^{1-1/\alpha}e^{-\delta t/\alpha}(M_t^i)^{1/\alpha}\,dt\big] = W_0^i$, whose value depends on the law of $\{M_t^i\}$, hence on the surprise threshold ε^i. Common wealth W_0^i forces type-dependent ζ_i, while common ζ_i forces type-dependent wealth—a genuine fork. We take the second branch: ε indexes a revision *rule*, not a preference or endowment we wish to price, so equal weights isolate the belief channel from Pareto-weight dispersion. This setup is the standard perpetual-youth reduced form (Yaari, 1965; Blanchard, 1985; Gârleanu and Panageas, 2015); the general heterogeneous-ζ kernel of Proposition 1 remains available.

Definition 7 (Competitive equilibrium): *A* competitive equilibrium *is a state-price density* $(\xi_t)_{t\geq 0}$ *and a consumption allocation* (c_t^i) *such that (i) each* c^i *maximizes* $\mathbb{E}^i\big[\int_0^\infty e^{-\delta t}u(c_t^i)\,dt\big]$ *subject to the static budget* $\mathbb{E}\big[\int_0^\infty \xi_t c_t^i\,dt\big] \leq \mathbb{E}\big[\int_0^\infty \xi_t e_t^i\,dt\big]$*, the expectation under the physical law* $\mathbb{P}$ *and* e^i *investor* i*'s endowment—the supporting endowment that implements the selected Pareto weights of Remark 3 (under heterogeneous beliefs a common multiplier implies type-dependent* e^i*), not a separate primitive of the economy; and (ii) the goods market clears,* $\int_0^1 c_t^i\,di = C_t$ *for every* t*.*

Proof of Proposition 1. Step 1 (the aggregator is finite and positive). Fix a path with $N_t < \infty$ disasters on $[0,t]$ (a.s., Assumption 1). By (C1) and the bounds $\underline{r} \leq r_i \leq \overline{r}$ and $\lambda_i \in [\underline{\lambda}, \overline{\lambda}]$,

$$e^{-(\overline{\lambda}-\lambda^\star)t}\,\underline{r}^{\,N_t} \;\leq\; M_t^i \;\leq\; e^{(\lambda^\star-\underline{\lambda})t}\,\overline{r}^{\,N_t} \qquad \text{for every } i, \tag{C2}$$

so M_t^i is bounded above and below by path-dependent constants uniform in i. Hence $(M_t^i)^{1/\alpha}$ is bounded uniformly in i, and with $\int_0^1 \zeta_i^{-1/\alpha}\,di < \infty$ (Assumption 3),

$$0 \;<\; \Xi_t = \int_0^1 \zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha}\,di \;\leq\; \big(e^{(\lambda^\star-\underline{\lambda})t}\overline{r}^{\,N_t}\big)^{1/\alpha}\int_0^1 \zeta_i^{-1/\alpha}\,di \;<\; \infty.$$

Step 2 (individual optimum). Markets are complete (Assumption 3), so investor i's dynamic problem is the static budget problem under the unique state-price density ξ (Cox and Huang, 1989; Karatzas et al., 1990). Writing the objective under $\mathbb{P}$ via $\mathbb{E}^i[\,\cdot\,] = \mathbb{E}[M_t^i\,\cdot\,]$, the Lagrangian $\mathbb{E}\big[\int_0^\infty \big(e^{-\delta t}u(c_t^i)M_t^i - \zeta_i\xi_t c_t^i\big)dt\big]$ is concave in c^i (as u is concave), so its pointwise (in (t,ω)) stationarity condition is necessary and sufficient:

$$e^{-\delta t}u'(c_t^i)\,M_t^i = \zeta_i\,\xi_t. \tag{C3}$$

With $u'(c) = c^{-\alpha}$, (C3) inverts to

$$c_t^i = \zeta_i^{-1/\alpha}\big(\xi_t e^{\delta t}\big)^{-1/\alpha}\big(M_t^i\big)^{1/\alpha}. \tag{C4}$$

Step 3 (market clearing fixes the kernel). Integrate (C4) over i and impose $\int_0^1 c_t^i\,di = C_t$. The factor $(\xi_t e^{\delta t})^{-1/\alpha}$ is constant in i, so

$$\begin{aligned}
\big(\xi_t e^{\delta t}\big)^{-1/\alpha}\int_0^1 \zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha}\,di &= C_t \\
\big(\xi_t e^{\delta t}\big)^{-1/\alpha}\,\Xi_t &= C_t \\
\big(\xi_t e^{\delta t}\big)^{-1/\alpha} &= C_t/\Xi_t \qquad \text{(C5)} \\
\xi_t e^{\delta t} &= \big(C_t/\Xi_t\big)^{-\alpha} = C_t^{-\alpha}\Xi_t^{\alpha} \\
\xi_t &= e^{-\delta t}C_t^{-\alpha}\Xi_t^{\alpha}, \qquad \text{(C6)}
\end{aligned}$$

which is (10); $\Xi_t \in (0,\infty)$ by Step 1, so ξ_t is well-defined and positive.

Step 4 (shares). Divide (C4) by C_t and substitute $(\xi_t e^{\delta t})^{-1/\alpha} = C_t/\Xi_t$ from (C5):

$$s_t^i = \frac{c_t^i}{C_t} = \frac{\zeta_i^{-1/\alpha}(C_t/\Xi_t)(M_t^i)^{1/\alpha}}{C_t} = \frac{\zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha}}{\Xi_t}, \tag{C7}$$

which is (11); $\int_0^1 s_t^i\,di = \Xi_t/\Xi_t = 1$ and $s_t^i > 0$, so s_t is a probability density in i.

Step 5 (existence and uniqueness). Equations (C6) and (C7) construct (ξ, c^i); by Step 2 each c^i is the (unique) individual optimum given ξ, and by Step 3 the goods market clears, so (ξ, c^i) is a competitive equilibrium (Definition 7). Uniqueness: at any equilibrium the FOC (C3) and clearing hold, and Steps 2–3 derive (10)–(11) from them with no freedom left once the ζ_i are fixed. The weights ζ_i are in turn pinned by the budget system. Although each agent's budget is evaluated under the common ξ, which co-moves with *every* ζ_j through market clearing, the equilibrium weights are nonetheless unique: this is the standard strictly monotone Negishi map from Pareto weights to net excess demands, invertible because u is strictly concave and the market is complete, so the weights—and hence (ξ, c^i)—are determined uniquely (Cox and Huang, 1989; Karatzas et al., 1990). Under the symmetric allocation $\zeta_i \equiv \zeta = 1$ (Assumption 3(iv); a weight normalization, not equal initial wealth, which under heterogeneous beliefs would make ζ_i type-dependent, Remark 3) ζ cancels from (C7).

Step 6 (the power mean is not the arithmetic mean). From (C7), $\zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha} = s_t^i \Xi_t$, so $M_t^i = \zeta_i\,(s_t^i \Xi_t)^\alpha$. Substituting into the share-weighted arithmetic mean $\langle M \rangle_t = \int_0^1 s_t^i M_t^i \, di$,

$$\langle M \rangle_t = \int_0^1 s_t^i \,\zeta_i\,(s_t^i \Xi_t)^\alpha \, di = \Xi_t^\alpha \int_0^1 \zeta_i\,(s_t^i)^{1+\alpha} \, di.$$

Under the equal-Pareto-weight selection $\zeta_i \equiv \zeta$ (the case carried downstream), the shares are price-irrelevantly ζ-free and

$$\langle M \rangle_t = \zeta\, \Xi_t^\alpha \int_0^1 (s_t^i)^{1+\alpha} \, di \;\geq\; \zeta\, \Xi_t^\alpha,$$

the inequality by Jensen applied to the convex map $x \mapsto x^{1+\alpha}$ ($\alpha > 0$) and the probability measure di on $[0,1]$, since $\int_0^1 (s_t^i)^{1+\alpha} di \geq (\int_0^1 s_t^i \, di)^{1+\alpha} = 1$, with equality iff s_t^i is constant in i—equivalently iff the M_t^i are equal across the cross-section. Hence the priced power-mean factor Ξ_t^α and the descriptive arithmetic mean $\langle M \rangle_t$ coincide (up to the price-irrelevant constant ζ) only at cross-sectional degeneracy, and otherwise differ, the arithmetic mean strictly larger; the kernel prices the former. $\square$

Proof of Proposition 2. Step 1 (the kernel's jump ratio). At a disaster at t of size $x = J_k$, the endowment falls by $C_t = C_{t-}e^{-x}$ (Assumption 1), so

$$\frac{C_t^{-\alpha}}{C_{t-}^{-\alpha}} = \left(e^{-x}\right)^{-\alpha} = e^{\alpha x}.$$

Each density jumps by its likelihood ratio, $M_t^i = M_{t-}^i\, r_i(x;t)$ (C1), so $(M_t^i)^{1/\alpha} = (M_{t-}^i)^{1/\alpha} r_i(x;t)^{1/\alpha}$, and by (10) and (C7),

$$\begin{aligned}
\frac{\Xi_t}{\Xi_{t-}} &= \frac{1}{\Xi_{t-}} \int_0^1 \zeta_i^{-1/\alpha} (M_t^i)^{1/\alpha}\, di \\
&= \frac{1}{\Xi_{t-}} \int_0^1 \zeta_i^{-1/\alpha} (M_{t-}^i)^{1/\alpha} r_i(x;t)^{1/\alpha}\, di \\
&= \int_0^1 \frac{\zeta_i^{-1/\alpha} (M_{t-}^i)^{1/\alpha}}{\Xi_{t-}}\, r_i(x;t)^{1/\alpha}\, di \qquad \text{(C8)} \\
&= \int_0^1 s_{t-}^i\, r_i(x;t)^{1/\alpha}\, di \;=\; g_t(x), \qquad \text{(C9)}
\end{aligned}$$

the third line moving the constant Ξ_{t-}^{-1} under the integral and the fourth using $s_{t-}^i = \zeta_i^{-1/\alpha}(M_{t-}^i)^{1/\alpha}/\Xi_{t-}$ (C7); this is (12). Multiplying, the state-price density jumps by

$$\frac{\xi_t}{\xi_{t-}} = \frac{C_t^{-\alpha}}{C_{t-}^{-\alpha}} \left(\frac{\Xi_t}{\Xi_{t-}}\right)^{\alpha} = e^{\alpha x}\, g_t(x)^{\alpha}, \qquad \text{(C10)}$$

the continuous parts of $C^{-\alpha}$ and Ξ^{α} contributing no jump.

Step 2 (risk-neutral compensator). Markets are complete (Assumption 3), so there is a unique equivalent martingale measure $\mathbb{Q}$ with density process $Z_t = \xi_t \exp(\int_0^t r_s\, ds)/\xi_0$. We verify the hypotheses of the change-of-measure theorem for marked point processes (Brémaud, 1981, VI, T2–T4). *(a) Z is a local martingale.* The equilibrium short rate is the process that deflates the state-price density to a local martingale, and this fixes the drift cancellation. Write the stochastic logarithm of the SDF (10) as

$$\frac{d\xi_t}{\xi_{t-}} = b_t\, dt + dm_t,$$

with b_t the predictable finite-variation drift coefficient—the Itô drift of $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^{\alpha}$, a functional of the cross-section through Ξ—and m a local martingale collecting the Brownian and compensated-jump parts. The equilibrium short rate is, by its defining no-arbitrage property, $r_t := -b_t$; this property is logically prior to its closed form (which we do not need here). Since $e^{\int_0^t r_s ds}$ is continuous and of finite variation it has zero covariation with ξ, so integration by parts gives

$$d\big(\xi_t e^{\int_0^t r_s ds}\big) = e^{\int_0^t r_s ds}\big(d\xi_t + \xi_{t-} r_t\, dt\big) = e^{\int_0^t r_s ds}\, \xi_{t-}\big((b_t + r_t)\, dt + dm_t\big) = e^{\int_0^t r_s ds}\, \xi_{t-}\, dm_t,$$

the drift vanishing by $r_t = -b_t$. Hence $\xi_t e^{\int_0^t r_s ds}$ is a positive $\mathbb{P}$-local-martingale, and $Z_t = \xi_t e^{\int_0^t r_s ds}/\xi_0$ is its unit normalization, with $Z_0 = 1$ and $\xi > 0$. The remaining and substantive task is to upgrade this to a *true* $(\mathcal{F}_t, \mathbb{P})$-martingale with $\mathbb{E}[Z_t] = 1$ (hence the density of $\mathbb{Q}$), which requires controlling *both* factors of Z, since ξ carries a continuous (Brownian) part as well as the disaster jumps. *Continuous part:* the diffusive loading of $\log \xi$ is $-\alpha\sigma\, dW_t$, constant because σ is constant (Assumption 1); the continuous factor of Z is then the Doléans exponential $\mathcal{E}(-\alpha\sigma W)_t$, a true martingale by Novikov's condition $\mathbb{E}[\exp(\frac{1}{2}\int_0^t (\alpha\sigma)^2 ds)] = \exp(\frac{1}{2}\alpha^2\sigma^2 t) < \infty$. *Jump part:* the bounded predictable jump ratio of (b)–(c) and the integrability bound $\int_0^t \int_{\mathcal{J}} |Y_s(x) - 1|\, \nu(ds, dx) \le (e^{\alpha \overline{J}}\overline{r} + 1)\, \lambda^\star t < \infty$ place the jump factor within Brémaud (1981, VI, T4). The two factors are driven by independent sources (W and μ), so their product Z is a true martingale; completeness independently yields a unique equivalent martingale measure, of which Z is the density. *(b)* The bank account $\exp(\int_0^t r_s\, ds)$ is continuous and of finite variation and ξ_0 is constant, so Z and ξ have the same jump ratio: at a disaster of size x, $Z_t/Z_{t-} = \xi_t/\xi_{t-} = e^{\alpha x} g_t(x)^\alpha =: Y_t(x)$ by (C10). *(c)* Y_t is strictly positive and predictable (it is built from the left limits s^i_{t-} and g_{t-}) and bounded on the compact $\mathcal{J}$, since $e^{\alpha x} \le e^{\alpha \overline{J}}$ and $0 < g_t(x) \le \overline{r}^{1/\alpha}$ by (12). Under (a)–(c) the theorem gives the $\mathbb{Q}$-compensator of μ as $\nu^{\mathbb{Q}}(dt, dx) = Y_t(x)\, \nu(dt, dx)$; with $\nu(dt, dx) = \lambda^\star\, dt\, F_J(dx)$ (3),

$$\nu^{\mathbb{Q}}(dt, dx) = \lambda^\star\, e^{\alpha x}\, g_t(x)^\alpha\, dt\, F_J(dx), \tag{C11}$$

which is (13); Y_t bounded makes $\nu^{\mathbb{Q}}$ a finite measure, so the change of measure is admissible.

Step 3 (size law). Using $F_J(dx) = f(x)\,dx$ (Assumption 1), the finite measure (C11) is $\nu^{\mathbb{Q}}(dt,dx) = \lambda^\star e^{\alpha x} g_t(x)^\alpha f(x)\,dt\,dx$. Factor it as intensity times a normalized size density, $\nu^{\mathbb{Q}}(dt,dx) = \lambda_t^{\mathbb{Q}}\,dt\,f_t^{\mathbb{Q}}(x)\,dx$, by integrating out x for the intensity and dividing for the density:

$$\begin{aligned}\lambda_t^{\mathbb{Q}} &= \int_{\mathcal{J}} \frac{\nu^{\mathbb{Q}}(dt,dx)}{dt} = \lambda^\star \int_{\mathcal{J}} e^{\alpha x} g_t(x)^\alpha f(x)\,dx = \lambda^\star\,\mathbb{E}_f\big[e^{\alpha J} g_t(J)^\alpha\big],\\ f_t^{\mathbb{Q}}(x) &= \frac{\lambda^\star e^{\alpha x} g_t(x)^\alpha f(x)}{\lambda_t^{\mathbb{Q}}} = \frac{e^{\alpha x} g_t(x)^\alpha f(x)}{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]},\end{aligned} \tag{C12}$$

where $\mathbb{E}_f[h] = \int_{\mathcal{J}} h\,f$; the denominator is finite and positive by the bounds in Step 2, so $f_t^{\mathbb{Q}}$ is a proper density on $\mathcal{J}$. This is (14) and proves (i).

Step 4 (the bridge). Multiply and divide $U_t = \lambda_t^{\mathbb{Q}}$ by $\mathbb{E}_f[e^{\alpha J}]$:

$$U_t = \lambda^\star\,\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha] = \underbrace{\lambda^\star\,\mathbb{E}_f[e^{\alpha J}]}_{U^{\mathrm{RA}}} \cdot \underbrace{\frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]}{\mathbb{E}_f[e^{\alpha J}]}}_{\kappa_t}, \tag{C13}$$

which is (15). The factor $U^{\mathrm{RA}} = \lambda^\star \mathbb{E}_f[e^{\alpha J}]$ contains only the physical $(\lambda^\star, F_J)$ and α; the factor κ_t contains the cross-section only, through g_t. This proves (ii).

Step 5 (belief-driven variation). Holding $(\alpha, \lambda^\star, F_J)$ fixed, U^{RA} is constant in t, so taking logs of (C13) and differentiating,

$$d\log U_t = d\log U^{\mathrm{RA}} + d\log \kappa_t = d\log \kappa_t,$$

which is (16). In (C12) the only cross-section-dependent object is g_t, so $f_t^{\mathbb{Q}}$, $\bar{U}_t$, and U_t are functionals of g_t alone—of the held models ϕ_i and shares s_{t-}^i of the cross-section (Proposition 9)—while $(\lambda^\star, F_J)$ stay fixed. This proves (iii). □

Remark 4 (The size channel can lift the binary magnitude ceiling (general case, not exercised)). This remark concerns the size channel, which the quantitative sections do not exercise (Scope, above). Under a two-point *size* menu $\{\theta_L, \theta_H\}$ at fixed intensity, $g_t(x)$ is a two-point mixture of the $1/\alpha$ powers $r(\phi, x)^{1/\alpha}$ of the two likelihood ratios, so the attainable risk-neutral tilt is bounded by the more fearful of the two. With a full-support size family $\{F_\theta\}$ whose held indices realize on an interval of Θ, $g_t(x)$ is a continuum-weighted aggregator that can tilt $f_t^{\mathbb{Q}}$ continuously across $\mathcal{J}$. Two caveats. First, this is a statement about the size indices θ_i; the continuum-support mechanism of Remark 2 concerns the calm-time *intensity* mode, not θ, so it does not by itself populate an interval of Θ—a size-dispersed newborn law would. Second, a continuum size family widens the *range* of attainable tilts but need not raise the supremum if the two-point menu already contains the most fearful model; the gain is continuity and dispersion-sensitivity—a comparison across designs, not an unboundedness claim (compactness of $\mathcal{J}$ and Θ keeps $f_t^{\mathbb{Q}}$ a proper density at all times).

Remark 5 (Which belief dynamics can move the fear factor). Two features of the bridge (15) bear on what kind of belief process can generate a time-varying fear factor.

(a) With constant fundamentals the fear factor is a pure belief object (proved). Volatility σ, the physical law $(\lambda^\star, F_J)$, and risk aversion α are constant (Assumptions 1 and 3: $\sigma, \lambda^\star, F_J$ from the former, α from the latter), so U^{RA} is constant and, by the bridge relation (16), $d\log U_t = d\log \kappa_t$: all movement in priced fear is belief movement. A representative agent, or any population holding correct beliefs, has $\kappa_t \equiv 1$ and a flat U_t. This fear factor is the structural counterpart of the unspanned left-tail factor of Andersen et al. (2015b), which moves the risk-neutral tail with no matching move in volatility or realized tail risk; here that property is exact and entirely belief-driven. What must be stripped out for a constant factor is *moving* beliefs, not merely heterogeneity: a correct-belief or fully merged population has κ_t constant ($\equiv 1$ when correct), but a single *learning* Bayesian—no heterogeneity at all—still moves the factor through her evolving posterior (Appendix I). Cross-sectional dispersion is what keeps κ_t from merging to a constant; it is not the only belief process that can.

(b) Beliefs that merge make it flat (standard theory and Remark 6). If agents update on the common public filtration, their predictive laws merge (Blackwell and Dubins, 1962)

and, under the identifiable family of Assumption 4, concentrate on the Kullback–Leibler model (Berk, 1966); the held likelihood ratios then share a common limit, κ_t converges to a constant, and the fear factor loses its time variation. This applies to a textbook single Bayesian and, as Remark 6 notes, to an infinitely-lived lazy agent alike: a time-varying fear factor requires beliefs that have not merged.

What is and is not claimed. We do *not* claim only lazy revision can move the factor. A Bayesian overlapping-generations population of as-yet-unconverged learners would also keep dispersion alive and update upward at disasters. The distinction is qualitative, not a theorem: lazy beliefs stay fixed within a tolerance and then jump discretely at rejection (Proposition 12), with scheduled relief in calm (Proposition 7), giving abrupt recurring spikes and persistent calm wedges—the regime-shift shape of the Bollerslev and Todorov (2011) fears series. The contrast with Bayesian learning is at the level of the individual belief path—fixed within tolerance, then a discrete jump, versus smooth between-disaster updating—not at disaster arrivals, where a rare-event Bayesian fear factor also jumps; and the persistence of the aggregate wedge owes as much to demographic turnover as to the learning rule. Non-representability here is relative to the named public filtration and the single-prior class (Section 3), not to Bayesians on enriched states. Whether the multiplier sits below one on average ($\kappa_t < 1$, the Backus et al. (2011) moderation—the inverted intensity $\lambda^\star \kappa_t$ below the physical $\lambda^\star$) and any quantitative match are calibration questions (Section 5).

Appendix D. Proofs for the revision hazard and the law of motion of the pricing state

Proposition 10 (Selection dynamics of the kernel and shares): *In the equilibrium of Proposition 1, between revisions (held models fixed):*

(i) in calm (no disaster) the aggregator and the shares drift by selection,

$$\frac{d\log\Xi_t}{dt} = -\tfrac{1}{\alpha}\big(\bar\lambda^S_t - \lambda^\star\big), \qquad \frac{d\log s^i_t}{dt} = \tfrac{1}{\alpha}\big(\bar\lambda^S_t - \lambda_i(t)\big), \tag{D1}$$

so the share of a below-average-intensity (complacent) investor grows and that of an above-average-intensity (fearful) investor shrinks;

(ii) at a disaster of size x the aggregator and the shares jump multiplicatively,

$$\frac{\Xi_t}{\Xi_{t-}} = g_t(x), \qquad \frac{s^i_t}{s^i_{t-}} = \frac{r_i(x;t)^{1/\alpha}}{g_t(x)}, \tag{D2}$$

so an investor whose held model fits the realized disaster better than the aggregate, $r_i(x;t)^{1/\alpha} > g_t(x)$, gains share.

Proof of Proposition 10. (i) Calm. Between disasters the product in (C1) is constant, so $M^i_t = \exp(-\int_0^t(\lambda_i(s-) - \lambda^\star)ds) \times (\text{const})$ and, with the held model fixed,

$$\frac{d\log M^i_t}{dt} = -\big(\lambda_i(t) - \lambda^\star\big), \qquad \frac{d\log(M^i_t)^{1/\alpha}}{dt} = -\tfrac{1}{\alpha}\big(\lambda_i(t) - \lambda^\star\big). \tag{D3}$$

Differentiating $\Xi_t = \int_0^1 \zeta_i^{-1/\alpha}(M^i_t)^{1/\alpha}\,di$ in t: the interchange of $\frac{d}{dt}$ and $\int_0^1 \cdot\,di$ is justified by dominated convergence, since on any compact t-interval both the integrand $\zeta_i^{-1/\alpha}(M^i_t)^{1/\alpha}$ and its t-derivative $\zeta_i^{-1/\alpha}(M^i_t)^{1/\alpha}\big(-\frac{1}{\alpha}(\lambda_i - \lambda^\star)\big)$ are bounded uniformly in i by $\zeta_i^{-1/\alpha}$ times path-constants (by (C2) and $\lambda_i \in [\underline{\lambda}, \overline{\lambda}]$), with $\int_0^1 \zeta_i^{-1/\alpha} di < \infty$ (Assumption 3). Using (D3) for $\frac{d}{dt}(M^i_t)^{1/\alpha} = (M^i_t)^{1/\alpha}(-\frac{1}{\alpha}(\lambda_i - \lambda^\star))$ and then $\zeta_i^{-1/\alpha}(M^i_t)^{1/\alpha} = s^i_t\Xi_t$ from (C7),

$$\begin{aligned}\frac{d\Xi_t}{dt} &= \int_0^1 \zeta_i^{-1/\alpha}(M^i_t)^{1/\alpha}\Big(-\tfrac{1}{\alpha}(\lambda_i - \lambda^\star)\Big)di \\ &= -\tfrac{1}{\alpha}\,\Xi_t\int_0^1 s^i_t(\lambda_i - \lambda^\star)\,di = -\tfrac{1}{\alpha}\,\Xi_t\big(\bar\lambda^S_t - \lambda^\star\big),\end{aligned} \tag{D4}$$

the last equality using $\int_0^1 s^i_t\lambda_i\,di = \bar\lambda^S_t$ and $\int_0^1 s^i_t\,di = 1$ (Definition 3); this gives the first equation of (D1). For the share, take logs of (C7): $\log s_t^i = -\frac{1}{\alpha}\log\zeta_i + \frac{1}{\alpha}\log M_t^i - \log\Xi_t$; the first term is constant in t, so

$$\frac{d\log s_t^i}{dt} = \tfrac{1}{\alpha}\frac{d\log M_t^i}{dt} - \frac{d\log\Xi_t}{dt}. \tag{D5}$$

By (D3), $\frac{1}{\alpha}\, d\log M_t^i/dt = -\frac{1}{\alpha}(\lambda_i - \lambda^\star)$; and dividing (D4) by Ξ_t, $d\log\Xi_t/dt = \Xi_t^{-1}\, d\Xi_t/dt = -\frac{1}{\alpha}(\bar\lambda_t^S - \lambda^\star)$. Substituting both into (D5),

$$\frac{d\log s_t^i}{dt} = -\tfrac{1}{\alpha}(\lambda_i - \lambda^\star) + \tfrac{1}{\alpha}(\bar\lambda_t^S - \lambda^\star) = \tfrac{1}{\alpha}\big(\bar\lambda_t^S - \lambda_i\big), \tag{D6}$$

the $\lambda^\star$ terms cancelling; this is the second equation of (D1).

(ii) Disaster. At a disaster of size x, $M_t^i = M_{t-}^i r_i(x;t)$ (C1), so $(M_t^i)^{1/\alpha} = (M_{t-}^i)^{1/\alpha} r_i(x;t)^{1/\alpha}$ and $\Xi_t/\Xi_{t-} = g_t(x)$ exactly as in (C9). Then from $s_t^i = \zeta_i^{-1/\alpha}(M_t^i)^{1/\alpha}/\Xi_t$, the constant $\zeta_i^{-1/\alpha}$ cancels in the ratio:

$$\frac{s_t^i}{s_{t-}^i} = \frac{(M_t^i)^{1/\alpha}/\Xi_t}{(M_{t-}^i)^{1/\alpha}/\Xi_{t-}} = \frac{(M_t^i)^{1/\alpha}}{(M_{t-}^i)^{1/\alpha}} \cdot \frac{\Xi_{t-}}{\Xi_t} = r_i(x;t)^{1/\alpha} \cdot \frac{1}{g_t(x)} = \frac{r_i(x;t)^{1/\alpha}}{g_t(x)},$$

which is (D2); the share grows iff $r_i(x;t)^{1/\alpha} > g_t(x)$. $\square$

Proof of Proposition 3, the no-finite-state claim. That the calm drift (D1) depends on S_t only through $\bar\lambda_t^S = \int_\Phi \lambda\, dS_t$ is immediate from (D4). For the disaster tilt, (12) written against S_t is (21), a linear map $S \mapsto g(\cdot) = \int_\Phi H(\cdot,\phi)\, S(d\phi)$ on signed measures, with kernel $H(x,\phi) = (\lambda/\lambda^\star)^{1/\alpha}(f_\theta(x)/f(x))^{1/\alpha}$. Since the factor $(\lambda/\lambda^\star)^{1/\alpha} > 0$ is a θ-independent scalar, $\operatorname{span}\{H(\cdot,\phi) : \phi \in \Phi\} = \operatorname{span}\{(f_\theta/f)^{1/\alpha} : \theta \in \Theta\}$, which by Assumption 4(v) is infinite-dimensional in $L^1(\mathcal{J}, f)$; hence the linear map $T : \eta \mapsto \int_\Phi H(\cdot,\phi)\, \eta(d\phi)$ has infinite-dimensional range.

Fix any finite collection of moment functionals $L_1, \ldots, L_n$ of the pricing state (each $L_j(\eta) = \int_\Phi \ell_j\, d\eta$ for a bounded ℓ_j), and adjoin the total-mass functional $L_0(\eta) = \int_\Phi d\eta$. Work in the space $\mathcal{V}$ of signed measures on Φ with bounded Lebesgue density, and let $\mathcal{N} = \{\eta \in \mathcal{V} : L_0(\eta) = L_1(\eta) = \cdots = L_n(\eta) = 0\}$, the intersection of $n+1$ hyperplanes, so $\operatorname{codim}_{\mathcal{V}} \mathcal{N} \le n+1$.

There is $\eta \in \mathcal{N}$ with $T\eta \not\equiv 0$. Suppose not, i.e. $T(\mathcal{N}) = \{0\}$, so $\mathcal{N} \subseteq \ker T$. Then $\operatorname{codim} \ker T \le \operatorname{codim} \mathcal{N} \le n+1$, and by rank–nullity $\dim \operatorname{range}(T|_{\mathcal{V}}) = \operatorname{codim} \ker T \le n+1$. But T has infinite-dimensional range already on bounded-density measures (the functions $(f_\theta/f)^{1/\alpha}$ are approximated in $L^1(\mathcal{J}, f)$ by T of bounded-density measures concentrated near each θ, and are linearly independent by (v)), a contradiction. Hence such an η exists; fix it, with bounded density and $\|\eta\|_\infty < \infty$.

Admissibility of the perturbation. Let $\mathcal{S}$ denote the *admissible instantaneous pricing states*—probability measures on Φ with a density bounded below on the compact Φ, the valid wealth-weighted held-model cross-sections. The directions η constructed above keep the perturbation inside $\mathcal{S}$ (next paragraph; the explicit m-parameter version is Proposition 11), so the statement is about the *instantaneous pricing map* $S \mapsto (U_t, f_t^{\mathbb{Q}})$ over $\mathcal{S}$: no finite collection of moments is a sufficient statistic for it, two states in $\mathcal{S}$ sharing $L_1, \dots, L_n$ yet pricing disasters differently. The dynamically reachable cross-sections—wealth-weighted pushforwards of the primitive type law G (Proposition 9)—form a subset of $\mathcal{S}$; the claim is no finite sufficient statistic for the pricing map on $\mathcal{S}$, not that the economy visits any particular separating pair.

The perturbation is a probability measure. For $0 < \varepsilon \le c/\|\eta\|_\infty$, the density of $S_t' = S_t + \varepsilon\eta$ is $\ge c - \varepsilon\|\eta\|_\infty \ge 0$, and its total mass is $1 + \varepsilon L_0(\eta) = 1$; so S_t' is a probability measure on Φ. By construction $L_j(S_t') = L_j(S_t)$ for $j = 1, \dots, n$ (as $L_j(\eta) = 0$), yet

$$g_t'(\cdot) - g_t(\cdot) = T(S_t' - S_t) = \varepsilon\, T\eta \not\equiv 0 \quad \text{in } L^1(\mathcal{J}, f).$$

Distinct g_t give distinct prices. The risk-neutral compensator obeys $\nu_t^{\mathbb{Q}}(dx) \propto e^{\alpha x} g_t(x)^\alpha f(x)\, dx$ (13). The map $y \mapsto y^\alpha$ is strictly increasing on $(0, \infty)$, so $g_t' \ne g_t$ on a set of positive f-measure forces $g_t'^{\alpha} \ne g_t^\alpha$ there, hence $e^{\alpha x} g_t'^{\alpha} f \ne e^{\alpha x} g_t^\alpha f$ on that set and $\nu_t^{\mathbb{Q}\prime} \ne \nu_t^{\mathbb{Q}}$ as measures; two finite measures that differ as measures differ in at least one of total mass or normalized shape, and U_t is the total mass while $f_t^{\mathbb{Q}}$ is the normalized shape, so $(U_t', f_t^{\mathbb{Q}\prime}) \ne (U_t, f_t^{\mathbb{Q}})$. Thus S_t and S_t' agree on $L_1, \dots, L_n$ but price disaster risk differently: no finite collection of moments is a sufficient statistic for $(U_t, f_t^{\mathbb{Q}})$. $\square$

Proposition 11 (Strengthening of Proposition 3: no continuous finite-dimensional sufficient statistic): *Fix t and the primitives. No* continuous *finite-dimensional statistic of the pricing state is sufficient for the priced disaster law: for every $n \geq 1$ and every continuous map $T : \mathcal{P}(\Phi) \to \mathbb{R}^n$ ($\mathcal{P}(\Phi)$ in the total-variation topology) there are admissible pricing states S, S' with $T(S) = T(S')$ yet $(U_t, f_t^{\mathbb{Q}}) \neq (U_t', f_t^{\mathbb{Q}\prime})$; equivalently, the instantaneous map $S \mapsto (U_t, f_t^{\mathbb{Q}})$ factors through no continuous $\mathbb{R}^n$-valued statistic. The continuity hypothesis is essential: under the weak topology $\mathcal{P}(\Phi)$ is Polish, hence a standard Borel space, so a measurable bijection with a subset of $\mathbb{R}$ furnishes a (discontinuous, non-constructive) measurable sufficient statistic of dimension one—the measurable structure here being the weak-Borel one, decoupled from the total-variation topology in which the continuity hypothesis is stated.*

Proof. Write $g_t = T_{\text{lin}} S$ for the linear map $T_{\text{lin}} : S \mapsto \int_\Phi H(\cdot, \phi)\, S(d\phi)$ with $H(x, \phi) = (\lambda/\lambda^\star)^{1/\alpha}(f_\theta(x)/f(x))^{1/\alpha}$, as in the proof of Proposition 3. The price determines the aggregator: if $f_t^{\mathbb{Q}\prime} = f_t^{\mathbb{Q}}$ then $g_t'^{\,\alpha} = c\, g_t^\alpha$ f-a.e. with $c = \mathbb{E}_f[e^{\alpha J} g_t'^{\,\alpha}]/\mathbb{E}_f[e^{\alpha J} g_t^\alpha]$, and $U_t' = U_t$ forces $c = 1$, whence $g_t' = g_t$ f-a.e. (both positive); so $(U_t, f_t^{\mathbb{Q}})$ and $g_t(\cdot)$ determine one another, and a statistic is sufficient for the price iff g_t factors through it.

Suppose some continuous $T : \mathcal{P}(\Phi) \to \mathbb{R}^n$ were sufficient, and put $m = n + 1$. By the spanning condition (Assumption 4(v)) choose $\phi_0, \ldots, \phi_m$ with $H(\cdot, \phi_0), \ldots, H(\cdot, \phi_m)$ linearly independent in $L^1(\mathcal{J}, f)$; the m differences $H(\cdot, \phi_k) - H(\cdot, \phi_0)$, $k = 1, \ldots, m$, are then linearly independent. Let S_0 have a continuous density bounded below by $c_0 > 0$ on the compact Φ (an admissible state), let β_j be a bounded-density probability measure concentrated near ϕ_j, and set $\eta_k := \beta_k - \beta_0$ (zero total mass). By continuity of H on the compact $\mathcal{J} \times \Phi$ (Assumption 4(i)), $T_{\text{lin}}\eta_k \to H(\cdot, \phi_k) - H(\cdot, \phi_0)$ in $L^1(\mathcal{J}, f)$ as the β_j concentrate, so for tight enough β_j the $\{T_{\text{lin}}\eta_k\}_{k=1}^m$ are linearly independent. On the open box $B = \{\varepsilon \in \mathbb{R}^m : \|\varepsilon\|_\infty < c_0 / \sum_k \|\eta_k\|_\infty\}$ the affine map $\varepsilon \mapsto S_\varepsilon := S_0 + \sum_{k=1}^m \varepsilon_k \eta_k$ takes values in $\mathcal{P}(\Phi)$—its density is $\geq c_0 - \sum_k |\varepsilon_k|\, \|\eta_k\|_\infty \geq c_0 - \|\varepsilon\|_\infty \sum_k \|\eta_k\|_\infty > 0$ (the $\sum_k$ bound, not a per-term one, is what keeps this nonnegative for $m > 1$) and its mass is 1 (the η_k have zero mass)—and is continuous in total variation. Along it $g_t(S_\varepsilon) = g_t(S_0) + \sum_k \varepsilon_k T_{\text{lin}}\eta_k$ is injective in ε, the $T_{\text{lin}}\eta_k$ being independent. Sufficiency then gives, for $\varepsilon, \varepsilon' \in B$,

$$T(S_\varepsilon) = T(S_{\varepsilon'}) \implies (U_t, f_t^{\mathbb{Q}}) \text{ coincide} \implies g_t(S_\varepsilon) = g_t(S_{\varepsilon'}) \implies \varepsilon = \varepsilon',$$

so $\varepsilon \mapsto T(S_\varepsilon)$ is a continuous injection of the open set $B \subseteq \mathbb{R}^m$, $m = n+1$, into $\mathbb{R}^n$. This contradicts Brouwer's invariance of domain, which forbids a continuous injection of a nonempty open subset of $\mathbb{R}^m$ into $\mathbb{R}^n$ when $m > n$. Hence no continuous finite-dimensional sufficient statistic exists.

For the final clause, equip $\mathcal{P}(\Phi)$ with the *weak* topology, under which—Φ being compact metric—it is Polish, hence a standard Borel space; the total-variation topology of the main argument is finer and, for uncountable Φ, not separable, so it is this weak-Borel structure that is standard. By the Borel isomorphism theorem $\mathcal{P}(\Phi)$ is then Borel-isomorphic to a subset of $\mathbb{R}$, and g_t is weak-Borel measurable—indeed weak-continuous, being integration of the bounded continuous kernel $H(\cdot, \phi)$ against the measure—so composing it with such an isomorphism writes it as a measurable function of one real coordinate, a measurable, one-dimensional sufficient statistic. It is discontinuous and non-constructive, which is why the continuity hypothesis is not cosmetic. □

Corollary 1 (Intensity channel: a scalar fear state): *If the size law is undisputed and correct—$f_{\theta_i} = f$ for all i, the held size densities equalling the objective f—then $r_i(x;t) = \lambda_i(t-)/\lambda^\star$ is independent of x,*

$$g_t(x) \equiv G_t := \int_\Phi \left(\tfrac{\lambda}{\lambda^\star}\right)^{1/\alpha} S_t(d\phi) = \int_0^1 s_t^i \left(\tfrac{\lambda_i(t)}{\lambda^\star}\right)^{1/\alpha} di, \tag{D7}$$

displayed at the generic state S_t; as a jump compensator G_t is read at the predictable left limit S_{t-}, the two agreeing off disasters. The risk-neutral size law then collapses to the fixed risk-

aversion tilt $f_t^{\mathbb{Q}}(x) = e^{\alpha x} f(x)/\mathbb{E}_f[e^{\alpha J}]$*, and the fear factor is the single scalar*

$$U_t = G_t^{\alpha}\, U^{\mathrm{RA}}. \tag{D8}$$

The size dispute is precisely what makes $f_t^{\mathbb{Q}}$ *move; absent it, beliefs price only the disaster* frequency U_t*, through the one scalar* G_t*.*

Proof of Corollary 1. With $f_{\theta_i} = f$ the likelihood ratio (C1) simplifies:

$$\begin{aligned} r_i(x;t) &= \frac{\lambda_i(t-)\, f_{\theta_i}(x)}{\lambda^\star\, f(x)} \\ &= \frac{\lambda_i(t-)\, f(x)}{\lambda^\star\, f(x)} && \text{(D9)} \\ &= \frac{\lambda_i(t-)}{\lambda^\star}, && \text{(D10)} \end{aligned}$$

line (D9) substituting $f_{\theta_i} = f$ and (D10) cancelling $f(x)$; r_i is independent of x. Substituting $r_i(x;t) = \lambda_i(t-)/\lambda^\star$ into (12),

$$g_t(x) = \int_0^1 s_{t-}^i \left(\tfrac{\lambda_i(t-)}{\lambda^\star}\right)^{1/\alpha} di =: G_t,$$

independent of x; this is (D7), the predictable aggregator evaluated at the left-limit S_{t-} (equal to S_t off disasters; (D7) displays it at the generic state). For the size law, start from (14) and use $g_t \equiv G_t$:

$$\begin{aligned} f_t^{\mathbb{Q}}(x) &= \frac{e^{\alpha x} g_t(x)^{\alpha} f(x)}{\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha}]} \\ &= \frac{e^{\alpha x} G_t^{\alpha} f(x)}{\mathbb{E}_f[e^{\alpha J} G_t^{\alpha}]} && \text{(D11)} \\ &= \frac{e^{\alpha x} G_t^{\alpha} f(x)}{G_t^{\alpha} \mathbb{E}_f[e^{\alpha J}]} && \text{(D12)} \\ &= \frac{e^{\alpha x} f(x)}{\mathbb{E}_f[e^{\alpha J}]}, && \text{(D13)} \end{aligned}$$

line (D11) substituting $g_t = G_t$, (D12) pulling the constant G_t^α out of the f-expectation, and (D13) cancelling it; the law is free of G_t. For the fear factor, from $U_t = \lambda_t^{\mathbb{Q}} = \lambda^\star \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ (14),

$$U_t = \lambda^\star\, \mathbb{E}_f[e^{\alpha J} G_t^\alpha] \tag{D14}$$

$$= G_t^\alpha\, \lambda^\star\, \mathbb{E}_f[e^{\alpha J}] \tag{D15}$$

$$= G_t^\alpha\, U^{\mathrm{RA}}, \tag{D16}$$

line (D14) substituting $g_t = G_t$, (D15) pulling the constant G_t^α out of the expectation, and (D16) using $U^{\mathrm{RA}} = \lambda^\star \mathbb{E}_f[e^{\alpha J}]$ (Definition 2); this is (D8). □

Corollary 2 (Finite menu: finite-dimensional pricing functionals): *If the menu is finite, $\Phi = \{\phi_1, \ldots, \phi_K\}$, then $S_t = (S_t^1, \ldots, S_t^K)$ lies in the simplex Δ^{K-1} and the* instantaneous *pricing objects $\bar{\lambda}_t^S$, $g_t(\cdot)$, $f_t^{\mathbb{Q}}$, and U_t are functions of this finite vector. (The vector is a sufficient statistic for the* instantaneous *pricing objects above; it is not a sufficient statistic for forward-looking prices such as the price–dividend ratio $v(\Psi_t)$, nor by itself a Markov state for the dynamics—see the proof.) Under the intensity channel of Corollary 1 with a two-point menu $\{\lambda_{lo}, \lambda_{hi}\}$, the instantaneous fear factor is a function of the single share S_t^{hi},*

$$U_t = \Big[S_t^{hi}\big(\tfrac{\lambda_{hi}}{\lambda^\star}\big)^{1/\alpha} + (1 - S_t^{hi})\big(\tfrac{\lambda_{lo}}{\lambda^\star}\big)^{1/\alpha}\Big]^\alpha U^{\mathrm{RA}}, \tag{D17}$$

recovering the scalar, intensity-only fear factor as a doubly-restricted special case.

Proof of Corollary 2. With $\Phi = \{\phi_1, \ldots, \phi_K\}$ finite the pricing state is the atomic measure $S_t = \sum_{k=1}^K S_t^k\, \delta_{\phi_k}$ with $(S_t^1, \ldots, S_t^K) \in \Delta^{K-1}$. Each aggregate is the integral of a fixed function against S_t, so it collapses to a finite sum of the shares:

$$\bar{\lambda}_t^S = \int_\Phi \lambda\, S_t(d\phi) = \sum_{k=1}^{K} S_t^k\, \lambda_k, \tag{D18}$$

$$g_t(x) = \int_\Phi r(\phi,x)^{1/\alpha}\, S_t(d\phi) = \sum_{k=1}^{K} S_t^k\, r(\phi_k,x)^{1/\alpha}, \tag{D19}$$

with $r(\phi,x) = \lambda f_\theta(x)/(\lambda^\star f(x))$ as in (12). Through (14), $f_t^{\mathbb{Q}}(x) = e^{\alpha x} g_t(x)^\alpha f(x)/\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ and $U_t = \lambda^\star \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ depend on the cross-section only through $g_t(\cdot)$, which by (D19) is a function of (S_t^k); hence $\bar{\lambda}_t^S, g_t(\cdot), f_t^{\mathbb{Q}}, U_t$ are functions of $(S_t^k) \in \Delta^{K-1}$.

Under the intensity channel (Corollary 1) with $K = 2$ and $\Phi = \{\lambda_{lo}, \lambda_{hi}\}$, $f_\theta = f$ gives $r(\phi_k,x)^{1/\alpha} = (\lambda_k/\lambda^\star)^{1/\alpha}$ (independent of x), so (D19) becomes

$$\begin{aligned} g_t(x) \equiv G_t &= S_t^{hi}\big(\tfrac{\lambda_{hi}}{\lambda^\star}\big)^{1/\alpha} + S_t^{lo}\big(\tfrac{\lambda_{lo}}{\lambda^\star}\big)^{1/\alpha} \\ &= S_t^{hi}\big(\tfrac{\lambda_{hi}}{\lambda^\star}\big)^{1/\alpha} + (1 - S_t^{hi})\big(\tfrac{\lambda_{lo}}{\lambda^\star}\big)^{1/\alpha}, \end{aligned} \tag{D20}$$

line (D20) substituting $S_t^{lo} = 1 - S_t^{hi}$ (the simplex constraint). Substituting (D20) into $U_t = G_t^\alpha U^{\mathrm{RA}}$ (D8) gives (D17).

The reduction is of the *instantaneous* pricing map only. The vector (S_t^k) summarizes current prices, but it is not by itself a Markov state: its own law of motion is not autonomous, because the calm switch flow out of each class depends on the within-class distribution of regime ages and record statistics, not on the shares alone (Proposition 3). The full Markov state therefore carries that within-class record structure, and the valuation operator acts on it rather than on (S_t^k) alone—a point made precise where the price level is computed. $\square$

Proposition 12 (The revision hazard as a deviance first-passage): *Fix an investor holding $\phi = (\lambda,\theta)$ with tolerance ε and since-adoption record $\mathcal{R} = (n,(x_1,\ldots,x_n))$ at age τ; let $D_{\phi,\tau}(\mathcal{R})$ be the deviance (7) and $r_{\phi,\tau}(\varepsilon)$ the acceptance threshold of (8). The revision time $\tau^{\mathrm{rev}} = \inf\{t : D_{\phi_i,\tau_t^i}(\mathcal{R}_t^i) \geq r_{\phi_i,\tau_t^i}(\varepsilon^i)\}$ is the first passage of the deviance across r, with:*

(i) (calm drift) *between disasters the record is fixed and the deviance drifts deterministically,*

$$\frac{\partial}{\partial \tau} D_{\phi,\tau}(\mathcal{R}) = \lambda - \frac{n}{\tau}, \tag{D21}$$

rising—the held model increasingly contradicted—iff the realized rate n/τ is below the held λ; with $\mathcal{R}$ fixed, D meets the moving threshold $r_{\phi,\tau}(\varepsilon)$ at a deterministic calm deadline, the record-$\mathcal{R}$ generalization of $\Delta^*(\phi, \varepsilon)$, whose $n = 0$ slice is exactly Proposition 7;

(ii) (disaster trigger) *a disaster arrives at the true rate $\lambda^\star$ (Assumption 1) with size $x \sim f$, appends x to the record, and triggers rejection iff the updated deviance leaves the typical set, so the disaster-triggered rejection intensity is*

$$\lambda^\star \, \pi(\phi, \tau, \mathcal{R}; \varepsilon), \qquad \pi(\phi, \tau, \mathcal{R}; \varepsilon) = \int_{\mathcal{J}} \mathbf{1}\big\{D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon)\big\} f(x)\, dx, \tag{D22}$$

a functional of the full record $\mathcal{R}$, not of the age τ alone;

(iii) (no closed form) *τ^{rev} is the first passage of the piecewise-deterministic Markov process on the record-resolved state $(\phi, \mathcal{R}, \tau)$—the scalar deviance alone is not Markov in the size-dispute model—with drift (D21) and jumps (D22); for a general regular family $\{F_\theta\}$ neither the threshold $r_{\phi,\tau}(\varepsilon)$ nor the first-passage law has an elementary form, and both are computed numerically from the process. A hazard in the low-dimensional state (n, τ)—the since-adoption count $n = n_t^i = N_t - N_{a^i}$ (5) and the age, rather than the full mark record—and a closed-form passage law obtain under the restriction of Corollary 1 (intensity-only), where $f_\theta \equiv f$ makes the deviance depend on the record only through the count n (so the rejection still reads the since-adoption count n, not the age alone). The finite menu of Corollary 2 reduces the* pricing *functionals to finite dimension but leaves the rejection hazard record-dependent—the marks x_j still enter the deviance through f_θ—so in general the rejection is governed by the full record.*

Proof of Proposition 12. (i) With the held model ϕ fixed and no disaster on the interval ending at age τ, the realized sizes and their MLE $\hat{\theta}$ are constant, so the size term of (7) is constant in τ and only the count term $C(\tau) := \tau\, d_P(n/\tau \,\|\, \lambda)$ varies. Substituting $a = n/\tau$ into $d_P(a\|\lambda) = a \log \frac{a}{\lambda} - a + \lambda$,

$$
\begin{aligned}
C(\tau) &= \tau\Big[\tfrac{n}{\tau} \log \tfrac{n/\tau}{\lambda} - \tfrac{n}{\tau} + \lambda\Big] \\
&= n \log \tfrac{n/\tau}{\lambda} - n + \lambda\tau && \text{(D23)} \\
&= n \log \tfrac{n}{\tau\lambda} - n + \lambda\tau, && \text{(D24)}
\end{aligned}
$$

line (D23) distributing the factor τ and (D24) using $\frac{n/\tau}{\lambda} = \frac{n}{\tau\lambda}$; with the convention $0 \log 0 = 0$ this holds also at $n = 0$, where $C(\tau) = \lambda\tau$. To differentiate, expand the logarithm $\log \frac{n}{\tau\lambda} = \log n - \log \tau - \log \lambda$:

$$C(\tau) = n \log n - n \log \tau - n \log \lambda - n + \lambda\tau.$$

Only the $-n \log \tau$ and $\lambda\tau$ terms depend on τ, and $D_{\phi,\tau}(\mathcal{R})$ differs from $C(\tau)$ by a τ-constant, so

$$
\begin{aligned}
\frac{\partial}{\partial\tau} D_{\phi,\tau}(\mathcal{R}) = C'(\tau) &= -\frac{n}{\tau} + \lambda && \text{(D25)} \\
&= \lambda - \frac{n}{\tau}, && \text{(D26)}
\end{aligned}
$$

line (D25) using $\partial_\tau(-n \log \tau) = -n/\tau$ and $\partial_\tau(\lambda\tau) = \lambda$ and (D26) reordering; this is (D21). The drift is positive—the held model increasingly contradicted—iff $\lambda > n/\tau$, i.e. the realized rate n/τ lies below the held λ. With $\mathcal{R}$ fixed, $D_{\phi,\tau}(\mathcal{R})$ meets the moving threshold $r_{\phi,\tau}(\varepsilon)$ at a deterministic calm deadline (the record-$\mathcal{R}$ generalization of $\Delta^*(\phi, \varepsilon)$); at the $n = 0$ slice $C(\tau) = \lambda\tau$ and $\partial_\tau D = \lambda > 0$, recovering the deterministic deadline $\Delta^*(\phi, \varepsilon)$ of Proposition 7.

(ii) Disasters arrive as a marked Poisson process with $(\mathcal{F}_t, \mathbb{P})$-compensator $\nu(dt, dx) = \lambda^\star\, dt\, f(x)\, dx$ (3) (Assumption 1). A disaster of size x at age τ appends x to the record, $\mathcal{R} \mapsto \mathcal{R} \oplus x = (n + 1, (x_1, \ldots, x_n, x))$, and by the pass/fail rule (8) the test rejects at that

instant iff the updated record leaves the typical set,

$$D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon).$$

The triggering marks form the set $B_{\phi,\tau,\mathcal{R}} := \{x \in \mathcal{J} : D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon)\}$, matching the closed-threshold convention of Definition 1; its boundary $\{x : D_{\phi,\tau}(\mathcal{R} \oplus x) = r_{\phi,\tau}(\varepsilon)\}$ is f-null (the mark law is continuous and D is continuous in x), so the open and closed conventions yield the same disaster-triggered intensity π, which is therefore unambiguous. Restricting the disaster random measure to $B_{\phi,\tau,\mathcal{R}}$ predictably thins it to a point process whose (record-dependent) intensity is the ν-mass of $B_{\phi,\tau,\mathcal{R}}$:

$$\int_{B_{\phi,\tau,\mathcal{R}}} \frac{\nu(dt, dx)}{dt} = \lambda^\star \int_{\mathcal{J}} \mathbf{1}\{x \in B_{\phi,\tau,\mathcal{R}}\}\, f(x)\, dx \tag{D27}$$

$$= \lambda^\star \int_{\mathcal{J}} \mathbf{1}\{D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon)\}\, f(x)\, dx \tag{D28}$$

$$= \lambda^\star\, \pi(\phi, \tau, \mathcal{R}; \varepsilon), \tag{D29}$$

line (D27) writing the restricted mass against $\nu = \lambda^\star\, dt\, f\, dx$, (D28) substituting the definition of $B_{\phi,\tau,\mathcal{R}}$, and (D29) the definition of π; this is the disaster-triggered rejection intensity (D22). It depends on $\mathcal{R}$ through the map $x \mapsto D_{\phi,\tau}(\mathcal{R} \oplus x)$, not on the age τ alone.

(iii) Between disasters D evolves by (D21); at disasters it jumps as in (ii); τ^{rev} is the first time D exceeds $r_{\phi,\tau}(\varepsilon)$—the first passage of this piecewise-deterministic Markov process. The threshold is the $(1-\varepsilon)$-quantile of $D_{\phi,\tau}$ under $Q_{\phi,\tau}$ (6), a Poisson–mark mixture with no elementary quantile for general $\{F_\theta\}$, and the first-passage law of a PDMP across a moving boundary is likewise not elementary; both are computed numerically. The state collapses to (n, τ) only when the size law is undisputed, and to a finite vector under a finite menu; in general the rejection is governed by the full record. □

Lemma 6 (Rejection hazard, switch hazard, and the switch flow): *Fix the public posterior ρ_t*

and its mode $\phi_t^* = \mathfrak{m}(\rho_t)$ (9). *For an investor in state* $(\phi, \mathcal{R}, \tau, \varepsilon)$*, let the* rejection hazard $H_t^{\mathrm{rej}}(d\tau' \mid \phi, \mathcal{R}, \tau, \varepsilon)$ *be the hazard (intensity) measure of the first-passage time* τ^{rev} *of Proposition 12—a disaster-triggered rate* $\lambda^\star \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D22) *together with* certain *rejection at the deterministic calm deadline* $\Delta^\star$*, so that* τ^{rev} *is the earlier of* $\Delta^\star$ *and the first disaster-triggered rejection, with* local conditional *survival—the no-state-change factor along a fixed record, not the marginal survival of the PDMP—* $\mathbb{P}(\tau^{\mathrm{rev}} > u) = \mathbf{1}\{u < \Delta^\star\} \exp\!\big(-\int_\tau^u \lambda^\star \pi \, dt'\big)$ *(a non-rejecting disaster appends a mark, updating both* π *and the next deadline* $\Delta^\star$*, after which the first passage continues as the piecewise-deterministic law of Proposition 12(iii)), dropping to zero at* $\Delta^\star$ *(the deterministic deadline is not represented as a unit atom of an ordinary exponential hazard). Write* $\beta_t(d\phi)$ *for the wealth-weighted rate at which holders of* ϕ *reach their scheduled calm deadlines at* t*. Then:*

(i) the switch hazard *is the rejection hazard restricted to the switch event* $\{\phi \neq \phi_t^*\}$*,*

$$H_t^{\mathrm{sw}}(d\tau' \mid \phi, \mathcal{R}, \tau, \varepsilon) = \mathbf{1}\{\phi \neq \phi_t^*\} \, H_t^{\mathrm{rej}}(d\tau' \mid \phi, \mathcal{R}, \tau, \varepsilon); \tag{D30}$$

(ii) the switch flow has two parts on different timescales: a continuous *calm switch flow*

$$\Phi_t^{\mathrm{sw}}(A) = \int_A \mathbf{1}\{\phi \neq \phi_t^*\} \, \beta_t(d\phi), \qquad A \subseteq \Phi, \tag{D31}$$

with β_t *non-atomic where deadlines stagger (and carrying a predictable atom where a positive* G*-mass synchronizes, which by Lemma 7 moves no traded price), and a disaster-triggered component whose predictable* intensity *(compensator, integrating over marks) is* $\lambda^\star \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D22)*; the realized state jump at a disaster epoch is set by the realized mark* x *and its triggered set, not by a continuous rate. Every law-of-motion term runs against the switch set* $\{\phi \neq \phi_t^*\}$*, never the raw rejection rate, so a self-re-adoption (*$\phi = \phi_t^*$*) renews the test window but leaves* ϕ*, hence* S_t*, unchanged.*

Proof of Lemma 6. (i) Switch hazard. By the re-selection map (9), after a rejection the held model is $\phi_t^* = \mathfrak{m}(\rho_t)$, which differs from the incumbent ϕ exactly on the switch event $\{\phi \neq \phi_t^*\}$. A rejection resets the record $\mathcal{R} \to \varnothing$ and age $\tau \to 0$ whether or not the held model moves, but the held model itself moves only on a switch; hence the held-model part of the rejection hazard is its restriction to $\{\phi \neq \phi_t^*\}$, which is (D30). A self-re-adoption $\phi = \phi_t^*$ renews the test window but leaves ϕ, and therefore S_t, unchanged.

(ii) Two timescales. For a single investor the calm-deadline first passage $\Delta^\star$ is deterministic—an atom of her revision law. Across investors the induced calm-deadline law is atomless: the tolerance law G is continuous (Assumption 2) and the calm deadline $\varepsilon \mapsto a + \Delta^\star(\phi, \varepsilon)$ is non-increasing in ε, and *strictly* so wherever the continuous size-deviance component of (7) renders the threshold $r_{\phi,\Delta}(\varepsilon)$ strictly monotone (Proposition 12); on such ranges the deadline map has G-null level sets, its pushforward places no positive mass at any single date, the wealth-weighted deadline-arrival rate $\beta_t(d\phi)$ is non-atomic, and the calm switch flow (D31) is a *continuous* finite-variation flow. On a flat level set of $\Delta^\star(\phi, \cdot)$ of positive G-mass—possible where the discrete count deviance dominates—the deadline-coincident revisers re-select simultaneously, contributing a *predictable* finite-variation jump rather than a Brownian increment; either way the flow is finite-variation, and the *traded* price is continuous across it (Lemma 7), so the no-arbitrage decomposition is unaffected. The disaster-triggered component, whose predictable intensity (compensator, integrating over marks) is $\lambda^\star \pi(\phi, \tau, \mathcal{R}; \varepsilon)$ (D22), can fire only at a disaster epoch; the realized jump is then set by the realized mark x, simultaneously for the whole triggered set, and is not a continuous rate. Both parts are restricted to $\{\phi \neq \phi_t^*\}$, so a self-re-adoption moves nothing. □

Remark 6 (Why a stationary fear cross-section needs demographic refresh). The re-selection (9) reads the second-order posterior ρ_t^i, updated on the *full* history (4), whereas the test reads only the since-adoption window: the two memories differ. Under the true DGP and the identifiable family of Assumption 4, ρ_t^i is consistent for the Kullback–Leibler model $\phi^\star$, so $\mathrm{MAP}(\rho_t^i) \to \phi^\star$; once an infinitely-lived investor holds $\phi^\star$ she re-selects near $\phi^\star$ at each rejection, and the held-model cross-section concentrates on $\phi^\star$—the fear dispersion dies out. A *nondegenerate* stationary fear cross-section is therefore not a property of the present primitives; it is sustained by demographic refresh—newborns who enter holding a *drawn* model $\phi_0 \sim \Pi_\phi$ with a reset test clock (Assumption 5(iv)), held until their first rejection—which an overlapping-generations birth–death structure supplies, added as an explicit primitive where the stationary cross-section is required. The refresh is in the *held models*, not the posteriors: each newborn inherits the (concentrated) common posterior ρ_t, so $\mathfrak{m}(\rho_t)$ stays common across cohorts, but holds her drawn $\phi_0 \neq \mathfrak{m}(\rho_t)$ until she revises, keeping a positive mass of wealth away from $\phi^\star$. The drawn ϕ_0 is an exogenous status-quo endowment, not a birth-time MAP re-initialization: a newborn carries her inherited held model until her own test rejects it, and the MAP rule (9) governs only re-selection at rejection, exactly as for an incumbent. This departure from MAP-initialization at birth is the OLG primitive on which the stationary dispersion rests. We do not assert stationarity from the present primitives. *(Not proved here; the posterior-consistency step and the stationary cross-section are formalized with the demographic primitive.)*

Proof of Proposition 3, the law of motion. (i) Between disasters and switches every held model ϕ_i is fixed. Writing S_t as the pushforward of the wealth measure $s_t^i\,di$ under $i \mapsto \phi_i$ (Definition 3), $\langle h, S_t\rangle = \int_0^1 h(\phi_i)\, s_t^i\, di$. The interchange of $\frac{d}{dt}$ and $\int_0^1 \cdot\, di$ is justified by dominated convergence: h is bounded and, on compact t-intervals, s_t^i and its t-derivative are bounded uniformly in i (as in Proposition 10, via (C2) and $\lambda_i \in [\underline{\lambda}, \overline{\lambda}]$). Then

$$\begin{aligned}
\frac{d}{dt}\langle h, S_t\rangle &= \int_0^1 h(\phi_i)\,\frac{ds_t^i}{dt}\,di \\
&= \int_0^1 h(\phi_i)\, s_t^i\, \frac{d\log s_t^i}{dt}\,di && \text{(D32)}\\
&= \int_0^1 h(\phi_i)\, s_t^i\, \tfrac{1}{\alpha}\big(\bar\lambda_t^S - \lambda_i\big)\,di && \text{(D33)}\\
&= \tfrac{1}{\alpha}\int_\Phi \big(\bar\lambda_t^S - \lambda\big)\, h(\phi)\, S_t(d\phi), && \text{(D34)}
\end{aligned}$$

line (D32) using $ds_t^i/dt = s_t^i\, d\log s_t^i/dt$, (D33) substituting the calm share drift (D6), and (D34) pulling out the constant $\frac{1}{\alpha}$ and rewriting the i-integral as a ϕ-integral against S_t (Definition 3);

this is (18).

(ii) At a calm-deadline switch, investor i's held model jumps from ϕ_i to $\phi_i' = \mathfrak{m}(\rho_t^i)$ (9). By Lemma 4 the second-order posterior is common, $\rho_t^i = \rho_t$, so $\phi_i' = \mathfrak{m}(\rho_t) = \phi_t^*$ for every investor revising at t. Her contribution to $\langle h, S_t \rangle$ changes by $s_t^i[h(\phi_t^*) - h(\phi_i)]$; the staggered deadlines spread across the atomless threshold law, so the aggregate rate of switching out of ϕ is the *continuous* calm switch flow $\Phi_t^{\mathrm{sw}}(d\phi)$ of Lemma 6 (D31) (restricted to $\{\phi \neq \phi_t^*\}$), and aggregating gives (19). Disaster-triggered switches do not enter here—they fire only at disaster epochs, in (iii).

(iii) At a disaster of size x, Proposition 10(ii) gives $s_t^i = s_{t-}^i\, r(\phi_i, x)^{1/\alpha}/g_t(x)$ (with $r_i(x;t) = r(\phi_i, x)$). Hence for any bounded h,

$$
\begin{aligned}
\langle h, S_t \rangle &= \int_0^1 h(\phi_i)\, s_t^i\, di \\
&= \int_0^1 h(\phi_i)\, s_{t-}^i\, \frac{r(\phi_i, x)^{1/\alpha}}{g_t(x)}\, di && \text{(D35)} \\
&= \frac{1}{g_t(x)} \int_0^1 h(\phi_i)\, r(\phi_i, x)^{1/\alpha}\, s_{t-}^i\, di && \text{(D36)} \\
&= \frac{1}{g_t(x)} \int_\Phi h(\phi)\, r(\phi, x)^{1/\alpha}\, S_{t-}(d\phi), && \text{(D37)}
\end{aligned}
$$

line (D35) substituting the share jump, (D36) pulling out the constant $g_t(x)^{-1}$, and (D37) rewriting the i-integral against S_{t-} (Definition 3). As this holds for every bounded h, it is the measure identity (20)—but only the *reweight* step (a) of Definition 4, in which h is still read at the pre-disaster model ϕ. The full post-disaster state applies the relocation step (d) on top: the holders the disaster triggers move to ϕ_t^* at the disaster epoch (a jump, not the continuous flow of (ii)), so for them h is read at ϕ_t^*. The triggered investors form the region $\mathcal{T}_t^x := \{(\phi, \mathcal{R}, \tau, \varepsilon) : D_{\phi,\tau}(\mathcal{R} \oplus x) \geq r_{\phi,\tau}(\varepsilon)\}$ (D22) in the *record-resolved* state—*not* a subset of Φ, since the trigger depends on the full state and two investors sharing ϕ can differ in it—so the relocation is read against Ψ_{t-}:

$$\langle h, S_t^x\rangle = \frac{1}{g_t(x)}\int r(\phi,x)^{1/\alpha}\Big[h(\phi)\,\mathbf{1}\{(\phi,\mathcal{R},\tau,\varepsilon)\notin\mathcal{T}_t^x\}+h(\phi_t^*)\,\mathbf{1}\{(\phi,\mathcal{R},\tau,\varepsilon)\in\mathcal{T}_t^x\}\Big]\,\Psi_{t-}\big(d(\phi,\mathcal{R},\tau,\varepsilon)\big);$$

the reweight (D37) is the special case $\mathcal{T}_t^x=\varnothing$ (no triggered switchers). Since $\mathcal{T}_t^x$ and ϕ_t^* are functionals of the record cross-section and the public posterior, the evolution does not reduce to finitely many coordinates of S_t (Proposition 3) □

Appendix E. Proofs for Section 4

A. Where each instantaneous price sits in the decomposition

The short rate, the price–dividend ratio, and the disaster premium are derived below; here we record where each sits in the decomposition of Proposition 5.

The short rate is pure $B_t(S_t)$. The instantaneous safe rate is $r_t=\delta+\alpha\mu_0-\frac{1}{2}\alpha(1+\alpha)\sigma^2+\bar\lambda_t^S-U_t$ (Proposition 13): a constant power-utility diffusive part plus the belief-driven disaster part $\bar\lambda_t^S-U_t$, both functionals of S_t, so the rate carries no survival-history residual and a Markov-in-cross-section benchmark matches it exactly. Beliefs enter only through the difference, so a fear wave lowers the rate precisely when the priced-tilt response dominates the mean-intensity response, $dU_t>d\bar\lambda_t^S$—the flight-to-safety of the lazy economy, purely belief-driven since $(\mu_0,\sigma,\lambda^\star,F_J)$ never move. The same fear factor that lowers the rate raises the disaster premium, so one channel delivers a low safe rate and a high premium together, the pairing the rare-disaster literature uses to address both puzzles (Barro, 2006). Under power utility this passes a volatile fear wave straight into r_t, against smooth real rates; recursive-preference disaster models keep r_t smooth (Wachter, 2013), a device we forgo, so the belief block must be disciplined (Section 5).

The price–dividend ratio carries the residual. The ratio is $v(\Psi_t)$ (Proposition 14), a Feynman–Kac functional of the full record-resolved state with no finite-dimensional reduction in S_t

(Proposition 3); in the decomposition it splits as $v_B(S_t) + \mathcal{H}_t^v(\Psi_t)$, the residual driven by the same staggered-revision channels as the forward-fear residual below and zero for any Markov benchmark. Its comparative static is *two-signed*: the valuation-growth rate reads the cross-section through two non-interchangeable aggregates—the *arithmetic* mean $\bar{\lambda}_t^S$ in the reallocation drift and the *power* mean $G_t^\alpha \le \bar{\lambda}_t^S/\lambda^\star$ in the priced-jump term—so whether a more fearful cross-section raises valuation growth and *expands* the price–dividend ratio, against the naive "higher discount, lower price" reading, is a quantitative condition met at our calibration, not a consequence of Jensen. A finite-menu Feynman–Kac equation and a first-order belief-dispersion expansion follow (Corollaries 3 and 4).

The premium loads on the fear factor, and predicts returns. The instantaneous excess return is $\alpha\sigma^2 + \mathrm{DRP}_t$ (Proposition 15): the diffusive premium $\alpha\sigma^2$ is constant, so all variation is the disaster premium, which loads on the fear factor $U_t = U^{\mathrm{RA}}\kappa_t$. Priced disaster fear therefore forecasts excess returns while carrying no information about diffusive volatility, pinned at the constant σ—the return-premium leg of the Andersen et al. (2015b) unspanned-tail finding, with the caveat that, σ constant, the non-prediction of volatility is an identity rather than a match. The premium's *level* runs through the price–dividend jump ratio ψ_t, hence $v(\Psi_t)$, so it too carries the survival-history residual.

Lemma 7 (Predictable revisions carry no price jump): *Let τ be a calm-deadline revision time—a predictable time with $\Delta N_\tau = 0$ at which the revising cohort's held model jumps from ϕ to the common destination $\phi_\tau^* = \mathfrak{m}(\rho_\tau)$ (Lemma 4), possibly for a positive G-mass synchronizing on a flat level set of $\Delta^\star(\phi,\cdot)$. Then* (i) *every belief density is continuous at τ, $M_\tau^i = M_{\tau-}^i$, so Ξ_t, the kernel ξ_t, and every consumption share s_t^i are continuous at τ and each reviser carries its pre-jump share with no instantaneous wealth transfer;* (ii) *the share-weighted held-model distribution S_t jumps in total variation, so the instantaneous functionals $g_t(\cdot)$, U_t, the short rate, and the* level *of the disaster premium carry a* predictable *jump at τ;* (iii) *the traded price is continuous, $P_{\tau-} = P_\tau$, equivalently $v(\Psi_{\tau-}, \rho_{\tau-}) = v(\Psi_\tau, \rho_\tau)$ on the record-resolved state of Definition 4, and P jumps only at the totally inaccessible disaster epochs. The conclusion uses* *no atomlessness of the deadline law.*

Proof. (i). In (C1) the factor $\exp(-\int_0^t(\lambda_i(s-)-\lambda^\star)\,ds)$ is absolutely continuous in t, and the product $\prod_{T_k\le t} r_i(J_k;T_k)$ acquires a factor only when t crosses a disaster epoch T_k; since $\Delta N_\tau = 0$, none is acquired at τ, and the revision changes $\lambda_i(\cdot-)$ and the forward likelihood ratios only on $\{s \ge \tau\}$—a Lebesgue-null contribution to M^i_τ. Hence $M^i_\tau = M^i_{\tau-}$: the held-model switch moves the predictable forward drift of M^i, not its level. (Equivalently, M^i is a true $(\mathcal{F}_t,\mathbb{P})$-martingale built from the predictable, left-continuous held-model compensator (Brémaud, 1981, VI, T4), and the driving Brownian-plus-Poisson filtration is quasi-left-continuous, so $\mathcal{F}_\tau = \mathcal{F}_{\tau-}$ and $M^i_\tau = \mathbb{E}[M^i_\tau \mid \mathcal{F}_{\tau-}] = M^i_{\tau-}$.) On a disaster-free neighbourhood of τ the ratio M^i_t/M^i_τ is bounded above and below uniformly in i (Λ compact), so dominated convergence gives $\Xi_t \to \Xi_\tau$ in (10); with C_t continuous in calm, $\xi_t = e^{-\delta t}C_t^{-\alpha}\Xi_t^\alpha$ and the shares (11) are continuous, so $s^i_\tau = s^i_{\tau-}$.

(ii). A positive mass relocates $\phi \to \phi^*_\tau$ while s^i is continuous, so $S_t = \int_0^1 s^i_t\,\delta_{\phi_i(t)}\,di$ jumps in total variation; the belief aggregator $g_t(x) = \int_\Phi r(\phi,x)^{1/\alpha} S_t(d\phi)$ and the functionals it determines inherit a predictable jump.

(iii). The equilibrium gain process is a $\mathbb{P}$-martingale (Proposition 1), so $\xi_t P_t = \mathbb{E}_t[\int_t^\infty \xi_s C_s\,ds]$. The dividend carries no mass at the single time τ, and by quasi-left-continuity $\mathcal{F}_\tau = \mathcal{F}_{\tau-}$ with $\rho_\tau = \rho_{\tau-}$ (the public posterior is continuous in calm), so the two conditional expectations coincide and $\xi_{\tau-}P_{\tau-} = \xi_\tau P_\tau$; dividing by the continuous, strictly positive ξ gives $P_{\tau-} = P_\tau$. Writing $P_t = C_t\,v(\Psi_t,\rho_t)$ with $C_\tau = C_{\tau-}$ yields the value-matching form. The destination $\phi^*_\tau = \mathfrak{m}(\rho_\tau)$ is read from ρ_τ, so the relocation map is $\mathcal{F}_{\tau-}$-measurable precisely because the forward-looking state carries ρ_t; and $\Psi_{\tau-}$ already encodes the imminent deadline (the ages and records that pin $\Delta^\star$), so the forward value already incorporates the scheduled relocation. The disaster epochs are totally inaccessible, so P jumps only there. $\square$

Proof of Proposition 5. The decomposition (22) itself is immediate and self-contained: $B_t = \mathbb{E}[F_t \mid S_t]$ is the L^2 projection of F_t onto the σ-algebra generated by S_t, so it is S_t-measurable, and the residual $\mathcal{H}_t = F_t - B_t$ has $\mathbb{E}[\mathcal{H}_t \mid S_t] = 0$ by the tower property and is measurable in the finer Ψ_t, since $S_t = \pi_\phi \Psi_t$. This holds for any equilibrium price object F_t and uses nothing beyond the definitions of the two states. The classification (i)–(iii) of which objects carry a zero versus a nonzero residual is read off the pricing results established later in this appendix; we record it here only to fix the organizing split, and it carries no logical weight in the present argument. *(i)* The fear factor and multiplier, the risk-neutral disaster law, and the short rate are S_t-measurable (Proposition 16), so each equals its own projection and $\mathcal{H}_t \equiv 0$. *(ii)* The price–dividend ratio and the premium level are functionals of the full Ψ_t, not autonomous in S_t (Proposition 14); and the forward fear factor has $F_t(0) = U_t$ measurable in S_t with right-hand slope $\partial_s^+ F_t(0) = \sigma(S_t) + U_t\, c_{\mathrm{sw}}(\Phi_t^{\mathrm{sw}}) + \lambda^\star \Theta_{\mathrm{sw}}(\Psi_t)$, whose last two terms are measurable in Ψ_t and not in S_t (Theorem 1), so $\mathcal{H}_t = s\,[U_t\, c_{\mathrm{sw}}(\Phi_t^{\mathrm{sw}}) + \lambda^\star \Theta_{\mathrm{sw}}(\Psi_t)] + o(s)$, zero at horizon zero and first order in s. *(iii)* For any benchmark whose pricing state is Markov in the current cross-section S_t, the conditional law of the future is by construction a functional of S_t, so every forward object $\mathbb{E}[\,\cdot \mid \mathcal{F}_t]$ equals its own S_t-projection and $\mathcal{H}_t \equiv 0$. The worked instance is Theorem 1(iv) with Appendix I: under Bayesian updating the per-investor state is the posterior, its own sufficient statistic, so the cross-section is Markov, the deadline flow Φ_t^{sw} and trigger propensity π are absent, and every forward object closes in S_t; an affine latent-state specification and a representative filtering economy are Markov in their own states by construction, with the same conclusion. □

Proposition 13 (The equilibrium short rate): *In the equilibrium of Proposition 1, the instantaneous risk-free rate is*

$$r_t = \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha(1+\alpha)\sigma^2 + \bar{\lambda}_t^S - U_t, \tag{E1}$$

with $\bar{\lambda}_t^S = \int_\Phi \lambda\, dS_t$ the wealth-weighted held intensity (Definition 3) and U_t the disaster-fear factor (14). The diffusive part $\delta + \alpha\mu_0 - \frac{1}{2}\alpha(1+\alpha)\sigma^2$ is the constant power-utility rate; the disaster part $\bar{\lambda}_t^S - U_t$ is belief-driven, and at correct beliefs ($\bar{\lambda}_t^S = \lambda^\star,\ U_t = U^{\mathrm{RA}}$) collapses to the rare-disaster precautionary term $-\lambda^\star(\mathbb{E}_f[e^{\alpha J}] - 1) < 0$.

Proof of Proposition 13. Take logs of $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^{\alpha}$ (10):

$$\log \xi_t = -\delta t - \alpha \log C_t + \alpha \log \Xi_t. \tag{E2}$$

By Assumption 1 (the jump multiplies C by e^{-x}, so $\Delta \log C = -x$),

$$d \log C_t = \bigl(\mu_0 - \tfrac{1}{2}\sigma^2\bigr) dt + \sigma\, dW_t - \int_{\mathcal{J}} x\, \mu(dt, dx), \tag{E3}$$

and by Proposition 10—the calm drift $d \log \Xi_t / dt = -\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^\star)$ (from (D4) divided by Ξ_t) and the disaster jump $\Xi_t / \Xi_{t-} = g_t(x)$ (D2), so $\Delta \log \Xi = \log g_t(x)$, with no diffusion since M^i carries none—

$$d \log \Xi_t = -\tfrac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^\star)\, dt + \int_{\mathcal{J}} \log g_t(x)\, \mu(dt, dx). \tag{E4}$$

Differentiating (E2) and substituting (E3)–(E4),

$$\begin{aligned} d \log \xi_t &= -\delta\, dt - \alpha\, d \log C_t + \alpha\, d \log \Xi_t \\ &= \bigl[-\delta - \alpha(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}_t^S - \lambda^\star)\bigr] dt - \alpha\sigma\, dW_t + \int_{\mathcal{J}} \bigl[\alpha x + \alpha \log g_t(x)\bigr] \mu(dt, dx), \end{aligned} \tag{E5}$$

the dt, dW, and μ terms collected from (E3)–(E4).

Now pass from $\log \xi$ to $\xi = e^{\log \xi}$ by Itô's formula for the exponential of a semimartingale. Write $L = \log \xi$; by (E5) its continuous martingale part L^c satisfies $dL_t^c = -\alpha\sigma\, dW_t$, so, with $d\langle W \rangle_t = dt$, its quadratic variation and the resulting Itô correction are

$$d\langle L^c \rangle_t = (-\alpha\sigma)^2\, d\langle W \rangle_t = \alpha^2 \sigma^2\, dt, \qquad \tfrac{1}{2}\, d\langle L^c \rangle_t = \tfrac{1}{2}\alpha^2\sigma^2\, dt, \tag{E6}$$

and its jump at a disaster of size x is $\Delta L_t = \alpha x + \alpha \log g_t(x)$ (the μ-integrand of (E5)). Itô's exponential formula then gives

$$\frac{d\xi_t}{\xi_{t-}} = dL_t^c + \bigl(\text{drift of } L\bigr) dt + \tfrac{1}{2}\, d\langle L^c \rangle_t + \int_{\mathcal{J}} \bigl(e^{\Delta L_t(x)} - 1\bigr)\, \mu(dt, dx), \tag{E7}$$

in which the jump factor evaluates as

$$e^{\Delta L_t} - 1 = e^{\alpha x + \alpha \log g_t(x)} - 1 = e^{\alpha x}\, e^{\alpha \log g_t(x)} - 1 = e^{\alpha x} g_t(x)^\alpha - 1$$

(consistent with (C10)). Collecting the dt terms of (E7)—the drift of L read from (E5) plus the correction (E6)—the continuous drift of $d\xi_t/\xi_t$ is therefore

$$\left(\frac{d\xi_t}{\xi_t}\right)^{\text{cont}}_{\text{drift}} = -\delta - \alpha(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}^S_t - \lambda^\star) + \tfrac{1}{2}\alpha^2\sigma^2. \tag{E8}$$

The mean jump rate is the jump factor integrated against the disaster compensator $\nu(dt, dx) = \lambda^\star\, dt\, f(x)\, dx$ (3):

$$\begin{aligned} \frac{1}{dt}\mathbb{E}_t\Big[\big(e^{\alpha x} g_t(x)^\alpha - 1\big)\,\mu\Big] &= \lambda^\star \int_{\mathcal{J}} \big(e^{\alpha x} g_t(x)^\alpha - 1\big) f(x)\, dx \\ &= \lambda^\star\, \mathbb{E}_f\big[e^{\alpha J} g_t(J)^\alpha\big] - \lambda^\star \qquad \text{(E9)} \\ &= U_t - \lambda^\star, \qquad \text{(E10)} \end{aligned}$$

line (E9) splitting the integral and (E10) using $U_t = \lambda^\star \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ (14). The safe rate is minus the total expected return on ξ, $r_t = -\frac{1}{dt}\mathbb{E}_t[d\xi_t/\xi_t]$, i.e. minus the sum of (E8) and (E10):

$$\begin{aligned} r_t &= \delta + \alpha(\mu_0 - \tfrac{1}{2}\sigma^2) - \tfrac{1}{2}\alpha^2\sigma^2 + (\bar{\lambda}^S_t - \lambda^\star) - (U_t - \lambda^\star) \\ &= \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha\sigma^2 - \tfrac{1}{2}\alpha^2\sigma^2 + \bar{\lambda}^S_t - U_t \qquad \text{(E11)} \\ &= \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha(1+\alpha)\sigma^2 + \bar{\lambda}^S_t - U_t, \qquad \text{(E12)} \end{aligned}$$

line (E11) expanding $\alpha(\mu_0 - \frac{1}{2}\sigma^2)$ and cancelling $\mp\lambda^\star$, and (E12) factoring $-\frac{1}{2}\alpha\sigma^2 - \frac{1}{2}\alpha^2\sigma^2 = -\frac{1}{2}\alpha(1+\alpha)\sigma^2$; this is (E1). At correct beliefs $\bar{\lambda}^S_t = \lambda^\star$ and $U_t = U^{\text{RA}} = \lambda^\star\mathbb{E}_f[e^{\alpha J}]$, so $\bar{\lambda}^S_t - U_t = -\lambda^\star(\mathbb{E}_f[e^{\alpha J}] - 1)$. □

Proposition 14 (The price–dividend ratio): *Let $v_t = P_t/C_t$ be the tree's price–dividend ratio,* *and let* Ψ_t *denote the* record-resolved pricing state*—the wealth-weighted distribution over per-investor states* $(\phi, \mathcal{R}, \tau, \varepsilon)$ *(held model, since-adoption record, regime age, tolerance), whose* ϕ*-marginal is the pricing state* S_t *(Definition 3). Then* $v_t = v(\Psi_t)$ *with*

$$v(\Psi_t) = \mathbb{E}_t\Big[\int_t^\infty e^{-\delta(s-t)}\Big(\tfrac{C_s}{C_t}\Big)^{1-\alpha}\Big(\tfrac{\Xi_s}{\Xi_t}\Big)^{\alpha}ds\Big]. \tag{E13}$$

The local expected growth rate m_t *of* $Y_t := C_t^{1-\alpha}\Xi_t^{\alpha}$ *is a function of the marginal* S_t *alone,*

$$m(S_t) = (1-\alpha)\mu_0 - \tfrac{1}{2}\alpha(1-\alpha)\sigma^2 - \big(\bar{\lambda}_t^S - \lambda^\star\big) + \lambda^\star\big(\mathbb{E}_f[e^{-(1-\alpha)J}g_t(J)^\alpha] - 1\big), \tag{E14}$$

but the valuation v *depends on the full* Ψ_t*: scale-freeness removes only the level* C_t*, and the conditional law of the future path* $\{S_s\}_{s\geq t}$*—hence of* $\{m(S_s)\}$ *and of* Ξ_s/Ξ_t*—is governed by the record-resolved dynamics of Proposition 3, which are Markov in the pair* (Ψ_t, ρ_t)*—the calm re-selection destination* $\phi_s^* = \mathfrak{m}(\rho_s)$ *being read from* ρ_s*—and* not *autonomous in* S_t*. Under the transversality condition*

$$\delta > \sup m := \sup\nolimits_S m(S) \tag{T}$$

—non-vacuous because each term of (E14) is a bounded functional of S_t *over the compact menu (*g_t *bounded,* $\mathcal{J}$ *compact), so* $-\infty < \inf_S m(S) \leq \sup_S m(S) < \infty$*—the integral (E13) converges,* $0 < v(\Psi_t) \leq (\delta - \sup m)^{-1} < \infty$*, and* v *is the unique bounded valuation. It is* not *semi-closed:* v *is a Feynman–Kac functional of the full record-resolved state* Ψ_t*, whose conditional law is Markov in* (Ψ_t, ρ_t) *and not autonomous in its marginal* S_t *(above), and even the instantaneous disaster tilt has no finite-dimensional sufficient statistic (Proposition 3); so* v *is not summarized by finitely many moments of* S_t*. In the decomposition of Proposition 5 the valuation splits as* $v(\Psi_t) = v_B(S_t) + \mathcal{H}_t^v(\Psi_t)$*: the cross-section part* $v_B(S_t) = \mathbb{E}[v \mid S_t]$ *is the valuation a Markov-in-cross-section benchmark assigns from the current beliefs, and the survival-history residual* $\mathcal{H}_t^v(\Psi_t)$ *is the discounted integral of the forward valuation-growth residual* $\mathbb{E}_t[m(S_{t+\tau}) \mid \Psi_t] - \mathbb{E}[m(S_{t+\tau}) \mid S_t]$*, driven by the same staggered-revision channels* $(\Phi_t^{\text{sw}}, \pi)$ *as the forward-*

fear residual of Theorem 1 and zero for any Markov benchmark. Capitalization is at the discount net of growth $\delta - m(S_t)$, not at a gross return. The comparative static is two-signed. In the intensity channel ($f_\theta \equiv f$, $g_t \equiv G_t = \mathbb{E}_{S_t}[(\lambda/\lambda^\star)^{1/\alpha}]$), (E14) reads

$$m(S_t) = const - (\bar{\lambda}_t^S - \lambda^\star) + \lambda^\star\big(\mathbb{E}_f[e^{-(1-\alpha)J}]\, G_t^\alpha - 1\big),$$

which carries the cross-section through two distinct *aggregates: the wealth-weighted* arithmetic *mean $\bar{\lambda}_t^S$ in the reallocation drift $-(\bar{\lambda}_t^S - \lambda^\star)$, and the* power *mean G_t^α in the priced-jump term, with $G_t^\alpha \le \bar{\lambda}_t^S/\lambda^\star$, strict by Jensen for $\alpha > 1$ unless λ is S_t-a.s. constant (26)—the two are* not *interchangeable. A more fearful cross-section raises both; whether the priced-jump (marginal-utility) term dominates the frequency drag—so that m* rises *and the constant-m ratio* expands*—is the quantitative condition above, met at our calibration; compression of* levered *equity would in any case require a leverage channel, absent here.*

Proof of Proposition 14. The valuation and its state. By (10), $\xi_u = e^{-\delta u} C_u^{-\alpha} \Xi_u^\alpha$, so

$$\begin{aligned}\frac{\xi_s C_s}{\xi_t C_t} &= \frac{e^{-\delta s} C_s^{-\alpha} \Xi_s^\alpha\, C_s}{e^{-\delta t} C_t^{-\alpha} \Xi_t^\alpha\, C_t} \\ &= e^{-\delta(s-t)} \Big(\frac{C_s}{C_t}\Big)^{1-\alpha} \Big(\frac{\Xi_s}{\Xi_t}\Big)^{\alpha}, \end{aligned} \tag{E15}$$

line (E15) using $C_s^{-\alpha} C_s = C_s^{1-\alpha}$ (likewise at t) and collecting the $e^{-\delta}$ and Ξ factors. The tree pays the dividend C_s, so by the pricing identity $P_t = \mathbb{E}_t[\int_t^\infty (\xi_s/\xi_t) C_s\, ds]$; inserting C_t/C_t and using (E15),

$$\begin{aligned} P_t &= C_t\, \mathbb{E}_t\Big[\int_t^\infty \frac{\xi_s C_s}{\xi_t C_t}\, ds\Big] \\ &= C_t\, \mathbb{E}_t\Big[\int_t^\infty e^{-\delta(s-t)} \Big(\tfrac{C_s}{C_t}\Big)^{1-\alpha} \Big(\tfrac{\Xi_s}{\Xi_t}\Big)^{\alpha} ds\Big]. \end{aligned}$$

Dividing by C_t gives $v_t = P_t/C_t =$ (E13). The integrand is homogeneous of degree zero in the level C_t, so v_t does not depend on C_t; it depends on the conditional law given $\mathcal{F}_t$ of the path $(C_s/C_t,\, \Xi_s/\Xi_t)_{s\geq t}$. The ratios C_s/C_t are functions of the primitive increments only (Assumption 1), independent of the cross-section. The ratios Ξ_s/Ξ_t are driven by the future cross-section path, whose conditional law is Markov in the record-resolved state Ψ_t and *not* autonomous in its ϕ-marginal S_t (Proposition 3: the switch flow depends on the within-class record distribution). Hence $v_t = v(\Psi_t)$.

The growth rate $m(S_t)$. Take logs of $Y_t = C_t^{1-\alpha}\Xi_t^{\alpha}$, $\log Y_t = (1-\alpha)\log C_t + \alpha \log \Xi_t$, and substitute (E3)–(E4):

$$\begin{aligned} d\log Y_t &= (1-\alpha)\, d\log C_t + \alpha\, d\log \Xi_t \\ &= \big[(1-\alpha)(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}_t^S - \lambda^\star)\big]\, dt + (1-\alpha)\sigma\, dW_t + \int_{\mathcal{J}} \big[-(1-\alpha)x + \alpha \log g_t(x)\big]\, \mu(dt, dx). \end{aligned} \tag{E16}$$

By Itô's formula for $Y = e^{\log Y}$, the continuous martingale part of (E16) is $(1-\alpha)\sigma\, dW_t$, with quadratic variation and Itô correction

$$\big((1-\alpha)\sigma\big)^2\, d\langle W\rangle_t = (1-\alpha)^2\sigma^2\, dt, \qquad \tfrac{1}{2}(1-\alpha)^2\sigma^2\, dt,$$

and the jump factor at a disaster of size x is $e^{-(1-\alpha)x + \alpha \log g_t(x)} - 1 = e^{-(1-\alpha)x} g_t(x)^\alpha - 1$; hence

$$\frac{dY_t}{Y_{t-}} = \Big[(1-\alpha)(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}_t^S - \lambda^\star) + \tfrac{1}{2}(1-\alpha)^2\sigma^2\Big] dt + (1-\alpha)\sigma\, dW_t + \int_{\mathcal{J}} \big(e^{-(1-\alpha)x} g_t(x)^\alpha - 1\big)\, \mu(dt, dx).$$

Take $\mathbb{E}_t[\cdot]/dt$ (conditioning on $\mathcal{F}_t$): the dW_t term has mean zero, and the random measure μ has compensator $\nu = \lambda^\star\, dt\, f\, dx$ (3), so

$$m(S_t) = \tfrac{1}{dt}\mathbb{E}_t\Big[\tfrac{dY_t}{Y_{t-}}\Big] = (1-\alpha)(\mu_0 - \tfrac{1}{2}\sigma^2) - (\bar{\lambda}_t^S - \lambda^\star) + \tfrac{1}{2}(1-\alpha)^2\sigma^2 + \lambda^\star \int_{\mathcal{J}} \big(e^{-(1-\alpha)x} g_t(x)^\alpha - 1\big) f(x)\, dx. \tag{E17}$$

The diffusion terms in (E17) collect as

$$(1-\alpha)(\mu_0 - \tfrac{1}{2}\sigma^2) + \tfrac{1}{2}(1-\alpha)^2\sigma^2 = (1-\alpha)\mu_0 + \tfrac{1}{2}\sigma^2\big[-(1-\alpha) + (1-\alpha)^2\big] \tag{E18}$$

$$= (1-\alpha)\mu_0 + \tfrac{1}{2}\sigma^2(1-\alpha)\big[(1-\alpha) - 1\big] \tag{E19}$$

$$= (1-\alpha)\mu_0 + \tfrac{1}{2}\sigma^2(1-\alpha)(-\alpha) \tag{E20}$$

$$= (1-\alpha)\mu_0 - \tfrac{1}{2}\alpha(1-\alpha)\sigma^2, \tag{E21}$$

(E18) factoring $\frac{1}{2}\sigma^2$ from the two variance terms, (E19) factoring $(1-\alpha)$ from the bracket, (E20) using $(1-\alpha) - 1 = -\alpha$, and (E21) multiplying out. The jump integral splits, using $\int_{\mathcal{J}} f\,dx = 1$, as

$$\begin{aligned}\lambda^\star \int_{\mathcal{J}} \big(e^{-(1-\alpha)x} g_t(x)^\alpha - 1\big) f(x)\,dx &= \lambda^\star \int_{\mathcal{J}} e^{-(1-\alpha)x} g_t(x)^\alpha f(x)\,dx - \lambda^\star \int_{\mathcal{J}} f(x)\,dx \\ &= \lambda^\star\, \mathbb{E}_f\big[e^{-(1-\alpha)J} g_t(J)^\alpha\big] - \lambda^\star \\ &= \lambda^\star\big(\mathbb{E}_f[e^{-(1-\alpha)J} g_t(J)^\alpha] - 1\big).\end{aligned} \tag{E22}$$

Substituting (E21) and (E22) into (E17),

$$m(S_t) = (1-\alpha)\mu_0 - \tfrac{1}{2}\alpha(1-\alpha)\sigma^2 - (\bar{\lambda}_t^S - \lambda^\star) + \lambda^\star\big(\mathbb{E}_f[e^{-(1-\alpha)J} g_t(J)^\alpha] - 1\big), \tag{E23}$$

which depends on the cross-section only through $\bar{\lambda}_t^S$ and g_t, i.e. through S_t; this is (E14).

Convergence, bounds, uniqueness. Set $Z_s = e^{-\sup m\,(s-t)} Y_s/Y_t$. Its local drift rate is $(m(S_s) - \sup m) Z_s \leq 0$ (by the definition of $\sup m$), and its jumps have finite mean (g_t is bounded by $\bar{r}^{1/\alpha}$ (12) and $\mathcal{J}$ is compact), so Z is a supermartingale and $\mathbb{E}_t[Y_s/Y_t] \leq e^{\sup m\,(s-t)}$. Therefore, under (T),

$$v(\Psi_t) \leq \int_t^\infty e^{-\delta(s-t)} e^{\sup m\,(s-t)}\,ds = \int_0^\infty e^{-(\delta - \sup m)u}\,du = \frac{1}{\delta - \sup m} < \infty, \tag{E24}$$

and $v > 0$ since the integrand is positive. *The lower bound is uniform.* Set $\inf m := \inf_S m(S) > -\infty$, finite because the jump integrand in (E23) is bounded ($g_t \in [\underline{r}^{1/\alpha}, \overline{r}^{1/\alpha}]$ by (12) and $\mathcal{J}$ is compact). Then $\widetilde{Z}_s = e^{-\inf m\,(s-t)} Y_s/Y_t$ has local drift $(m(S_s) - \inf m)\widetilde{Z}_s \geq 0$, a submartingale, so $\mathbb{E}_t[Y_s/Y_t] \geq e^{\inf m\,(s-t)}$ and

$$v(\Psi_t) \geq \int_t^{\infty} e^{-\delta(s-t)} e^{\inf m\,(s-t)}\,ds = \frac{1}{\delta - \inf m} > 0, \tag{E25}$$

a positive lower bound independent of Ψ_t (it uses only $\inf m$ and δ). With (E24) this makes $\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$ bounded above and bounded below away from zero—the bound used in Proposition 15 and the excess-volatility result of the Internet Appendix. *Uniqueness among bounded valuations.* If v_1, v_2 are bounded solutions of the pricing identity, then for every horizon $T > t$ their difference $D := v_1 - v_2$ satisfies $D(\Psi_t) = e^{-\delta(T-t)}\mathbb{E}_t[(Y_T/Y_t)\,D(\Psi_T)]$; bounding $|D| \leq \|D\|_\infty$ and using $\mathbb{E}_t[Y_T/Y_t] \leq e^{\sup m\,(T-t)}$ (the supermartingale bound above),

$$|D(\Psi_t)| \leq \|D\|_\infty\, e^{-(\delta - \sup m)(T-t)} \xrightarrow[T\to\infty]{} 0 \quad \text{under (T)},$$

so $D \equiv 0$ and the bounded valuation is unique. That the bounded valuation depends on the full record-resolved state Ψ_t, not on finitely many moments of S_t, has two sources: the instantaneous disaster tilt is already infinite-dimensional in S_t (Proposition 3), and the law of motion is non-autonomous—the switch flow and revision deadlines are functionals of Ψ_t (Propositions 12, 3)—so the forward expectation cannot collapse to a function of S_t alone. $\square$

Corollary 3 (Finite menu: instantaneous reduction, valuation on the record-resolved state): *Under the finite menu of Corollary 2, the instantaneous growth rate is a function of the finite share vector, $m_t = m(S_t)$ with $S_t \in \Delta^{K-1}$, and the price–dividend ratio solves the Feynman–Kac equation*

$$[\,\delta - m(S_t)\,]\,v(\Psi_t) = 1 + (\mathcal{A}_\Psi v)(\Psi_t), \tag{E26}$$

with $\mathcal{A}_\Psi$ the pricing *generator of the record-resolved state Ψ_t—the state generator of Proposi-*

tion 3 with the disaster term reweighted by the payoff-growth *jump of $Y_t = C_t^{1-\alpha}\Xi_t^\alpha$, namely $e^{-(1-\alpha)x}g_t(x)^\alpha$ (not the state-price-density jump $e^{\alpha x}g_t(x)^\alpha$), so that it carries the Y-weighted disaster-jump contribution $\lambda^\star\mathbb{E}_f[e^{-(1-\alpha)J}g_t(J)^\alpha\,(v(\Psi_t^J) - v(\Psi_{t-}))]$ rather than the bare state jump (proof in Appendix E). Even with K finite, Ψ_t is infinite-dimensional—the since-adoption records and ages are continuous—so (E26) does* not *reduce to a finite-dimensional-state equation in $S \in \Delta^{K-1}$ unless the cross-section is Markov in the shares alone, i.e. the revision/switch flow is autonomous in S. The lazy-revision rule violates this: the switch flow depends on the within-class record distribution (Proposition 12). The linear Feynman–Kac operator equation $[\delta - m(S)]v(S) = 1 + (\mathcal{A}_S v)(S)$ on the continuous simplex Δ^{K-1} holds only in the selection-only specialization with no record-dependent switching, where Ψ_t collapses to S_t; it is an operator equation on a continuous state space, not a finite system of algebraic equations—for $K = 2$, a scalar-state equation for a function of S^{hi} over a continuum.*

Proof of Corollary 3. Under the finite menu (Corollary 2), $\bar{\lambda}_t^S$ and g_t are functions of the share vector (eqs. (D18)–(D19)), so $m_t = m(S_t)$ with $S_t \in \Delta^{K-1}$. Because ξ prices the consumption claim, $e^{-\delta t}Y_t\,v(\Psi_t) + \int_0^t e^{-\delta s}Y_s\,ds$ is a martingale, where $Y_t = C_t^{1-\alpha}\Xi_t^\alpha$ has local growth $m(S_t)$ (E14). Setting the drift of $e^{-\delta t}Y_t\,v(\Psi_t)$ plus the running term $e^{-\delta t}Y_t\,dt$ to zero and dividing by $e^{-\delta t}Y_t$,

$$-\delta\,v(\Psi_t) + m(S_t)\,v(\Psi_t) + (\mathcal{A}_\Psi v)(\Psi_t) + 1 = 0.$$

Here $\mathcal{A}_\Psi$ is the *pricing* (Y-weighted) generator of Ψ_t. Writing $\mathcal{A}_\Psi = \mathcal{A}_\Psi^{\mathrm{c}} + \mathcal{A}_\Psi^{\mathrm{dis}}$ with $\mathcal{A}_\Psi^{\mathrm{c}}$ the finite-variation selection/switch drift and the disaster part

$$\mathcal{A}_\Psi^{\mathrm{dis}}v = \lambda^\star\int_{\mathcal{J}} e^{-(1-\alpha)x}g_t(x)^\alpha\big(v(\Psi_t^x) - v(\Psi_{t-})\big)\,f(x)\,dx,$$

the kernel jump $Y_t/Y_{t-} = e^{-(1-\alpha)x}g_t(x)^\alpha$ weights the post-jump valuation, so the simultaneous Y–$v(\Psi)$ disaster covariation in the Itô product rule is carried *inside* $\mathcal{A}_\Psi$. Equivalently, relative to the physical state generator $\mathcal{A}_\Psi^P v = \mathcal{A}_\Psi^{\mathrm{c}}v + \lambda^\star\int_{\mathcal{J}}(v(\Psi_t^x) - v(\Psi_{t-}))f\,dx$, the pricing generator adds the covariation correction $\lambda^\star \int_{\mathcal{J}} (e^{-(1-\alpha)x} g_t(x)^\alpha - 1)(v(\Psi_t^x) - v(\Psi_{t-})) f\, dx$. Multiplying through by -1 and collecting the two terms in $v(\Psi_t)$,

$$\delta\, v(\Psi_t) - m(S_t)\, v(\Psi_t) = 1 + (\mathcal{A}_\Psi v)(\Psi_t), \qquad \text{i.e.} \qquad [\,\delta - m(S_t)\,]\, v(\Psi_t) = 1 + (\mathcal{A}_\Psi v)(\Psi_t),$$

which is the Feynman–Kac equation (E26). The generator $\mathcal{A}_\Psi$ acts on functions of Ψ_t: the selection drift and the disaster reweighting are autonomous in S_t, but the switch term carries the within-class record distribution (Proposition 3(ii) through the hazard of Proposition 12), so $(\mathcal{A}_\Psi v)$ is in general not a function of S alone; moreover Ψ_t—with its continuous since-adoption records and ages—is infinite-dimensional even for finite K. If the switch flow is suppressed (selection-only), Ψ_t collapses to $S_t \in \Delta^{K-1}$, $\mathcal{A}_\Psi = \mathcal{A}_S$ is the finite-dimensional share generator read off the selection drift and disaster reweighting of Proposition 3, and (E26) becomes a finite-dimensional-state operator equation on the simplex, $[\delta - m(S)]v(S) = 1 + (\mathcal{A}_S v)(S)$ for $S \in \Delta^{K-1}$—an integro-differential equation in the $K-1$ share coordinates, not a finite linear system, since v remains a function on Δ^{K-1}—solvable under (T); for $K = 2$ a scalar integro-differential equation in S^{hi}. □

Corollary 4 (First-order belief-heterogeneity expansion around the homogeneous economy): *Consider the homogeneous benchmark in which all investors hold the common model* $\phi_0 = (\lambda_0, \theta_0)$*—because at* t *they share the common held model* ϕ_0*, either as the common posterior mode under Bayesian updating from a shared prior (no tolerance threshold) or under a shared tolerance with synchronized switching (one threshold)—so the instantaneous cross-section is the point mass* $S_t = \delta_{\phi_0}$ *and* Ψ_t *is degenerate. Perturb it by a small wealth-weighted dispersion,* $S_t = \delta_{\phi_0} + \varepsilon\, \eta$*, with* η *an* admissible dispersion*—*$\eta = \nu - \mu\, \delta_{\phi_0}$ *for a finite positive measure* ν *on* Φ *of mass* $\mu = \nu(\Phi)$ *(weight shifted off the benchmark)—so* $\int_\Phi \eta = 0$ *and* $S_t = (1-\varepsilon\mu)\, \delta_{\phi_0} + \varepsilon\, \nu$ *is a probability measure for* $0 \le \varepsilon \le 1/\mu$*. Write* $\langle \psi \rangle_\eta := \int_\Phi \psi\, d\eta$*,* $g_0(x) := r(\phi_0, x)^{1/\alpha}$*, and*

$g_1(x) := \langle r(\cdot,x)^{1/\alpha}\rangle_\eta$. *Then, to first order in* ε,

$$g_t(x) = g_0(x) + \varepsilon\, g_1(x) \qquad \text{(exact)}, \tag{E27}$$

$$\bar\lambda_t^S = \lambda_0 + \varepsilon\,\langle\lambda\rangle_\eta \qquad \text{(exact)}, \tag{E28}$$

$$U_t = U_0 + \varepsilon\, U_1 + O(\varepsilon^2), \qquad U_0 = \lambda_0 \mathbb{E}_{\theta_0}[e^{\alpha J}], \quad U_1 = \alpha\lambda^\star \mathbb{E}_f\big[e^{\alpha J} g_0^{\alpha-1} g_1\big],$$

$$r_t = r_0 + \varepsilon\,(\langle\lambda\rangle_\eta - U_1) + O(\varepsilon^2), \qquad r_0 = \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha(1+\alpha)\sigma^2 + \lambda_0 - U_0, \tag{E29}$$

where $\mathbb{E}_{\theta_0}$ *is under* f_{θ_0}. *The price–dividend ratio satisfies* $v_t = v_0 + O(\varepsilon)$ *with the homogeneous value* $v_0 = (\delta - m_0)^{-1}$ *and* $m_0 = (1-\alpha)\mu_0 - \frac{1}{2}\alpha(1-\alpha)\sigma^2 - (\lambda_0 - \lambda^\star) + \lambda^\star(\mathbb{E}_f[e^{-(1-\alpha)J} g_0^\alpha] - 1)$. *The zeroth order is the representative-agent rare-disaster economy under belief* ϕ_0*; the first-order terms are the leading correction from belief heterogeneity, linear in the dispersion* η.

Proof of Corollary 4. Instantaneous objects. The aggregator g_t and the mean intensity $\bar\lambda_t^S$ are linear in S_t (Definitions 2, 3). Substituting $S_t = \delta_{\phi_0} + \varepsilon\eta$,

$$\begin{aligned} g_t(x) &= \int_\Phi r(\phi,x)^{1/\alpha}\big(\delta_{\phi_0} + \varepsilon\eta\big)(d\phi) \\ &= r(\phi_0,x)^{1/\alpha} + \varepsilon\int_\Phi r(\phi,x)^{1/\alpha}\eta(d\phi) = g_0(x) + \varepsilon\, g_1(x), \\ \bar\lambda_t^S &= \int_\Phi \lambda\big(\delta_{\phi_0} + \varepsilon\eta\big)(d\phi) = \lambda_0 + \varepsilon\,\langle\lambda\rangle_\eta, \end{aligned} \tag{E30}$$

both exact (the functionals are linear, and δ_{ϕ_0} evaluates at ϕ_0); these are (E27)–(E28). For $U_t = \lambda^\star \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ (14), expand the power by Taylor's theorem ($g_0 > 0$ and continuous on the compact $\mathcal{J}$, g_1 bounded), uniformly in x:

$$\begin{aligned} g_t(x)^\alpha &= \big(g_0(x) + \varepsilon g_1(x)\big)^\alpha \\ &= g_0(x)^\alpha + \alpha\, g_0(x)^{\alpha-1}\,\varepsilon\, g_1(x) + O(\varepsilon^2). \end{aligned}$$

Integrating against $\lambda^\star e^{\alpha x} f(x)\,dx$,

$$
\begin{aligned}
U_t &= \lambda^\star \mathbb{E}_f[e^{\alpha J} g_0^\alpha] + \varepsilon\,\alpha\lambda^\star \mathbb{E}_f[e^{\alpha J} g_0^{\alpha-1} g_1] + O(\varepsilon^2) \\
&= U_0 + \varepsilon\, U_1 + O(\varepsilon^2),
\end{aligned} \tag{E31}
$$

and the zeroth-order term simplifies with $g_0^\alpha = r(\phi_0,\cdot)$ and $r(\phi_0, x) = \lambda_0 f_{\theta_0}(x)/(\lambda^\star f(x))$:

$$
\begin{aligned}
U_0 = \lambda^\star \mathbb{E}_f[e^{\alpha J} r(\phi_0, J)] &= \lambda^\star \int_{\mathcal{J}} e^{\alpha x} \frac{\lambda_0 f_{\theta_0}(x)}{\lambda^\star f(x)} f(x)\,dx \\
&= \lambda_0 \int_{\mathcal{J}} e^{\alpha x} f_{\theta_0}(x)\,dx = \lambda_0 \mathbb{E}_{\theta_0}[e^{\alpha J}].
\end{aligned}
$$

Substituting (E30) and (E31) into the short rate (E1),

$$
\begin{aligned}
r_t &= \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha(1+\alpha)\sigma^2 + \big(\lambda_0 + \varepsilon\langle\lambda\rangle_\eta\big) - \big(U_0 + \varepsilon U_1\big) + O(\varepsilon^2) \\
&= r_0 + \varepsilon\big(\langle\lambda\rangle_\eta - U_1\big) + O(\varepsilon^2),
\end{aligned}
$$

with $r_0 = \delta + \alpha\mu_0 - \frac{1}{2}\alpha(1+\alpha)\sigma^2 + \lambda_0 - U_0$; this is (E29).

Valuation. At the frozen homogeneous belief ($\varepsilon = 0$, ϕ_0 held fixed) $g_t = g_0$ is constant, so by (E14) the growth rate is the constant $m_0 = (1-\alpha)\mu_0 - \frac{1}{2}\alpha(1-\alpha)\sigma^2 - (\lambda_0 - \lambda^\star) + \lambda^\star(\mathbb{E}_f[e^{-(1-\alpha)J} g_0^\alpha] - 1)$. Because the primitive increments are i.i.d. (Assumption 1) and the jump factor $e^{-(1-\alpha)x} g_0(x)^\alpha$ is time-invariant and independent of Y—a fixed function of the size x, not a constant in x—its compensator contribution $\lambda^\star(\mathbb{E}_f[e^{-(1-\alpha)J} g_0^\alpha] - 1)$ is constant, so the increment dY_s/Y_s has conditional mean $m_0\,ds$ independent of Y_s/Y_t, so $\mathbb{E}_t[Y_s/Y_t]$ (conditioning on $\mathcal{F}_t$) solves

$$
\frac{d}{ds}\mathbb{E}_t[Y_s/Y_t] = m_0\,\mathbb{E}_t[Y_s/Y_t], \qquad \mathbb{E}_t[Y_t/Y_t] = 1,
$$

whose solution is

$$
\mathbb{E}_t[Y_s/Y_t] = e^{m_0(s-t)}. \tag{E32}
$$

Since $(C_s/C_t)^{1-\alpha}(\Xi_s/\Xi_t)^{\alpha} = Y_s/Y_t \geq 0$, Tonelli lets $\mathbb{E}_t$ pass inside the ds-integral of (E13), and

$$\begin{aligned}
v_0 &= \int_t^{\infty} e^{-\delta(s-t)}\, \mathbb{E}_t[Y_s/Y_t]\, ds \\
&= \int_t^{\infty} e^{-\delta(s-t)} e^{m_0(s-t)}\, ds \qquad \text{(E33)} \\
&= \int_0^{\infty} e^{-(\delta-m_0)u}\, du \qquad \text{(E34)} \\
&= \frac{1}{\delta - m_0}, \qquad \text{(E35)}
\end{aligned}$$

line (E33) by (E32), (E34) substituting $u = s - t$ and combining $e^{-\delta u}e^{m_0 u} = e^{-(\delta-m_0)u}$, and (E35) evaluating the integral, which converges because $\delta - m_0 > 0$ ($m_0 \leq \sup m < \delta$ under (T)). For $\varepsilon > 0$ the perturbed path of the *full* record-resolved state $\Psi_t(\varepsilon)$ is an ε-perturbation of the benchmark, and all three of its drivers are stable in ε: the selection drift (18) and the disaster reweighting are Lipschitz in the cross-section (Proposition 3), g_t enters through the bounded factor (12), and—the piece the marginal S_t alone does not control—the record-dependent *switch flow* carries only the $O(\varepsilon)$ mass that the perturbation displaces off the benchmark cohort. However the relocation *times* of that mass differ between the perturbed and benchmark economies, it is the same $O(\varepsilon)$ total mass that relocates, so its contribution to $\|\Psi_t(\varepsilon) - \Psi_t(0)\|_{\mathrm{TV}}$ is $O(\varepsilon)$ at every t, with no continuity of the calm-deadline map in the tolerance required (the count atoms make $\Delta^\star(\phi, \cdot)$ a step function of the tolerance, but the displaced mass is bounded whatever its switch times). With all three drivers thus $O(\varepsilon)$-stable, the total-variation rate is $\|\Psi_t(\varepsilon) - \Psi_t(0)\|_{\mathrm{TV}} = O(\varepsilon)$ uniformly on compact time intervals. The valuation is in turn Lipschitz in the state in total variation. Two initial states within $\|\Psi - \Psi'\|_{\mathrm{TV}}$ admit a coupling of the two economies sharing all but an $O(\|\Psi - \Psi'\|_{\mathrm{TV}})$ fraction of investors—identical primitives drive identical trajectories on the shared part—so the induced laws of the forward payoff path $\{(C_s/C_t)^{1-\alpha}(\Xi_s/\Xi_t)^{\alpha}\}_{s\geq t}$ differ in total variation by $O(\|\Psi - \Psi'\|_{\mathrm{TV}})$; with the integrand of (E13) bounded by the two-sided bound (E24)–(E25), this transfers to $|v(\Psi) - v(\Psi')| \leq L\, \|\Psi - \Psi'\|_{\mathrm{TV}}$ for a finite L. Composing the two Lipschitz transfers—with (E24) dominating the time integrand for dominated convergence—gives $v_t = v_0 + O(\varepsilon)$. The $O(\varepsilon)$ correction solves the Feynman–Kac equation (E26) linearized at $\varepsilon = 0$, which is not closed form in general (Proposition 3). □

Proposition 15 (Equity premium and disaster premium): *In the equilibrium of Proposition 1, the tree's instantaneous expected excess return decomposes into a diffusive and a disaster part,*

$$\mathbb{E}_t[dR_t^e] - r_t\,dt = \underbrace{\alpha\sigma^2}_{\textit{diffusive}}\,dt + \mathrm{DRP}_t\,dt, \tag{E36}$$

where, with the tree's disaster return $\Delta R_t^e(x) = e^{-x}\psi_t(x) - 1$ and $\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$ the price–dividend jump ratio (the record-resolved state moving from Ψ_{t-} to its post-disaster value Ψ_t^x), the disaster risk premium is the gap between the physical and risk-neutral expected disaster returns,

$$\mathrm{DRP}_t = \lambda^\star\,\mathbb{E}_f\big[\Delta R_t^e(J)\big] - U_t\,\mathbb{E}_{f_t^{\mathbb{Q}}}\big[\Delta R_t^e(J)\big]. \tag{E37}$$

The diffusive premium $\alpha\sigma^2$ is constant. The disaster premium is priced by the risk-neutral disaster law $(U_t, f_t^{\mathbb{Q}})$—the fear factor of Definition 2—so, holding $(\alpha, \lambda^\star, F_J)$ fixed, its belief-driven variation runs through $(U_t, f_t^{\mathbb{Q}})$. In the intensity channel (shape fixed) the U_t-loading component of this variation runs through U_t alone (Proposition 2); in the general joint model the premium also moves with the shape $f_t^{\mathbb{Q}}$ and the state-dependent return jump. Even in the intensity channel, though, the premium's level *and part of its variation run through the jump ratio ψ_t, hence through $v(\Psi_t)$ (Proposition 14), which varies with the full cross-section and is not summarized by U_t. In the decomposition of Proposition 5,* $\mathrm{DRP}_t = B_t^{\mathrm{DRP}}(S_t) + \mathcal{H}_t^{\mathrm{DRP}}(\Psi_t)$*: the cross-section part carries the S_t-measurable fear-factor intensity, and the survival-history residual enters through the price–dividend jump ratio ψ_t and $v(\Psi_t)$ (Proposition 14), zero for any Markov benchmark.*

Proof of Proposition 15. The no-arbitrage relation. The consumption claim priced by ξ has gain process $\xi_t P_t + \int_0^t \xi_s C_s\,ds$, a $\mathbb{P}$-martingale (Proposition 1). Conditioning on $\mathcal{F}_t$, its drift is zero:

$$\mathbb{E}_t[d(\xi_t P_t)] + \xi_t C_t\,dt = 0. \tag{E38}$$

By Itô's product rule for semimartingales,

$$d(\xi_t P_t) = \xi_{t-}\,dP_t + P_{t-}\,d\xi_t + d[\xi, P]_t. \tag{E39}$$

Divide (E39) by $\xi_{t-}P_{t-}$:

$$\frac{d(\xi_t P_t)}{\xi_{t-}P_{t-}} = \frac{dP_t}{P_{t-}} + \frac{d\xi_t}{\xi_{t-}} + \frac{d[\xi, P]_t}{\xi_{t-}P_{t-}}. \tag{E40}$$

The cum-dividend return is $dR^e_t = dP_t/P_{t-} + (C_t/P_t)\,dt$, so $dP_t/P_{t-} = dR^e_t - (C_t/P_t)\,dt$; substitute this into (E40), take $\mathbb{E}_t[\cdot]$, and use (E38) divided by the left limit $\xi_{t-}P_{t-}$ (i.e. $\mathbb{E}_t[d(\xi_t P_t)/(\xi_{t-}P_{t-})] = -(C_t/P_{t-})\,dt$ to first order, the price jump entering the martingale part, not the drift):

$$\mathbb{E}_t[dR^e_t] - \frac{C_t}{P_{t-}}dt + \mathbb{E}_t\Big[\frac{d\xi_t}{\xi_{t-}}\Big] + \mathbb{E}_t\Big[\frac{d[\xi, P]_t}{\xi_{t-}P_{t-}}\Big] = -\frac{C_t}{P_{t-}}dt.$$

Cancel $-(C_t/P_t)\,dt$ from both sides, then use $\mathbb{E}_t[d\xi_t/\xi_t] = -r_t\,dt$ (Proposition 13) and $\mathbb{E}_t[d[\xi, P]_t/(\xi_{t-}P_{t-})] = \mathrm{Cov}_t(d\xi_t/\xi_t, dR^e_t)$ (the predictable covariation, the $(C_t/P_t)dt$ yield term being finite-variation and contributing none):

$$\mathbb{E}_t[dR^e_t] - r_t\,dt = -\,\mathrm{Cov}_t\Big(\frac{d\xi_t}{\xi_t},\, dR^e_t\Big).$$

Predictability and the jump structure. Placing the price jump in the martingale part of (E38) requires the jumps of P to be *inaccessible*, and they are. Between disasters the belief densities M^i are continuous—they jump only at the points of the disaster measure N, by (C1)—so the kernel ξ and, by Lemma 7, the price P are continuous across every predictable revision time, including a synchronized calm deadline at which a positive G-mass re-selects on a flat level set. There the share-weighted distribution S_t carries a predictable finite-variation jump, but the *traded* price does not: the re-selecting cohort relocates to the continuous public destination $\phi^*_t = \mathfrak{m}(\rho_t)$, an anticipated move that leaves the forward-looking value unchanged (Lemma 7(iii)).

The only jumps in P are at disasters, totally inaccessible stopping times of the Poisson filtration, so (E38) holds without any atomlessness restriction on the deadline law.

The two covariation channels. The martingale part of $d\xi_t/\xi_t$ is $-\alpha\sigma\, dW_t$ plus the compensated jump $\int_{\mathcal{J}}(e^{\alpha x} g_t(x)^\alpha - 1)(\mu - \nu)(dt, dx)$ ((E5), (C10)); the martingale part of dR_t^e is $\sigma\, dW_t$ (the record-resolved state Ψ_t has no Brownian part by Proposition 3, so the deterministic $v(\Psi_t)$ is diffusion-free and the sole Brownian source is $\sigma\, dW_t$ from C_t) plus the compensated jump $\int_{\mathcal{J}} \Delta R_t^e(x)(\mu - \nu)(dt, dx)$ with $\Delta R_t^e(x) = e^{-x}\psi_t(x) - 1$. The Brownian and Poisson sources are independent, so the covariation splits into a diffusion and a jump part. The diffusion part, using $d\langle W\rangle_t = dt$,

$$\mathrm{Cov}_t(-\alpha\sigma\, dW_t,\ \sigma\, dW_t) = -\alpha\sigma \cdot \sigma\, d\langle W\rangle_t = -\alpha\sigma^2\, dt.$$

For the jump part, the predictable covariation of two compensated integrals $\int H(\mu - \nu)$ and $\int K(\mu - \nu)$ against the same random measure is $\int HK\,\nu$, valid for predictable H, K with $\int H^2\nu, \int K^2\nu < \infty$ (Brémaud, 1981); here $H = e^{\alpha x} g_t^\alpha - 1$ and $K = \Delta R_t^e$ are bounded (g_t bounded by (12), ψ_t bounded since v is bounded above and below away from zero under (T), $\mathcal{J}$ compact), so with $\nu = \lambda^\star\, dt\, f\, dx$,

$$\mathrm{Cov}_t^{\mathrm{jump}} = \lambda^\star \int_{\mathcal{J}} (e^{\alpha x} g_t(x)^\alpha - 1)\, \Delta R_t^e(x)\, f(x)\, dx\, dt.$$

Hence the premium is $-\,\mathrm{Cov}_t = -(\text{diffusion}) - (\text{jump})$:

$$\frac{\mathbb{E}_t[dR_t^e]}{dt} - r_t = \alpha\sigma^2 - \lambda^\star \int_{\mathcal{J}} (e^{\alpha x} g_t(x)^\alpha - 1)\, \Delta R_t^e(x)\, f(x)\, dx. \tag{E41}$$

Split the integral over the factor $(e^{\alpha x} g_t^\alpha - 1)$:

$$-\lambda^\star \int_{\mathcal{J}} (e^{\alpha x} g_t^\alpha - 1)\Delta R_t^e f\, dx = \lambda^\star \int_{\mathcal{J}} \Delta R_t^e f\, dx - \lambda^\star \int_{\mathcal{J}} e^{\alpha x} g_t^\alpha \Delta R_t^e f\, dx$$
$$= \lambda^\star\, \mathbb{E}_f[\Delta R_t^e] - U_t\, \mathbb{E}_{f_t^{\mathbb{Q}}}[\Delta R_t^e], \tag{E42}$$

line (E42) using $\int \Delta R_t^e f\, dx = \mathbb{E}_f[\Delta R_t^e]$ and the risk-neutral disaster compensator $\lambda^\star e^{\alpha x} g_t(x)^\alpha f(x) = \nu_t^{\mathbb{Q}}(dx)/dt = U_t f_t^{\mathbb{Q}}(x)$ (14), so that $\lambda^\star \int e^{\alpha x} g_t^\alpha \Delta R_t^e f\, dx = U_t \int \Delta R_t^e f_t^{\mathbb{Q}}\, dx = U_t \mathbb{E}_{f_t^{\mathbb{Q}}}[\Delta R_t^e]$. Substituting (E42) into (E41) gives the diffusive term $\alpha\sigma^2$ and $\mathrm{DRP}_t = \lambda^\star \mathbb{E}_f[\Delta R_t^e] - U_t \mathbb{E}_{f_t^{\mathbb{Q}}}[\Delta R_t^e]$, which are (E36)–(E37). □

Proposition 16 (State reduction to the wealth-weighted distribution): *In the equilibrium of Proposition 1, every* instantaneous *equilibrium price quantity is a measurable functional of the pricing state S_t (Definition 3) alone: the belief aggregator $g_t(\cdot)$ (21), the kernel calm drift and disaster jump ratio (D1)–(D2), the fear factor U_t and multiplier κ_t and the risk-neutral disaster law $f_t^{\mathbb{Q}}$ (14)–(15), the short rate r_t (E1), the local valuation growth $m(S_t)$ (E14), and the diffusive premium $\alpha\sigma^2$ (E36). Equivalently, any two record-resolved states Ψ_t, Ψ_t' (Definition 4) with a common held-model marginal $\pi_\phi \Psi_t = \pi_\phi \Psi_t' = S_t$ yield identical values of all these quantities at t: the since-adoption records, regime ages, surprise thresholds, and second-order posteriors enter instantaneous prices only through S_t. These prices factor through the aggregator $g_t(\cdot)$, itself a functional of S_t coarser than the full measure; what fails is any* finite-dimensional *reduction—by Proposition 3 no finite collection of linear moments of S_t determines $g_t(\cdot)$. The forward-looking objects do* not *reduce to S_t: the price–dividend ratio $v(\Psi_t)$ (E13), the price–dividend jump ratio ψ_t, and hence the* level *of the disaster premium DRP_t (E37) are functionals of the full record-resolved state Ψ_t (Proposition 14).*

Proof of Proposition 16. Each instantaneous quantity is, by its defining display, a functional of S_t.

(a) By (21), $g_t(x) = \int_\Phi (\lambda/\lambda^\star)^{1/\alpha} \big(f_\theta(x)/f(x)\big)^{1/\alpha}\, S_t(d\phi)$ is the integral of a fixed (t,Ψ)-independent kernel against S_t; so the whole function $g_t(\cdot)$ depends on Ψ_t only through $S_t = \pi_\phi \Psi_t$.

(b) By Definition 3, $\bar{\lambda}_t^S = \int_\Phi \lambda \, S_t(d\phi)$. Hence the calm drift (D1), $-\frac{1}{\alpha}(\bar{\lambda}_t^S - \lambda^\star)$, and the disaster jump ratio (D2), $g_t(x)$, are functionals of S_t by (a).

(c) By (14)–(15), $U_t = \lambda^\star \, \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$, $\kappa_t = U_t/U^{\mathrm{RA}}$, and $f_t^{\mathbb{Q}}(x) = e^{\alpha x} g_t(x)^\alpha f(x)/\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ are fixed functionals of $g_t(\cdot)$, hence of S_t by (a).

(d) By (E1), $r_t = \delta + \alpha\mu_0 - \frac{1}{2}\alpha(1+\alpha)\sigma^2 + \bar{\lambda}_t^S - U_t$, and by (E14) $m(S_t)$, are functionals of $\bar{\lambda}_t^S$ and $g_t(\cdot)$ by (b)–(c); the diffusive premium $\alpha\sigma^2$ (E36) is constant.

The displays in (a)–(d) involve Ψ_t only through S_t; the coordinates $(\mathcal{R}, \tau, \varepsilon)$ of Definition 4 appear in none of them. Therefore $\pi_\phi \Psi_t = \pi_\phi \Psi_t'$ forces equality of every quantity in (a)–(d), the stated equivalence. That the instantaneous prices admit no *finite-dimensional* reduction—though they factor through the coarser aggregator $g_t(\cdot)$—is Proposition 3: the map $S \mapsto g_t(\cdot)$ in (a) is into an infinite-dimensional image (by the spanning condition, the family $\{(f_\theta/f)^{1/\alpha}\}$ is linearly independent), so no finite collection of linear moments of S_t is sufficient. For the forward-looking objects, $v(\Psi_t) = \mathbb{E}_t[\int_t^\infty e^{-\delta(s-t)}(C_s/C_t)^{1-\alpha}(\Xi_s/\Xi_t)^\alpha ds]$ (E13) is a conditional expectation of a functional of the future path $\{S_s\}_{s \geq t}$, whose conditional law is governed by the record-resolved generator of Proposition 3, Markov in Ψ_t and not autonomous in S_t (Proposition 14); two states with equal S_t but different Ψ_t have different future S-path laws, hence different v. The jump ratio $\psi_t = v(\Psi_t^J)/v(\Psi_{t-})$ and the premium level (E37) inherit this dependence. □

B. Dynamics around a disaster

Proof of Proposition 6. (i) Calm erosion. Between disasters the calm derivative is the sum of the selection drift (18) and the calm switch flow (19) (Proposition 3(i)–(ii)). The selection part gives, for bounded h,

$$\frac{d}{dt}\langle h, S_t\rangle = \tfrac{1}{\alpha}\int_\Phi (\bar\lambda_t^S - \lambda)\, h(\phi)\, S_t(d\phi) \tag{E43}$$

$$= \tfrac{1}{\alpha}\Big[\bar\lambda_t^S \int_\Phi h\, dS_t - \int_\Phi \lambda\, h\, dS_t\Big] \tag{E44}$$

$$= \tfrac{1}{\alpha}\big[\mathbb{E}_{S_t}[\lambda]\,\mathbb{E}_{S_t}[h] - \mathbb{E}_{S_t}[\lambda h]\big] \tag{E45}$$

$$= -\tfrac{1}{\alpha}\big(\mathbb{E}_{S_t}[\lambda h] - \mathbb{E}_{S_t}[\lambda]\,\mathbb{E}_{S_t}[h]\big) \tag{E46}$$

$$= -\tfrac{1}{\alpha}\,\mathrm{Cov}_{S_t}(\lambda, h), \tag{E47}$$

where (E43) is (18); (E44) distributes $(\bar\lambda_t^S - \lambda)h = \bar\lambda_t^S h - \lambda h$ and splits the integral; (E45) writes $\int h\, dS_t = \mathbb{E}_{S_t}[h]$, $\int \lambda h\, dS_t = \mathbb{E}_{S_t}[\lambda h]$ and substitutes $\bar\lambda_t^S = \mathbb{E}_{S_t}[\lambda]$; (E46) factors -1; and (E47) is the definition of covariance; the calm switch flow (19) adds $\int_\Phi [h(\phi_t^*) - h(\phi)]\, \Phi_t^{\mathrm{sw}}(d\phi)$. Setting $h(\phi) = \lambda$, the selection part is $-\frac{1}{\alpha}\,\mathrm{Var}_{S_t}(\lambda)$ and the switch part is $\int_\Phi (\lambda(\phi_t^*) - \lambda)\, \Phi_t^{\mathrm{sw}}(d\phi)$, which is ≤ 0 on the disaster-free-since-adoption part of the switch set—there the scheduled relief of Proposition 7 gives $\lambda(\phi_t^*) \le \lambda$ (and $\Phi_t^{\mathrm{sw}} \ge 0$)—and non-positive on the rest under the monotone-reselection condition, so

$$\frac{d\bar\lambda_t^S}{dt} = -\tfrac{1}{\alpha}\,\mathrm{Var}_{S_t}(\lambda) + \int_\Phi \big(\lambda(\phi_t^*) - \lambda\big)\, \Phi_t^{\mathrm{sw}}(d\phi), \tag{E48}$$

the selection term non-positive ($\mathrm{Var}_{S_t}(\lambda) \ge 0$, $\alpha > 0$) and the switch term non-positive under that condition; this is (23). Equation (E48) is the absolutely continuous (dt-rate) part of the calm derivative; where a positive G-mass synchronizes at a calm deadline, Φ_t^{sw} carries a predictable atom and $\bar\lambda_t^S$ a predictable jump, which by Lemma 7 moves no traded price.

(ii) Disaster reweight. At a disaster of size x, before triggered switches, the cross-section

reweights by $\widetilde{S}_t^x(d\phi) = r(\phi, x)^{1/\alpha} g_t(x)^{-1} S_{t-}(d\phi)$ (20), with $g_t(x) = \mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]$ (12). Hence

$$\begin{aligned}
\bar{\lambda}_t^S &= \int_\Phi \lambda\, S_t(d\phi) \\
&= \int_\Phi \lambda\, r(\phi, x)^{1/\alpha} g_t(x)^{-1} S_{t-}(d\phi) && \text{(E49)} \\
&= \frac{\mathbb{E}_{S_{t-}}[\lambda\, r(\cdot, x)^{1/\alpha}]}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]}, && \text{(E50)}
\end{aligned}$$

(E49) substituting the reweight and (E50) pulling out the constant $g_t(x)^{-1}$ and writing numerator and denominator ($= g_t(x)$) as expectations under S_{t-}. Subtract $\bar{\lambda}_{t-}^S = \mathbb{E}_{S_{t-}}[\lambda]$:

$$\begin{aligned}
\bar{\lambda}_t^S - \bar{\lambda}_{t-}^S &= \frac{\mathbb{E}_{S_{t-}}[\lambda\, r(\cdot, x)^{1/\alpha}]}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]} - \mathbb{E}_{S_{t-}}[\lambda] \\
&= \frac{\mathbb{E}_{S_{t-}}[\lambda\, r(\cdot, x)^{1/\alpha}] - \mathbb{E}_{S_{t-}}[\lambda]\, \mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]} && \text{(E51)} \\
&= \frac{\mathrm{Cov}_{S_{t-}}(\lambda,\, r(\cdot, x)^{1/\alpha})}{\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]}, && \text{(E52)}
\end{aligned}$$

(E51) over the common denominator and (E52) the covariance definition; this is (24). To sign the numerator, expand the covariance with an independent copy. Let (λ, ϕ) and (λ', ϕ') be two independent draws from S_{t-} and write $k := r(\phi, x)^{1/\alpha}$, $k' := r(\phi', x)^{1/\alpha}$; all expectations below are under S_{t-}. Then

$$\begin{aligned}
\mathbb{E}\big[(\lambda - \lambda')(k - k')\big] &= \mathbb{E}[\lambda k] - \mathbb{E}[\lambda k'] - \mathbb{E}[\lambda' k] + \mathbb{E}[\lambda' k'] && \text{(E53)} \\
&= \mathbb{E}[\lambda k] - \mathbb{E}[\lambda]\, \mathbb{E}[k'] - \mathbb{E}[\lambda']\, \mathbb{E}[k] + \mathbb{E}[\lambda' k'] && \text{(E54)} \\
&= 2\, \mathbb{E}[\lambda k] - 2\, \mathbb{E}[\lambda]\, \mathbb{E}[k] && \text{(E55)} \\
&= 2\ \mathrm{Cov}_{S_{t-}}(\lambda, k), && \text{(E56)}
\end{aligned}$$

(E53) expanding the product inside the expectation, (E54) factoring the cross terms $\mathbb{E}[\lambda k'] = \mathbb{E}[\lambda]\mathbb{E}[k']$ and $\mathbb{E}[\lambda' k] = \mathbb{E}[\lambda']\mathbb{E}[k]$ by independence of the two draws, (E55) using that the draws are identically distributed ($\mathbb{E}[\lambda' k'] = \mathbb{E}[\lambda k]$, $\mathbb{E}[\lambda'] = \mathbb{E}[\lambda]$, $\mathbb{E}[k'] = \mathbb{E}[k]$), and (E56) the covariance definition. Dividing (E56) by 2 and writing $k = r(\cdot, x)^{1/\alpha}$,

$$\mathrm{Cov}_{S_{t-}}\big(\lambda, r(\cdot, x)^{1/\alpha}\big) = \tfrac{1}{2}\mathbb{E}\big[(\lambda - \lambda')\big(r(\phi, x)^{1/\alpha} - r(\phi', x)^{1/\alpha}\big)\big]. \tag{E57}$$

In the intensity channel $r(\phi, x) = \lambda/\lambda^\star$, so $r(\phi, x)^{1/\alpha} = (\lambda/\lambda^\star)^{1/\alpha}$ is non-decreasing in λ; the two factors in (E57) then carry the common sign of $\lambda - \lambda'$, their product is ≥ 0 pointwise, and the expectation is ≥ 0, so $\mathrm{Cov}_{S_{t-}}(\lambda, r(\cdot, x)^{1/\alpha}) \geq 0$, strict iff $\mathrm{Var}_{S_{t-}}(\lambda) > 0$. Thus the reweight raises $\bar{\lambda}^S$. For the relocation mode: by the proof of Lemma 4 the intensity log-posterior is $\ell_t^\Lambda(\lambda) = \log \rho_0^\Lambda(\lambda) - \lambda t + N_t \log \lambda$, and at a disaster the count N_t rises by one at fixed t. We show the intensity mode $\lambda_t^{\mathrm{m}} = \min \arg\max_\lambda \ell_t^\Lambda$ is non-decreasing in N_t directly, by revealed preference. Write $\ell^{(N)}(\lambda) := \log \rho_0^\Lambda(\lambda) - \lambda t + N \log \lambda$ and let λ_N, λ_{N+1} be the corresponding min-maximisers. Optimality at each count gives

$$\ell^{(N)}(\lambda_N) \geq \ell^{(N)}(\lambda_{N+1}), \qquad \ell^{(N+1)}(\lambda_{N+1}) \geq \ell^{(N+1)}(\lambda_N).$$

Adding the two inequalities and regrouping,

$$\big[\ell^{(N+1)}(\lambda_{N+1}) - \ell^{(N)}(\lambda_{N+1})\big] - \big[\ell^{(N+1)}(\lambda_N) - \ell^{(N)}(\lambda_N)\big] \ \geq \ 0. \tag{E58}$$

Since $\ell^{(N+1)}(\lambda) - \ell^{(N)}(\lambda) = \log \lambda$ for every λ, the left-hand side of (E58) equals $\log \lambda_{N+1} - \log \lambda_N$; hence $\log \lambda_{N+1} \geq \log \lambda_N$, and since log is strictly increasing, $\lambda_{N+1} \geq \lambda_N$. Thus the intensity mode of ϕ_t^* is non-decreasing in N_t: a disaster raises N_t by one, so the common destination ϕ_t^* weakly rises. This is a statement about the destination across counts, not about the switchers' starting point: the triggered revisers relocate *from* their held intensities *to* $\lambda(\phi_t^*)$, so the relocation adds to the reweight (E52) exactly when $\lambda(\phi_t^*)$ exceeds the wealth-weighted mean held intensity of the switchers—the complacent-switcher condition, which holds in the stationary regime when the newborn law Π_ϕ is complacent relative to $\phi^\star$ (Proposition 18). The reweight (E57) raises $\bar{\lambda}^S$ unconditionally, so the fast-in does not rely on this amplifier.

(iii) Fear factor. By the bridge relation (16), $d\log U_t = d\log \kappa_t$, and $U_t = \lambda^\star \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ (14). In calm the held models are fixed and $g_t(x) = \langle r(\cdot,x)^{1/\alpha}, S_t\rangle$, so (E47) with $h = r(\cdot,x)^{1/\alpha}$ gives

$$\frac{dg_t(x)}{dt} = -\tfrac{1}{\alpha}\operatorname{Cov}_{S_t}\!\big(\lambda, r(\cdot,x)^{1/\alpha}\big). \tag{E59}$$

Differentiate $U_t = \lambda^\star \int_{\mathcal{J}} e^{\alpha x} g_t(x)^\alpha f(x)\,dx$ in t. The t-derivative of the integrand is $\alpha e^{\alpha x} g_t(x)^{\alpha-1}\,dg_t(x)/dt$, bounded uniformly on the compact $\mathcal{J}$: $g_t(x) = \int_0^1 s_t^i r_i(x)^{1/\alpha} di \in [\underline{r}^{1/\alpha}, \overline{r}^{1/\alpha}]$ with $0 < \underline{r} = \min_{\mathcal{J}\times\Phi} r \le \overline{r} = \max_{\mathcal{J}\times\Phi} r < \infty$ (continuity and positivity of the densities on the compact $\mathcal{J}\times\Phi$, Assumption 4), and dg_t/dt is bounded by (E59); so dominated convergence justifies differentiation under the integral, and

$$\begin{aligned}
\frac{dU_t}{dt} &= \lambda^\star \int_{\mathcal{J}} e^{\alpha x}\, \alpha\, g_t(x)^{\alpha-1}\, \frac{dg_t(x)}{dt}\, f(x)\,dx \\
&= \lambda^\star \int_{\mathcal{J}} e^{\alpha x}\, \alpha\, g_t(x)^{\alpha-1} \Big(-\tfrac{1}{\alpha}\operatorname{Cov}_{S_t}(\lambda, r(\cdot,x)^{1/\alpha})\Big) f(x)\,dx && \text{(E60)}\\
&= -\lambda^\star \int_{\mathcal{J}} e^{\alpha x} g_t(x)^{\alpha-1} \operatorname{Cov}_{S_t}(\lambda, r(\cdot,x)^{1/\alpha})\, f(x)\,dx, && \text{(E61)}
\end{aligned}$$

(E60) substituting (E59) and (E61) cancelling $\alpha\cdot\frac{1}{\alpha}$. Divide by $U_t = \lambda^\star \mathbb{E}_f[e^{\alpha J} g_t^\alpha]$ (chain rule $d\log U_t/dt = (dU_t/dt)/U_t$):

$$\frac{d\log U_t}{dt} = -\frac{\mathbb{E}_f[e^{\alpha J} g_t(J)^{\alpha-1} \operatorname{Cov}_{S_t}(\lambda, r(\cdot,J)^{1/\alpha})]}{\mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]},$$

which is (25). In the intensity channel $\operatorname{Cov}_{S_t}(\lambda, r(\cdot,x)^{1/\alpha}) \ge 0$ for every x (E57), and $e^{\alpha J}, g_t^{\alpha-1}, g_t^\alpha > 0$, so both expectations are ≥ 0 and the ratio is ≥ 0; hence $d\log U_t/dt \le 0$, strict iff $\operatorname{Var}_{S_t}(\lambda) > 0$. At a disaster of realised size x_0, the reweight $S_t = r(\cdot,x_0)^{1/\alpha} g_{t-}(x_0)^{-1} S_{t-}$ moves g to

$$\begin{aligned}
g_t(x) &= \mathbb{E}_{S_t}[r(\cdot,x)^{1/\alpha}] \\
&= \frac{\mathbb{E}_{S_{t-}}[r(\cdot,x)^{1/\alpha}\, r(\cdot,x_0)^{1/\alpha}]}{\mathbb{E}_{S_{t-}}[r(\cdot,x_0)^{1/\alpha}]}, && \text{(E62)}
\end{aligned}$$

(E62) substituting the reweight into $\mathbb{E}_{S_t}$. In the intensity channel this equals $\mathbb{E}_{S_t}[(\lambda/\lambda^\star)^{1/\alpha}]$, which rose by (ii) (the same reweight raises the mean of the non-decreasing $(\lambda/\lambda^\star)^{1/\alpha}$, by (E57)); so $g_t(x)$ rises for every x and $U_t = \lambda^\star \mathbb{E}_f[e^{\alpha J} g_t^\alpha]$ jumps up. The disaster premium loads on U_t (Proposition 15) and inherits the direction; its level also involves ψ_t, hence the valuation operator v (Proposition 14).

(iv) Staggered recovery. The drifts of (i),(iii) hold on any disaster- and switch-free interval. By Proposition 7 a calm spell ends at the deadline $\Delta^\star(\phi, \varepsilon)$ (B4), where the reviser re-selects the mode prevailing at her deadline; by Lemma 4 the mode is non-increasing through a disaster-free spell, so her held intensity weakly falls—a downward step in $\bar{\lambda}^S$ and, in the intensity channel, in U_t. By Proposition 8 the deadlines are weakly ordered by ε, so the steps are staggered, occurring at distinct times where the deadline map is non-constant; and by Proposition 7 $\Delta^\star$ is finite and increases as $\varepsilon \to 0$ or the held intensity $\lambda(\phi) \to 0$ (the dependence on $\lambda^\star$ is indirect, through which held models survive). The jump of (ii) is instantaneous (fast in); the drift of (i),(iii) and the staggered steps act over the deadline horizon (slow out). $\square$

Appendix F. Demographic turnover and the overlapping-generations economy

Assumption 5 (Demographic turnover: perpetual youth): *We extend Assumption 1 with overlapping generations in the manner of Yaari (1965) and Blanchard (1985), as used for heterogeneous agents by Gârleanu and Panageas (2015):*

(i) each living investor faces a constant death hazard $\Omega > 0$—an independent Poisson clock, the same for all and independent of the aggregate path;

(ii) newborns arrive at mass Ω per unit time (population mass constant), each drawn independently with type $(\varepsilon, \phi_0) \sim \Pi$, a fixed law on $(0,1) \times \Phi$ with ϕ_0-marginal Π_ϕ—the surprise threshold $\varepsilon \in (0,1)$ as in Assumption 2, so the typical-set test of Definition 1 is well defined

for every newborn type;

(iii) (allocation of cohorts) *the symmetric allocation of Assumption 3(iv) applies to every cohort, so the* effective *cohort multiplier is common (no preference- or endowment-dispersion is priced) and each newborn enters with no since-birth record ($M = 1$ at birth). Demographic turnover is then a primitive share-replacement rule: the newborn cohort carries aggregate consumption share $\Omega\,dt$, distributed over held models by the type law Π_ϕ. If formally decentralized (Appendix F), the underlying birth weight ζ_t is the state-dependent value that implements this $\Omega\,dt$ share, keeping the aggregator Ξ_t free of deterministic calendar-time growth. We take this share-replacement law as the* primitive*; no result below requires its decentralization, which Appendix F supplies only to exhibit the state-dependent Negishi weight ζ_t that supports it. Death is actuarially fairly annuitized (Yaari, 1965), so Ω does not enter any survivor's effective discount; the turnover acts on the share simplex and is aggregate-resource-neutral by construction, $\langle 1, S_t \rangle \equiv 1$.*

(iv) (newborn information). *A newborn inherits the common second-order posterior ρ_t prevailing at her birth date—the public object of Lemma 4, a deterministic functional of the aggregate record, shared by every living investor—and enters holding her drawn model ϕ_0 with a reset test state: regime age $\tau = 0$ and empty since-adoption record ($n = 0$). Inheriting ρ_t keeps the re-selection destination $\mathfrak{m}(\rho_t)$ common across cohorts (Lemma 4, Proposition 8); resetting the held model and test clock is what refreshes the cross-section, since young cohorts hold ϕ_0 and have not yet revised. The birth holding is a* boundary condition *distinct from the post-rejection adoption rule of Definition 1: at birth the held model is* drawn *from Π_ϕ, not set to the posterior mode $\mathfrak{m}(\rho_t)$, and the MAP rule governs only subsequent rejections. This birth-specific endowment—not unconcentrated newborn posteriors—is what sustains the stationary dispersion. Disagreement is thus carried by the held models, not by posterior dispersion. Economically, ρ_t is a* public statistic*—a deterministic function of the public record, on the same footing as an equilibrium price—so*

this clause is the standard common-information convention: every investor, newborn or incumbent, conditions on the same public posterior, while the only datum that resets at birth is the agent's private adoption clock. The primitives of Assumptions 1–4 do not by themselves fix a newborn's posterior; this selects the common-information reading, a specification choice within the existing information structure rather than a new economic primitive.

We now specialize to a finite *menu* $\Phi = \{\phi_1, \dots, \phi_K\}$ *(the setting of Corollary 2), an additional restriction for the stationary and quantitative exercise—in general* Φ *is a compact continuum (Assumption 4), not a finite set.*

Antecedent and difference. Perpetual youth is Blanchard's (1985) continuous-time overlapping-generations device, built on Yaari (1965); Gârleanu and Panageas (2015) use it precisely to keep heterogeneity from washing out and to obtain a nondegenerate stationary equilibrium, with the dead's wealth redistributed to the living. We inherit that structure and its purpose, and differ in the dimension of heterogeneity (the surprise threshold ε and the initial held model ϕ_0, not risk aversion) and the environment (CRRA with disaster jumps and lazy revision, not recursive-preference diffusion). The convention in (iii) is the Gârleanu and Panageas reduced form, adopted to avoid the annuity bookkeeping of the full Yaari (1965) market; it makes the demographic effect on the cross-section a single replacement term.

Under Assumption 5(iii) the law of motion of the pricing state (Proposition 3) gains exactly one term. Over $[t, t+dt]$ the population mass is held at one by balanced birth and death: a fraction $\Omega\,dt$ dies (proportionally in wealth, so the redistribution to survivors leaves their relative shares—hence the normalized measure—unchanged), while newborn mass $\Omega\,dt$ enters with type law Π_ϕ. The post-turnover normalized cross-section is therefore the mixture of the surviving S_t (mass $1-\Omega\,dt$) and the newborn Π_ϕ (mass $\Omega\,dt$):

$$S_{t+dt}\big|_{\text{demog}} = (1-\Omega\,dt)\,S_t + \Omega\,dt\,\Pi_\phi, \qquad \text{hence} \qquad dS_t\big|_{\text{demog}} = S_{t+dt}\big|_{\text{demog}} - S_t = \Omega\,(\Pi_\phi - S_t)\,dt. \tag{F1}$$

Pairing the measure identity (F1) with any bounded h,

$$\frac{d}{dt}\langle h, S_t\rangle\Big|_{\text{demog}} = \Omega\big(\langle h, \Pi_\phi\rangle - \langle h, S_t\rangle\big), \tag{F2}$$

added to the selection, switch, and disaster channels of (18)–(20): a mean-reversion of the cross-section toward the newborn law Π_ϕ at rate Ω.

The replacement is resource-neutral by construction: it operates in consumption-share space, where the inflow $\Omega\,\Pi_\phi\,dt$ and outflow $\Omega\,S_t\,dt$ of (F1) preserve $\int_{\text{living}} s_t^i\,di = 1$, so goods continue to clear, $\int_{\text{living}} c_t^i\,di = C_t$, with no second claim on the Lucas tree. The share-endowment birth rule of Assumption 5(iii) is the Gârleanu and Panageas (2015) reduced form for a Pareto-weight (human-capital) convention—a newborn receives a consumption share, not an additional financial claim funded out of the dead's annuitized wealth—which is what makes the demographic effect a single share-replacement term rather than a wealth-accounting transfer between the dead, the survivors, and the newborns. Formally the demographic turnover is a primitive sharing rule on the simplex; if decentralized, it is the Negishi construction of Appendix F with a *state-dependent* cohort weight ζ_t that implements the $\Omega\,dt$ share assignment, priced under fair annuities, so that the death hazard cancels from every survivor's Euler equation (F6) and enters the equilibrium only through the composition of the living. The share law (F1) is in turn invariant to the death-share bookkeeping—whether the dead's wealth-proportional share is annuitized to survivors and the block then diluted by the newborn inflow, or the dead's share is replaced directly by newborns, both give $(1-\Omega\,dt)S_t + \Omega\,dt\,\Pi_\phi$ to first order in dt—which is what licenses writing the demographic effect as the single replacement term (F2) without committing to either ledger.

Proposition 17 (The kernel under demographic turnover): *Under Assumptions 1–4 and 5, an overlapping-generations equilibrium (Appendix F) exists, and its state-price density retains the form of Proposition 1,*

$$\xi_t = e^{-\delta t}\, C_t^{-\alpha}\, \Xi_t^{\alpha}, \qquad \Xi_t = \int_{\text{living}} \widetilde{\psi}_i^{-1/\alpha} \big(M_t^i\big)^{1/\alpha}\, di, \tag{F3}$$

now the $\frac{1}{\alpha}$-power mean of beliefs over the living *population, with a* common *effective cohort multiplier $\widetilde{\psi}_i \equiv \zeta$. Demographic turnover is a primitive share-replacement law on the simplex (Assumption 5(iii), eq. (F2)): newborns enter carrying an aggregate consumption share $\Omega\, dt$ drawn from Π_ϕ, the dead's wealth-proportional share is released, and $\langle 1, S_t\rangle \equiv 1$ is preserved. If formally decentralized, the planner assigns each newborn cohort the state-dependent birth weight ζ_t that implements this $\Omega\, dt$ assignment; because the inflow and outflow of weight then balance exactly, the effective multiplier in Ξ_t is common across cohorts and $\Xi_t = \int_{\text{living}} \zeta^{-1/\alpha} (M_t^i)^{1/\alpha}\, di$ has no deterministic calendar-time growth—the inflow/outflow balance removes the birth-date drift, while invariance of its* law *is the ergodicity of the cross-section (Proposition 18), not the L^1 bound alone. The explicit $e^{-\delta t}$ in (F3) is therefore untouched and the riskless-rate* level *keeps its $+\delta$ term: (E1) holds verbatim in the turnover economy. A deterministic common* underlying *weight would instead let newborns enter at a birth-date-growing relative weight $e^{\delta b/\alpha}$, making their share fluctuate as $1/\Xi_t$, growing Ξ_t at $e^{\delta t/\alpha}$, and cancelling the $e^{-\delta t}$—a perpetual-youth δ-reduction in the rate (Gârleanu and Panageas, 2015); but that birth rule is incompatible with the constant-$\Omega\, dt$ share inflow the demographic law requires, so we do not adopt it.*[10] *The death hazard Ω does not enter the time preference of the kernel—the fair annuity of Assumption 5(iii) offsets mortality in each survivor's Euler equation—and enters the equilibrium only through the composition of the living, i.e. through the demographic channel (F2) of the cross-section. The bridge relation (Proposition 2) and the selection drift (18)—instantaneous relations of the kernel to the cross-section—hold verbatim in the turnover economy; the cross-section dynamics*

[10]Pinning the newborn cohort to a share-neutral replacement flow precludes the overlapping-generations δ-reduction in the rate *level*; the low safe rate in calm rests entirely on the fear-factor mechanism $\bar{\lambda}_t^S - U_t$, not on a demographic accounting offset.

are those of Proposition 3 augmented by (F2)*, so the calm erosion of Proposition 6(i) acquires a mean-reversion of $\bar\lambda_t^S$ toward the newborn mean at rate Ω, while the disaster jump and staggered recovery are unchanged in form.*

Finite lives change *who* sits in the power mean. With fair annuities the mortality risk is fully insured, so each survivor optimises as if infinitely-lived at rate δ (the annuity offsets the death hazard Ω in the Euler equation); clearing on the exogenous Lucas endowment returns the kernel $\xi_t = e^{-\delta t} C_t^{-\alpha} \Xi_t^{\alpha}$ over the living cross-section, which turnover keeps non-degenerate. The aggregate Ξ_t is positive a.s. and bounded in L^1: each M^i is a true $\mathbb{P}$-martingale (Appendix C), so by Fubini $\mathbb{E}_{\mathbb{P}}[\Xi_t] = \int_0^\infty \Omega e^{-\Omega a}\, \mathbb{E}_{\mathbb{P}}[(M_a^i)^{1/\alpha}]\, da$. For $\alpha > 1$ the map $x \mapsto x^{1/\alpha}$ is concave, so Jensen gives $\mathbb{E}_{\mathbb{P}}[(M_a^i)^{1/\alpha}] \le 1$ and $\mathbb{E}_{\mathbb{P}}[\Xi_t] \le 1$; for $0 < \alpha < 1$ it is convex, the sharp bound need not hold, but Ξ_t stays in L^1 by the finite exponential moments below. The bound is in expectation, not pathwise, since $(M_a^i)^{1/\alpha}$ can exceed 1 on reinforced paths, and $\Xi_t > 0$ a.s. The newborn cohorts ($a \approx 0$, $M \approx 1$) keep Ξ_t non-degenerate, with finite inverse moments (the disaster count over a fixed newborn window is Poisson, hence has finite exponential moments), and the demographic weight $\Omega e^{-\Omega a}$ controls the remote-age tail; the kernel keeps its form under the share-neutral replacement, and the stationary cross-section makes Ξ_t stationary in calendar time. The whole apparatus of Sections 2–C, including the finite-menu fear factor (D17), therefore transfers to the stationary economy of Proposition 18 without re-derivation; turnover supplies the stationary S_t these results take as input. The level formula (E1) is unchanged: with Ξ_t stationary the explicit $e^{-\delta t}$ survives, so r_t^f keeps its $+\delta$ term, and turnover enters the rate only through the demographic mean-reversion of $\bar\lambda_t^S$ and U_t in (F2), which smooths its realized path—the stationary, non-volatile safe rate of the perpetual-youth economy (Gârleanu and Panageas, 2015).

Primitives. Under Assumption 5 an investor born at b with type (ε, ϕ_0) and belief $\mathbb{P}^i$ (change-of-measure M^i) chooses consumption to solve

$$\max\ \mathbb{E}^i\Big[\int_b^\infty e^{-\delta(s-b)}\mathbf{1}\{T_i > s\}\, u(c_s^i)\, ds\Big] = \max\ \mathbb{E}^i\Big[\int_b^\infty e^{-(\delta+\Omega)(s-b)}\, u(c_s^i)\, ds\Big], \qquad u(c) = \tfrac{c^{1-\alpha}}{1-\alpha}, \tag{F4}$$

where $T_i \sim \mathrm{Exp}(\Omega)$ is her death time (Assumption 5(i)) and the second equality takes the expectation over T_i, independent of the path. Markets are complete in the aggregate risks (Assumption 3) with state-price density ξ; under the fair annuity of Assumption 5(iii) a survival-contingent unit of consumption at s costs $\xi_s\, e^{-\Omega(s-b)}$ (the market price under $\mathbb{P}$ times the survival probability), so the market value of the consumption plan—the budget—is taken under the pricing measure $\mathbb{P}$,

$$\mathbb{E}\Big[\int_b^\infty \frac{\xi_s}{\xi_b}\, e^{-\Omega(s-b)}\, c_s^i\, ds\,\Big|\, \mathcal{F}_b\Big] \le W_b^i, \tag{F5}$$

with birth wealth W_b^i the value the birth endowment of Assumption 5(iii) implies at the birth date b (type-dependent under heterogeneous beliefs, not per-capita), in date-b consumption units and $\mathcal{F}_b$-measurable (Assumption 5(ii)–(iii)); ξ_b is its date-0 price, and the normalization by ξ_b is absorbed into the cohort multiplier $\widetilde{\psi}_i$ in the proof, so the dating affects no equilibrium object. An *overlapping-generations equilibrium* is an SPD ξ and allocations $\{c^i\}$ such that every living investor solves (F4)–(F5) and goods clear, $\int_{\text{living}} c_t^i\, di = C_t$.

Proof of Proposition 17. Individual optimum. Markets are complete in the aggregate risks (Assumption 3), so investor i's dynamic problem reduces to the static budget problem under the unique state-price density ξ (the martingale method, valid for complete markets and a concave, increasing utility; Cox and Huang, 1989; Karatzas et al., 1990). Writing the objective (F4) under $\mathbb{P}$ via $\mathbb{E}^i[\,\cdot\,] = \mathbb{E}[M_s^i\,\cdot\,]$ and the budget (F5) under $\mathbb{P}$, the Lagrangian

$$\mathbb{E}\Big[\int_b^\infty \Big(e^{-(\delta+\Omega)(s-b)} u(c_s^i)\, M_s^i - \psi_i\, \xi_s\, e^{-\Omega(s-b)}\, c_s^i\Big) ds\Big]$$

is concave in c^i (u concave), so its pointwise (in (s,ω)) stationarity condition is necessary and sufficient:

$$
\begin{aligned}
e^{-(\delta+\Omega)(s-b)}\, u'(c^i_s)\, M^i_s &= \psi_i\, \xi_s\, e^{-\Omega(s-b)}, \\
e^{-\delta(s-b)}\, u'(c^i_s)\, M^i_s &= \psi_i\, \xi_s, && \text{(F6)} \\
e^{-\delta(s-b)}\, (c^i_s)^{-\alpha}\, M^i_s &= \psi_i\, \xi_s, && \text{(F7)}
\end{aligned}
$$

line (F6) cancelling the common mortality factor $e^{-\Omega(s-b)}$ (the Yaari (1965) insurance result: the survival weight in the objective and the annuity price offset), and (F7) substituting $u'(c) = c^{-\alpha}$. Solve (F7) for c^i_s in one step each:

$$
\begin{aligned}
(c^i_s)^{-\alpha} &= \psi_i\, \xi_s\, e^{\delta(s-b)}/M^i_s \\
c^i_s &= \psi_i^{-1/\alpha}\big(\xi_s e^{\delta(s-b)}\big)^{-1/\alpha}(M^i_s)^{1/\alpha} \\
&= \widetilde{\psi}_i^{-1/\alpha}\left(\xi_s e^{\delta s}\right)^{-1/\alpha}(M^i_s)^{1/\alpha}, \qquad \widetilde{\psi}_i := \psi_i\, e^{-\delta b}, && \text{(F8)}
\end{aligned}
$$

line (F8) writing $e^{\delta(s-b)} = e^{\delta s} e^{-\delta b}$ and absorbing the birth-time factor $e^{-\delta b}$ into $\widetilde{\psi}_i$ (so $(e^{-\delta b})^{-1/\alpha}$ joins $\psi_i^{-1/\alpha}$). The birth endowment of Assumption 5(iii) is the state-dependent value that implements the $\Omega\, dt$ share assignment of (F2); it makes the effective multiplier $\widetilde{\psi}_i \equiv \zeta$ common across cohorts, so Ξ_s carries no deterministic calendar-time growth and the explicit $e^{-\delta s}$ in (F12) is retained—no birth-date growth, hence no δ-cancellation in the rate level (invariance of its *law* is the separate ergodicity claim of Proposition 18).

Market clearing fixes the kernel. Impose $\int_{\text{living}} c^i_s\, di = C_s$ with $\Xi_s := \int_{\text{living}} \widetilde{\psi}_i^{-1/\alpha}(M^i_s)^{1/\alpha}\, di$;

the factor $(\xi_s e^{\delta s})^{-1/\alpha}$ is constant in i, so

$$\left(\xi_s e^{\delta s}\right)^{-1/\alpha} \int_{\text{living}} \widetilde{\psi}_i^{-1/\alpha} (M_s^i)^{1/\alpha} di = C_s$$

$$\left(\xi_s e^{\delta s}\right)^{-1/\alpha} \Xi_s = C_s \tag{F9}$$

$$\left(\xi_s e^{\delta s}\right)^{-1/\alpha} = C_s/\Xi_s \tag{F10}$$

$$\xi_s \, e^{\delta s} = \left(C_s/\Xi_s\right)^{-\alpha} = C_s^{-\alpha} \Xi_s^{\alpha} \tag{F11}$$

$$\xi_s = e^{-\delta s} \, C_s^{-\alpha} \, \Xi_s^{\alpha}, \tag{F12}$$

(F9) substituting the definition of Ξ_s, (F10) dividing by Ξ_s (> 0 as in Step 1 of Proposition 1, the bounds (C2) holding cohort by cohort), (F11) raising to the $-\alpha$, and (F12) multiplying by $e^{-\delta s}$. This is (F3), identical in form to (10) with the integral over the living. The death hazard appears nowhere in (F12): it cancelled at (F6) and enters the equilibrium only through which investors are alive at s—the demographic channel (F2)—and through the cohort multipliers $\widetilde{\psi}_i$ fixed at birth. Existence and uniqueness follow as in Proposition 1 (strict concavity of u and the complete-market budget), applied cohort by cohort. Since (F3) is the kernel of Proposition 1 evaluated at the living cross-section, the instantaneous relations of the kernel to the cross-section—the bridge (Proposition 2) and the selection drift (18)—hold verbatim with S_t the living cross-section, whose dynamics are those of Proposition 3 augmented by the demographic term (F2). □

Appendix G. The stationary cross-section

Proposition 18 (Stationary cross-section under demographic turnover): *Under Assumptions 1–4, the finite menu (Assumption 5), and the demographic turnover of Assumption 5:*

(i) (Age structure.) *For $\Omega > 0$ the age distribution of the living is stationary with density $\Omega e^{-\Omega a}$ on $a \geq 0$ (at $\Omega = 0$ there is no turnover and no such stationary age law).*

(ii) (No refresh $\Rightarrow$ concentration.) *If $\Omega = 0$, the second-order posterior is consistent,* $\mathrm{MAP}(\rho_t^i) \to \phi^\star$ *a.s. (Berk, 1966); every re-selection (9) then returns $\phi^\star$ and the held-model cross-section converges to the single vertex $\delta_{\phi^\star}$. (In the selection-only reduction with re-selection suppressed every vertex of Δ^{K-1} is absorbing; under the full re-selecting dynamics, by contrast, only $\delta_{\phi^\star}$ is invariant, since any other vertex is left at the next rejection—Remark 6.)*

(iii) (Refresh $\Rightarrow$ unique nondegenerate stationary cross-section.) *If $\Omega > 0$, the cohort-resolved cross-section Ψ_t—the wealth measure over (held model $\phi \in \Phi$, age $a \geq 0$, since-adoption record $\mathcal{R}$)—is, in the limiting regime where the public posterior has concentrated (*$\mathrm{MAP}(\rho_t) = \phi^\star$*, so the re-selection destination is calendar-invariant), a time-homogeneous measurable functional of the marked disaster point process N on the* two-sided *stationary line and the stationary demographic-and-type structure (Proposition 9 with Assumption 5). On that two-sided process Ψ_t is a stationary, ergodic process with a unique stationary law Ψ^∞, to which the economy from any finite prior converges—in the bounded-Lipschitz (Wasserstein-1) metric, geometrically at a rate proportional to Ω and* not *in total variation (Step 5)—as* $\mathrm{MAP}(\rho_t) \to \phi^\star$*; the pricing state $S_t \in \Delta^{K-1}$ (the ϕ-marginal of Ψ_t) and its instantaneous functionals (U_t, $\bar{\lambda}_t^S$), together with the price–dividend ratio $v(\Psi_t)$, the jump ratio ψ_t, and the disaster premium DRP_t—these last functionals of the* full *Ψ_t through $\psi_t(x) = v(\Psi_t^x)/v(\Psi_{t-})$ (Proposition 14), and measurable and integrable rather than necessarily continuous—inherit unique stationary laws, with unconditional moments equal to long-run time averages (Birkhoff). Ψ^∞ places mass on at least two models whenever Π_ϕ does; the dependence of Ψ_t on the remote past (and on any initial cross-section) decays geometrically, at a rate proportional to Ω.*

The turnover does exactly one thing to the mechanism and one thing to the mathematics. To the mechanism (eq. F2): it adds a constant pull of the cross-section back toward the newborn dispersion Π_ϕ, at rate Ω, which is precisely the force that offsets the selection-and-consistency concentration of Remark 6—young investors, whose *held models* have not yet moved to $\phi^\star$—

they inherit the already-concentrated common posterior but enter with a drawn held model—are continually reborn. To the mathematics: with the menu finite and the inflow of fresh types constant, the cross-section becomes a time-homogeneous functional of a stationary, ergodic disaster clock, so it is itself stationary and ergodic, and the unconditional moments the calibration needs are well-defined and computable as time averages. The parameter that governs the stationary *dispersion*—and hence, through Proposition 6, the amplitude of the synchronized revision wave—is the newborn *joint* law Π_ϕ over birth held models *and* thresholds, together with the turnover rate Ω: more dispersed births leave more disagreement, the threshold law sets how long birth beliefs survive before revision (through the rejection and deadline times), and Ω governs how quickly fresh cohorts replace the selected cross-section. This is the sense in which, as anticipated, stationarity rests on the assumed joint law of threshold and intensity at birth.

Two beliefs, two clocks. The refresh and the synchronization live on different objects, and the stationary economy keeps them apart. The common second-order posterior ρ_t (over the menu, the source of the re-selection destination) is a public functional of the record, inherited by every cohort (Assumption 5(iv)); by part (ii) it has concentrated on the fixed pseudo-true model, $\mathfrak{m}(\rho_t) \equiv \phi^\star$, so in the stationary regime the common destination is constant and carries no remote-past dependence. The per-agent state $(\phi, a, \mathcal{R})$ is reset at birth and its dependence on the past decays geometrically, at a rate proportional to Ω (Step 5 of the proof). Non-degeneracy is therefore a held-model phenomenon: a stationary age-mixture of ϕ_0-holding young and $\phi^\star$-absorbed old. The fast-in jump at a disaster is correspondingly carried by the reweighting channel—wealth flowing to the higher-intensity holders (Proposition 6(ii))—and by relocation to the fixed $\phi^\star$, not by a jumping destination; the destination's disaster-time movement of Lemma 4 is a transient, finite-record feature. The *direction* of that jump is a restriction, not an identity: relocation raises mean intensity only when the newborn law Π_ϕ is complacent relative to the destination—$\lambda(\phi^\star)$ above the Π_ϕ-mean intensity—the stationary counterpart of the "on average complacent" calibration (Section A), a parameter condition on Π_ϕ rather than a consequence of the primitives.

Two boundaries, kept sharp. First, stationarity delivers the invariant *distribution* Ψ^{∞}; it does not by itself close the price level. Even under the finite menu the price–dividend ratio $v(\Psi_t)$ solves the Feynman–Kac operator equation (E26) on the record-resolved state Ψ_t (Corollary 3), collapsing to a finite-dimensional-state operator equation in the shares only in the selection-only specialization with no record-dependent switching; with a continuous menu v likewise solves the analogous infinite-dimensional Feynman–Kac valuation problem of Proposition 14 (the displayed operator equation (E26) is the finite-menu form of Corollary 3). So the invariant-distribution statement and the price-level statement are distinct. Second, the turnover modifies the equilibrium kernel only through this cross-section: Proposition 17 showed the kernel form is preserved, with Ω entering the dynamics of Ξ_t (and hence r_t^f) but not the time preference, so the pricing of Sections 2–C transfers to the stationary economy: the realized record-resolved state Ψ_t is then *distributed* according to the invariant law Ψ^{∞} of Proposition 18, and the state-contingent pricing formulas are evaluated at the realized Ψ_t (with marginal S_t), not at the invariant law itself.

Proof of Proposition 18. (i) Age structure. Population mass is constant: births of mass Ω per unit time balance deaths at hazard Ω. A living agent of age a was born a units ago and has survived the death clock with probability $e^{-\Omega a}$. The birth flow a units ago was Ω, so the mass of agents currently of age in $[a, a+da]$ is the birth flow times the survival probability,

$$\Omega\, e^{-\Omega a}\, da, \tag{G1}$$

and the total living mass is

$$\int_0^{\infty} \Omega e^{-\Omega a}\, da = \left[-e^{-\Omega a}\right]_{a=0}^{a=\infty} = 0 - (-1) = 1,$$

consistent with unit population. The age density $\Omega e^{-\Omega a}$ does not depend on t: it is stationary

(Blanchard, 1985).

(ii) No refresh. By Assumption 4 the family is identifiable and, by Lemma 2, the Kullback–Leibler projection $\phi^\star$ of the true law onto Φ exists and is unique (Assumption 4(iv)). The second-order posterior ρ_t^i updates by Bayes on the full history (4). Under Assumption 1 the homogeneous Poisson process has stationary *independent* increments and the marks and diffusion increments are i.i.d., so the record splits into the i.i.d. sequence of unit-interval blocks $X_k =$ (the marked-Poisson record and diffusion increment on $(k-1, k]$), $k = 1, 2, \dots$; the full-history log-likelihood (4) is the sum of the block log-likelihoods, and the posterior at integer $t = n$ is the Bayes posterior from n i.i.d. observations $X_1, \dots, X_n$. Berk (1966)'s theorem applies to exactly this i.i.d. setting: for an identifiable dominated family observed i.i.d., with finite Kullback–Leibler divergence to the family, the posterior concentrates on the divergence-minimising parameter. Its three hypotheses hold block by block: identifiability (Assumption 4), i.i.d. blocks (stationary independent increments, Assumption 1), and finite divergence—the per-block divergence is the unit-time rate $\mathcal{K}(\phi)$ of (A2), finite with unique minimiser $\phi^\star$ (Lemma 2, (A3)). Hence $\rho_n^i \Rightarrow \delta_{\phi^\star}$ along integers. For non-integer t, the log-likelihood ratio of any $\phi \neq \phi^\star$ against $\phi^\star$ is the integer-time sum $-n\big(\mathcal{K}(\phi) - \mathcal{K}(\phi^\star)\big) + o(n)$ ($n = \lfloor t \rfloor$, $\mathcal{K}(\phi) > \mathcal{K}(\phi^\star)$ strictly by the unique minimiser) plus the contribution of the partial block $(\lfloor t \rfloor, t]$, which is a.s. finite (finitely many disaster marks on a bounded interval, Assumption 1, with r bounded); the $O(n)$ term dominates, so the ratio still diverges to $-\infty$ and $\rho_t^i \Rightarrow \delta_{\phi^\star}$ and $\mathrm{MAP}(\rho_t^i) \to \phi^\star$ for all t; on the finite menu the MAP is an argmax over finitely many vertices, and the divergence is uniform across them: each of the $K-1$ rivals has log-posterior-ratio to $\phi^\star$ tending to $-\infty$ a.s., so their maximum does too, and there is an a.s.-finite *lock time* τ_0 with $\mathrm{MAP}(\rho_t) = \phi^\star$ *exactly* for all $t \geq \tau_0$ (under a continuum of models this is convergence, not finite-time equality)—the *public-posterior lock time*, a property of the common record shared by the no-refresh dynamics here and the refresh economy of part (iii). Once an investor's MAP is $\phi^\star$, the common re-selection destination (9) (Lemma 4) is $\phi^\star$, so every rejection returns $\phi^\star$ and her held model is absorbed there. With $\Omega = 0$ the demographic term (F2) is absent, so no other model receives inflow.

Convergence to $\delta_{\phi^\star}$ is driven by re-selection and disaster reweighting, *not* by the calm selection drift (18): between disasters selection favors below-average-intensity (complacent) models and can move wealth *away* from $\phi^\star$, but every rejection returns the held model to $\phi^\star$, so each holder is re-selected there at its next rejection and $S_t \to \delta_{\phi^\star}$. The unique invariant law is therefore $\delta_{\phi^\star}$ itself: the other vertices of Δ^{K-1} are not invariant, since any holder there is re-selected to $\phi^\star$ at its next rejection (the selection-only contrast above)—the formalisation promised in Remark 6.

(iii) Refresh. Step 1 (functional representation). Proposition 9 established, for a *single* cohort sharing one path, that each investor's state is the deterministic measurable image $\mathsf{T}_t(\varepsilon)$ of her type and the cross-section the pushforward $\mathsf{T}_{t\#}G$—no law of large numbers, there being no idiosyncratic randomness to average. That argument applies cohort by cohort here, with two extensions it accommodates without change to its logic: the type is now the *joint* birth draw $(\varepsilon, \phi_0) \sim \Pi$ rather than a common ϕ_0—the revision times (B4) and the path-determined re-selection destinations (Lemma 4) are deterministic measurable functions of (ε, ϕ_0) exactly as they were of ε alone, so each cohort's contribution is a pushforward $\mathsf{T}^a_{t\#}\Pi$ over its since-birth segment $N|_{[t-a,t]}$—and the cross-section carries a *biological-age* coordinate (constant, hence suppressed, in the single-cohort case) recording survival and wealth. Under Assumption 5 the population is the wealth-weighted age-mixture of such cohorts—age density $\Omega e^{-\Omega a}$ by (G1), birth type Π. An agent of age a alive at t was born at $t-a$ with $(\varepsilon, \phi_0) \sim \Pi$ and has evolved her held model, record, and wealth share under the marked segment $N|_{[t-a,t]} = \{(T_k, J_k) : t-a < T_k \le t\}$ by the revision rule (Proposition 12) and the share dynamics (Proposition 10); the marks J_k enter through the likelihood ratios and the share jumps (C8), so it is the marked process, not the bare count, that drives her trajectory. The *destination* of any revision she makes is the common mode $\mathfrak{m}(\rho_t)$, inherited identically by every living cohort (Assumption 5(iv), Lemma 4); by part (ii) it has concentrated on the fixed $\phi^\star$, so it contributes no separate since-birth dependence. Her belief change-of-measure M^i, hence her share $s^i = \zeta_i^{-1/\alpha}(M^i)^{1/\alpha}/\Xi$ (11), is then a functional of the marked disaster history alone (the diffusion is undisputed), and the birth multiplier ζ_i is common across cohorts because the equal-weight normalization of Assump-

tion 3 assigns every cohort the common effective cohort multiplier ζ (Assumption 5(iii) with the share-implementing birth endowment), not because birth wealth is equal, cancelling from s^i. Aggregating over the stationary age-and-type structure—each age-a cohort contributing its Π-distributed, $N|_{[t-a,t]}$-evolved states with its *wealth* weight $\Omega e^{-\Omega a}\,(M_a)^{1/\alpha}/\Xi_t$ (the demographic mass $\Omega e^{-\Omega a}$ times the belief share of Proposition 17, not the bare age density)—gives

$$\Psi_t = \mathfrak{F}\big(N|_{(-\infty,t]}\big).$$

The aggregation over $a \in [0,\infty)$ converges: each cohort contributes a probability measure on the finite Φ weighted by its wealth share in $[0,1]$, and the age weight integrates to one ($\int_0^\infty \Omega e^{-\Omega a}\,da = 1$ by (G1)), so Ψ_t is a well-defined probability measure and $\mathfrak{F}$ is measurable and independent of calendar time (rule, DGP, and demographics are time-homogeneous).

Step 2 (stationarity and ergodicity). We establish stationarity and ergodicity by the *factor* method—representing Ψ_t as a measurable functional of the stationary ergodic disaster process—rather than by exhibiting an infinitesimal generator for the measure-valued birth–death dynamics and verifying a Doeblin/minorisation condition; the factor route is the natural one for a pricing state that is itself non-Markov (Proposition 3), and the stationary moments the calibration needs are time averages, for which ergodicity alone suffices. N is a homogeneous Poisson process; its law is invariant under the time shift $\theta_h : N|_{(-\infty,t]} \mapsto N|_{(-\infty,t+h]}$, and the shift is ergodic—a process with stationary independent increments is mixing, hence ergodic, under time translation. We construct the invariant law on the *two-sided* stationary marked Poisson measure $N|_{(-\infty,t]}$: on it the public posterior, informed by the infinite past, has already concentrated ($\mathrm{MAP}(\rho_t) = \phi^\star$ identically, Berk), so the re-selection destination $\mathfrak{m}(\rho_t)$ is calendar-time invariant and $\mathfrak{F}$ is genuinely time-homogeneous. A time-homogeneous measurable functional of the stationary ergodic two-sided process is stationary and ergodic (a measurable factor of an ergodic shift is ergodic), so $\Psi_t = \mathfrak{F}(\theta_t N)$ is stationary and ergodic with law Ψ^∞. *Uniqueness* and convergence from an arbitrary one-sided initialisation rest not on factor inheritance alone but on the demographic coupling, and are proved in Step 5 by an explicit synchronous coupling: the cohort weight $\Omega e^{-\Omega a}$ forgets the remote past at rate Ω, and the one-sided economy couples to the two-sided regime as $\mathrm{MAP}(\rho_t) \to \phi^\star$ (Berk; at the Bernstein–von Mises rate of Lemma 3(iii)).

Step 3 (inheritance and moments). $S_t = \pi_\phi(\Psi_t)$, the ϕ-marginal, is a continuous linear functional of Ψ_t; the instantaneous objects U_t, $f_t^{\mathbb{Q}}$, and $\bar{\lambda}_t^S$ are functionals of S_t (Proposition 1, Proposition 2), finite-dimensional on Δ^{K-1} under the finite menu, while the price–dividend ratio $v(\Psi_t)$ *and* the disaster-premium *level* DRP_t are functionals of the full record-resolved state Ψ_t (Propositions 14, 15, 16), bounded and measurable in Ψ_t—continuous off the deadline and trigger thresholds of the revision rule, where they may jump, but not assumed continuous. Each is thus a bounded measurable functional of Ψ_t and inherits a unique stationary law—the premium as a functional of the stationary Ψ_t, not as a finite-dimensional function of S_t alone. These functionals are bounded under (T): U_t and $\bar{\lambda}_t^S$ are continuous on the compact Δ^{K-1}; $0 < v(\Psi_t) \leq (\delta - \sup m)^{-1}$ by (E24), so by (E25) the jump ratio $\psi_t = v(\Psi_t^J)/v(\Psi_{t-})$ is bounded above and below away from zero; and DRP_t (E37) is then bounded, loading on the bounded U_t and on the bounded disaster return $\Delta R_t^e = e^{-J}\psi_t - 1$. All are integrable, so the Birkhoff ergodic theorem gives, for any such g,

$$\frac{1}{T}\int_0^T g(\Psi_t)\,dt \xrightarrow{\text{a.s.}} \mathbb{E}_{\Psi^\infty}[g] \qquad (T \to \infty),$$

so the unconditional, calendar-time moments are well-defined and equal these limits. Disaster-conditional *event-study* averages—means taken over disaster epochs—are not calendar-time averages; for the stationary, ergodic marked process N they are the Palm expectation $\mathbb{E}^0_{\Psi^\infty}[g]$, and the Palm–Campbell ergodic theorem gives the corresponding $\mathbb{P}^0$-a.s. convergence of disaster-epoch averages to it. The two limits coincide only for functionals that do not condition on an arrival.

Step 4 (non-degeneracy). Fix $\eta > 0$ small. The newborns of age in $[0, \eta]$ alive at t carry a wealth fraction $\Omega\eta + o(\eta)$ (Assumption 5(ii)–(iii)), distributed over models by Π_ϕ up to an $O(\eta)$ revision correction; hence S_t places mass at least $c\,\Pi_\phi(\{\phi\})$ on each $\phi \in \operatorname{supp}\Pi_\phi$, for some $c > 0$ uniform in t, and so does Ψ^∞. If $|\operatorname{supp}\Pi_\phi| \geq 2$, at least two models carry positive stationary mass: Ψ^∞ is nondegenerate, in contrast to (ii).

Step 5 (convergence from an arbitrary initialisation, by synchronous coupling). We prove that the law of the one-sided economy started from any initial cross-section Ψ_0 converges to Ψ^∞, and we identify the mode. Couple the two economies on a common driving path: run the one-sided economy from Ψ_0 at $t = 0$ and the stationary two-sided economy $\Psi_t^\infty = \mathfrak{F}(\theta_t N)$ on the *same* marked Poisson process N, sharing the entire post-zero birth stream—birth epochs, types $(\varepsilon, \phi_0) \sim \Pi$, and death clocks. The economies then differ only in their pre-zero composition: the one-sided economy carries the initial cohorts encoded in Ψ_0, the two-sided economy the cohorts born in $(-\infty, 0]$.

Destination lock. The public re-selection destination is the posterior MAP $\mathfrak{m}(\rho_t)$ (9), which by part (ii) *locks* on the finite menu at the a.s.-finite public-posterior lock time: each rival vertex's log-posterior-ratio to $\phi^\star$ diverges to $-\infty$ a.s., so with finitely many vertices the MAP equals $\phi^\star$ exactly for all t beyond that time. The two-sided posterior, informed by the infinite past, is locked at every t; the one-sided posterior from ρ_0 differs only by the bounded pre-zero log-likelihood (finitely many pre-zero marks, r bounded), which moves its lock time by an a.s.-finite amount, to some τ_0. For $t \geq \tau_0$ both destinations equal $\phi^\star$, and every revision in either economy sends the reviser there.

Common cohorts coincide. A cohort born at $b \geq \tau_0$ has, in both economies, the same birth type, the same since-birth segment $N|_{[b,t]}$, and the same destination $\phi^\star$ at every revision; its held model, since-adoption record, age, and unnormalised belief weight $(M^i)^{1/\alpha}$ are therefore *identical* across the two economies. Cohorts born in $[0, \tau_0)$ may have revised to differing destinations before τ_0 and so can differ until their next revision—after which both economies send them to $\phi^\star$—or their death.

Distance bound. The two cross-sections share the post-τ_0 cohorts' contribution as an *identical* unnormalised measure; they differ only through three wealth groups—the pre-zero initial cohorts, mass W_t^{init}; the pre-zero infinite-past cohorts, mass W_t^{∞}; and the $[0, \tau_0)$ cohorts still holding a pre-lock model, mass W_t^{pre}—and the renormalisation these induce in Ξ_t. For any test function h on the per-investor state space that is 1-Lipschitz with $\|h\|_\infty \le 1$, the identical post-τ_0 part cancels in $\langle h, \Psi_t\rangle - \langle h, \Psi_t^\infty\rangle$ up to that common renormalisation, and each offending group contributes at most twice its wealth share—the pre-zero groups through their own mass and the Ξ_t renormalisation, the pre-lock group through its differing copy in each economy (the pre-lock mass cancels in $\Xi_t - \Xi_t^\infty$)—so

$$d_{\text{BL}}(\Psi_t, \Psi_t^\infty) = \sup_h \big|\langle h, \Psi_t\rangle - \langle h, \Psi_t^\infty\rangle\big| \;\le\; 2\big(W_t^{\text{init}} + W_t^{\infty} + W_t^{\text{pre}}\big). \tag{G2}$$

Decay of the bound. W_t^∞ is the wealth share of two-sided cohorts older than t—the old-cohort tail

$$W_{>a}(t) = \frac{1}{\Xi_t}\int_a^\infty \Omega e^{-\Omega a'}\,\overline{(M^{a'})^{1/\alpha}}\,da' \tag{G3}$$

at $a = t$, with $\overline{(M^{a'})^{1/\alpha}}$ the age-a' cohort average of $(M^i)^{1/\alpha}$. Each M^i is a true $\mathbb{P}$-martingale (Appendix C), so for $\alpha > 1$ Jensen gives $\mathbb{E}_{\mathbb{P}}[(M^i_{a'})^{1/\alpha}] \le 1$, non-increasing in a'; the denominator has no *deterministic* positive lower bound—a recent disaster cluster can depress the young cohorts' $(M^i)^{1/\alpha}$—but $1/\Xi_t$ has uniformly bounded inverse moments: $\Xi_t \ge m_0(t) := \int_0^\delta \Omega e^{-\Omega a'}(M_{a'})^{1/\alpha}\,da'$, and the disaster count over the fixed window $[t-\delta, t]$ is Poisson$(\lambda^\star\delta)$ with finite moment generating function, so $\sup_t \mathbb{E}_{\mathbb{P}}[m_0(t)^{-q}] < \infty$ for every $q \ge 1$. Combining $\mathbb{E}_{\mathbb{P}}\big[\int_a^\infty \Omega e^{-\Omega a'}(M_{a'})^{1/\alpha}da'\big] \le e^{-\Omega a}$ (Jensen) with this inverse-moment bound (Hölder, then truncation) gives $\mathbb{E}_{\mathbb{P}}[W_{>a}(t)] \le C\,e^{-c\Omega a}$ for some $c \in (0,1]$, so $\mathbb{E}_{\mathbb{P}}[W_t^\infty] \le Ce^{-c\Omega t}$. The one-sided group W_t^{init} has the Ψ_0-dependent age profile of the initial cohorts, which need *not* be the stationary $\Omega e^{-\Omega a'}$; but those cohorts have survived from before 0 to t (death hazard Ω, survival $\le e^{-\Omega t}$) and carry the same $\mathbb{P}$-martingale weights with $\mathbb{E}_{\mathbb{P}}[(M^i_t)^{1/\alpha}] \le 1$, so the identical inverse-moment estimate gives $\mathbb{E}_{\mathbb{P}}[W_t^{\text{init}}] \le Ce^{-c\Omega t}$ as well. The third group decays too, by the *same* wealth-share route, not a headcount bound: a $[0, \tau_0)$ cohort still pre-lock at t has age $\geq t - \tau_0$, so *conditional on* the lock time τ_0 its wealth share is bounded by the old-cohort tail $W_{>(t-\tau_0)}(t)$ of (G3), $\mathbb{E}_{\mathbb{P}}[W_t^{\text{pre}} \mid \tau_0] \leq Ce^{-c\Omega(t-\tau_0)}$. The lock time is the last-passage time above zero of the negative-drift log-posterior-ratio walk, whose increments are light-tailed (bounded r, Poisson marks), so $\mathbb{E}_{\mathbb{P}}[e^{c\Omega\tau_0}] < \infty$ after possibly shrinking c; the outer expectation then gives $\mathbb{E}_{\mathbb{P}}[W_t^{\text{pre}}] \leq C'e^{-c\Omega t}$. (Cohorts that revise after τ_0 leave this group for the common $\phi^\star$ part; those that have not, decay with age.) Combining the three groups through (G2), $\mathbb{E}_{\mathbb{P}}[d_{\text{BL}}(\Psi_t, \Psi_t^\infty)] \leq C''e^{-c\Omega t}$, and by Markov's inequality $d_{\text{BL}}(\Psi_t, \Psi_t^\infty) \to 0$ in probability, geometrically at rate $c\Omega$ (a rate proportional to the demographic rate Ω; the lock-time moment caps c below 1, though W_t^∞ and W_t^{init} alone decay at rate Ω).

Mode and uniqueness. The coupled measures themselves converge: because the post-τ_0 part enters both as a common measure, the bound (G2) holds a fortiori in total variation, $\|\Psi_t - \Psi_t^\infty\|_{\text{TV}} \leq 2(W_t^{\text{init}} + W_t^\infty + W_t^{\text{pre}}) \to 0$, with $d_{\text{BL}} \leq \|\cdot\|_{\text{TV}}$ (this is the total-variation distance between the coupled *realisations* Ψ_t, Ψ_t^∞ on a common probability space, not between their laws). Through $\mathrm{W}_1\big(\mathrm{Law}(\Psi_t), \Psi^\infty\big) \leq \mathbb{E}_{\mathbb{P}}[d_{\text{BL}}(\Psi_t, \Psi_t^\infty)] \leq C''e^{-c\Omega t}$ this gives convergence of the *law* of Ψ_t to Ψ^∞ in the Wasserstein-1 (bounded-Lipschitz) metric on laws over $\mathcal{P}(\Phi \times \mathbb{R}_+ \times \mathcal{R})$, geometrically at rate $c\Omega$—the asymptotic-coupling route to Wasserstein ergodicity of Hairer et al. (2011). We do *not* claim total-variation convergence of the law in the Meyn and Tweedie (2009) sense: the synchronous coupling never sets $\Psi_t = \Psi_t^\infty$ (a positive, if vanishing, wealth of distinct-atom cohorts always differs), so it is not a successful coupling and yields no minorisation, and the measure-valued birth–death dynamics are not shown to satisfy a Doeblin condition in total variation—which the stationary moments do not require, being long-run *time averages* delivered by the Birkhoff theorem of Step 3 under integrability alone, a route that needs no continuity and no control of the law beyond ergodicity. Because the two-sided regime Ψ^∞ is independent of Ψ_0, every initialisation converges to the *same* limit, so the stationary law is unique; the dependence of Ψ_t, and of every functional of Step 3, on the initial cross-section and the remote past decays geometrically, at a rate proportional to Ω. □

Appendix H. The option surface against its benchmarks

The benchmark decomposition of Proposition 5 sorts the disaster-option literature by which side of the surface each model occupies: every documented feature reads the current cross-section, the level any Markov pricing state reproduces, while the term-structure slope reads the survival records only lazy revision produces.

Table III: What the option surface shows, and what reproduces it.

Object on the surface (anchor work)	**What it reads**	**Reproduced by**
Thin far tail on average (Backus et al., 2011)	the current cross-section: complacency $\kappa_t < 1$	any Markov or Bayesian benchmark
Investor-fears wedge, spiking after crises (Bollerslev and Todorov, 2011; Andersen et al., 2015b)	a current-state factor	a Markov latent state
Fast-in, slow-out jump premium (self-exciting models)	a post-crisis spike, slow reversion	a clustering intensity
Leverage effect, volatility dynamics (Dew-Becker et al., 2025)	a smoothly filtered intensity	its own Markov pricing state
Raw path dependence of implied volatility, *weakening* with maturity (Guyon and Lekeufack, 2023; Andrès et al., 2024)	the recent path, already in the cross-section	any model: the cross-section spans it
Slope of the tail term structure, *growing* with maturity (this paper)	**the records under which beliefs have survived**	**no belief-sufficient model**

Appendix I. A representative-agent Bayesian benchmark

This appendix records the benchmark against which the lazy economy is read: a *single* representative agent who updates a common prior by Bayes' rule and prices under the resulting predictive law, with no surprise threshold, no lazy holding, and—being one agent—no cross-section. It isolates what the main mechanism adds by removing all three. All objects are derived here from the same primitives (Assumptions 1, 3, and 4); the verbal comparison with the body is made only after each benchmark object is derived.

The benchmark economy

Endow the economy of Assumption 1 with a single representative agent with common CRRA utility $u(c) = c^{1-\alpha}/(1-\alpha)$ and discount δ, who consumes the aggregate endowment C_t. For a model $\phi = (\lambda, \theta) \in \Phi$ let $M_t^\phi := \frac{d\mathbb{P}^\phi}{d\mathbb{P}}\big|_{\mathcal{F}_t}$ be its belief density (C1), a strictly positive $(\mathcal{F}_t, \mathbb{P})$-martingale with $\mathbb{E}[M_t^\phi] = 1$ (Appendix C, verbatim, with ϕ in place of the held model). The agent holds the common second-order prior ρ_0 of Assumption 2 and prices under the *Bayesian predictive measure*

$$\mathbb{P}^B := \int_\Phi \mathbb{P}^\phi \, \rho_0(d\phi), \qquad Z_t^B := \frac{d\mathbb{P}^B}{d\mathbb{P}}\Big|_{\mathcal{F}_t} = \int_\Phi M_t^\phi \, \rho_0(d\phi). \tag{I1}$$

Lemma 8 (The Bayesian belief density): *Z^B is a strictly positive $(\mathcal{F}_t, \mathbb{P})$-martingale with $Z_0^B = 1$ and $\mathbb{E}[Z_t^B] = 1$, and the implied posterior is the common posterior of (4): writing $\rho_t^B(d\phi) := M_t^\phi \rho_0(d\phi)/Z_t^B$,*

$$\rho_t^B = \rho_t \ \textit{(of (4))}, \qquad \frac{Z_t^B}{Z_{t-}^B}\bigg|_{J=x} = \frac{\int_\Phi M_{t-}^\phi \, r(\phi, x)\, \rho_0(d\phi)}{\int_\Phi M_{t-}^\phi \, \rho_0(d\phi)} = \mathbb{E}_{\rho_{t-}^B}\big[r(\cdot, x)\big], \tag{I2}$$

the posterior-mean disaster likelihood ratio, with $r(\phi, x) = \lambda f_\theta(x)/(\lambda^\star f(x))$ as in (20).

Proof. Each M^ϕ is a positive $\mathbb{P}$-martingale with $\mathbb{E}[M_t^\phi] = 1$ (Appendix C); by Tonelli the mixture $Z_t^B = \int_\Phi M_t^\phi \rho_0(d\phi) \geq 0$ has, for $s < t$, $\mathbb{E}[Z_t^B \mid \mathcal{F}_s] = \int_\Phi \mathbb{E}[M_t^\phi \mid \mathcal{F}_s]\rho_0(d\phi) = \int_\Phi M_s^\phi \rho_0(d\phi) = Z_s^B$ (the interchange justified by positivity and $\mathbb{E}\int M_t^\phi \rho_0(d\phi) = \int \mathbb{E}[M_t^\phi]\rho_0(d\phi) = 1 < \infty$), so Z^B is a martingale with $\mathbb{E}[Z_t^B] = 1$; $Z_0^B = \int M_0^\phi \rho_0 = 1$, and strict positivity holds because each $M_t^\phi > 0$ (Appendix C) and ρ_0 has full support (Assumption 2). The normalized measure $\rho_t^B(d\phi) = M_t^\phi \rho_0(d\phi)/Z_t^B$ is, by (C1) and Bayes' rule, proportional to $\rho_0(\phi)e^{-\lambda t}\lambda^{N_t}\prod_k f_\theta(J_k)$, which is (4); hence $\rho_t^B = \rho_t$. At a disaster of size x, $M_t^\phi = M_{t-}^\phi r(\phi, x)$ (C1), so $Z_t^B/Z_{t-}^B = \int M_{t-}^\phi r(\phi, x)\rho_0 / \int M_{t-}^\phi \rho_0 = \mathbb{E}_{\rho_{t-}^B}[r(\cdot, x)]$, which is (I2). □

The benchmark kernel and fear factor

Proposition 19 (Benchmark state-price density and fear factor): *In the representative-agent Bayesian economy the unique state-price density is*

$$\xi_t^B = e^{-\delta t} C_t^{-\alpha} Z_t^B, \tag{I3}$$

and the benchmark risk-neutral disaster law tilts the physical law by risk aversion and by the predictable (pre-jump) posterior-mean likelihood ratio: with $\zeta_t^B(x) := \mathbb{E}_{\rho_{t-}^B}[r(\cdot, x)]$,

$$\nu^{\mathbb{Q},B}(dt, dx) = \lambda^\star e^{\alpha x}\zeta_t^B(x)\, dt\, F_J(dx), \qquad U_t^B = U^{\mathrm{RA}}\kappa_t^B, \qquad \kappa_t^B = \frac{\mathbb{E}_f[e^{\alpha J}\zeta_t^B(J)]}{\mathbb{E}_f[e^{\alpha J}]}. \tag{I4}$$

Proof. The agent values an $\mathcal{F}_t$-payoff X at $\mathbb{E}^{\mathbb{P}}[\, e^{-\delta t}u'(C_t)\,(d\mathbb{P}^B/d\mathbb{P})X\,] = \mathbb{E}^{\mathbb{P}^B}[e^{-\delta t}C_t^{-\alpha}X]$ (change of measure by (I1)), so the state-price density under $\mathbb{P}$ is $\xi_t^B = e^{-\delta t}u'(C_t)Z_t^B = e^{-\delta t}C_t^{-\alpha}Z_t^B$; market clearing is automatic (one agent consumes C_t) and uniqueness is the strict concavity of u. This is the single-agent case of (10): with one belief the aggregator $\Xi_t = (Z_t^B)^{1/\alpha}$ (the multiplier cancels), so $\Xi_t^\alpha = Z_t^B$ and the $\frac{1}{\alpha}$- and α-powers compose to the identity—the power-mean distortion of (10) is absent. The ξ^B jump ratio at a disaster of size x is, by (I3), $C_t^{-\alpha}/C_{t-}^{-\alpha}$ times Z_t^B/Z_{t-}^B, i.e. $e^{\alpha x}\zeta_t^B(x)$ by (I2); the change-of-measure theorem (Appendix C, hypotheses verified there: Z^B a positive martingale, ζ_t^B predictable, strictly positive, and bounded by $\bar{r}$ via (C1)) gives $\nu^{\mathbb{Q},B} = e^{\alpha x}\zeta_t^B(x)\nu$, the first equation of (I4). Integrating the size out, $U_t^B = \lambda^\star \mathbb{E}_f[e^{\alpha J}\zeta_t^B(J)] = U^{\mathrm{RA}}\kappa_t^B$ with κ_t^B as displayed, exactly as in Proposition 2 but with $\zeta_t^B = \mathbb{E}_{\rho_{t-}^B}[r(\cdot,\cdot)]$ in place of g_t^α. □

Proposition 20 (Merging: the benchmark wedge has no persistent variation): *Under Assumptions 1 and 4, as $t \to \infty$ the posterior concentrates on the Kullback–Leibler model, $\rho_t^B \Rightarrow \delta_{\phi^\star}$ $\mathbb{P}$-a.s. (with $\phi^\star = (\lambda^\star, \theta^\star)$ of (A3)), so*

$$\zeta_t^B(x) \longrightarrow r(\phi^\star, x) = \frac{f_{\theta^\star}(x)}{f(x)} \quad \textit{for each } x, \qquad \kappa_t^B \longrightarrow \kappa_\infty^B := \frac{\mathbb{E}_f[e^{\alpha J} f_{\theta^\star}(J)/f(J)]}{\mathbb{E}_f[e^{\alpha J}]}, \tag{I5}$$

a constant; if the size family is correctly specified ($f_{\theta^\star} = f$) then $\zeta_t^B \to 1$, $\kappa_\infty^B = 1$, and $U_t^B \to U^{\mathrm{RA}}$. The approach is smooth and shrinking: the predictive moments converge at the rare-disaster rate, $\zeta_t^B(x) - r(\phi^\star, x) = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big)$, so the increments of $\log U_t^B$—its calm drift and its disaster-jump size—vanish in probability at that rate, $d \log U_t^B \to 0$.

Proof. Concentration follows from Berk's theorem applied directly to the single posterior $\rho_t^B = \rho_t$ (Lemma 8). Three facts hold. The record splits into i.i.d. unit-interval blocks by the stationary independent increments of Assumption 1; the family is identifiable with a unique Kullback–Leibler minimiser $\phi^\star$ (Assumption 4(iii)–(iv), Lemma 2); and the per-block divergence $\mathcal{K}(\phi)$ is finite, with that unique minimiser (Lemma 2). These are exactly the hypotheses of Berk (1966), whose theorem—for an identifiable dominated family observed i.i.d. with finite divergence to the family—gives posterior concentration on the divergence-minimising parameter, $\rho_t^B \Rightarrow \delta_{\phi^\star}$ $\mathbb{P}$-a.s. (The same three facts drive the wealth-weighted concentration of Proposition 18(ii); here they are applied to the single posterior, requiring none of the demographic machinery.) Since $x \mapsto r(\phi, x)$ is continuous and bounded on the compact $\mathcal{J} \times \Phi$ (Assumption 4(i)), weak convergence of ρ_t^B (hence of its left limit ρ_{t-}^B) to $\delta_{\phi^\star}$ gives $\zeta_t^B(x) = \mathbb{E}_{\rho_{t-}^B}[r(\cdot, x)] \to r(\phi^\star, x)$ for each x, and bounded convergence gives $\kappa_t^B \to \kappa_\infty^B$ of (I5); $r(\phi^\star, x) = (\lambda^\star/\lambda^\star)(f_{\theta^\star}(x)/f(x)) = f_{\theta^\star}(x)/f(x)$, which is 1 when $f_{\theta^\star} = f$. The rate is the posterior-contraction (Bernstein–von Mises) counterpart of Lemma 3(iii): under the smoothness of Assumption 4(vi) the posterior contracts on the Kullback–Leibler point $\phi^\star$ at the maximum-likelihood rate—by the misspecified Bernstein–von Mises theorem (Kleijn and van der Vaart, 2012) when $f_{\theta^\star} \neq f$, and the classical one (van der Vaart, 1998) when $f_{\theta^\star} = f$—itself governed by the disaster count $N_t \sim \text{Poisson}(\lambda^\star t)$, so the posterior-predictive mean ζ_t^B concentrates at order $(\lambda^\star t)^{-1/2}$—this requires the contraction rate, not merely the maximum-likelihood rate that Lemma 3(iii) supplies; hence $d \log U_t^B = d \log \kappa_t^B \to 0$ at that rate. □

What the lazy economy adds

With the benchmark objects derived, the contrast is exact and lives in four places.

(i) No endogenous disagreement. The benchmark has one agent and no surprise threshold, so there is no cross-section and nothing to disperse; $\text{Var}(\cdot) = 0$ identically. The lazy economy manufactures a non-degenerate cross-section from an identical start through the threshold heterogeneity and the revision rule (Proposition 4)—disagreement that is a *result*, not a primitive, and that the single Bayesian cannot have.

(ii) Merging kills the wedge; lazy revision need not. Here κ_t^B converges to the constant κ_∞^B (Proposition 20): the belief-driven *variation* of the fear factor dies out as the predictive laws merge (Blackwell and Dubins, 1962; Kalai and Lehrer, 1993), and with correct specification the wedge relaxes to the frictionless benchmark U^{RA}. This is the single-agent face of Remark 5(b). The lazy economy escapes it only through demographic turnover, which keeps the cross-section non-degenerate and sustains a *stationary, fluctuating* wedge (Proposition 18); an infinitely-lived lazy agent alone merges just as the Bayesian does (Remark 6).

(iii) Shrinking-and-merging versus threshold dynamics. The benchmark Bayesian belief is itself càdlàg—it jumps at each disaster and moves continuously between—but its variation *shrinks and merges*, vanishing at rate $(\lambda^\star t)^{-1/2}$ (Proposition 20). The lazy economy differs not by carrying jumps the benchmark lacks, but in two structural ways: *individual* held beliefs

stay flat within a tolerance and then jump at a threshold-triggered rejection, and the *aggregate* fear factor does not vanish—it drifts in calm through wealth-reallocating selection and, under turnover, settles at a non-degenerate stationary cross-section with persistent wedges (Propositions 6, 18). The contrast is shrinking-and-merging versus threshold holding/switching with a non-vanishing stationary spread, not continuity versus jumps.

(iv) Arithmetic posterior mean versus wealth-weighted power mean. The benchmark prices the posterior *arithmetic* mean $\zeta_t^B = \mathbb{E}_{\rho_{t-}^B}[r(\cdot, x)]$ (Proposition 19); the lazy economy prices the wealth-weighted $\frac{1}{\alpha}$-*power* mean $g_t(x)^\alpha = \left(\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]\right)^\alpha$ (Proposition 1, Proposition 2). The power-mean gap below the wealth-weighted arithmetic mean—generic under disagreement, and the source of complacency in calm spells—is intrinsically a heterogeneity phenomenon: it closes (power mean equals arithmetic mean) for the single agent and is therefore invisible in the benchmark.

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